

Uniform excess charge for Coulomb molecules and the outer radius of neutral atoms

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Abstract

We prove that a molecule with M fixed nuclei of real charges at least one and total nuclear charge Z strictly binds at most $Z + CM$ electrons, where C is universal. The result holds for the full nonrelativistic Coulomb Hamiltonian with two spin states and arbitrary distinct nuclear positions. For neutral atoms with integer nuclear charge $Z \geq 1$, the first ionization energy and, for every ground state, the radius outside which one half of an electron remains are each bounded above and below by positive universal constants.

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1 Introduction

We prove a uniform bound on the number of excess electrons that fixed Coulomb nuclei can strictly bind. If there are M nuclei of total charge Z , that number is at most CM , with a constant independent of their charges and positions. For a neutral atom with integer nuclear charge, we also prove that the cost of removing one electron and the radius containing all but one half of an electron each stay between positive universal constants. These conclusions address the number of bound electrons, the first ionization energy, and the location of the outer electronic mass. Their proofs share the local screening estimates developed in this paper.

1.1 The model and the results

Fix $M \geq 1$, distinct positions $R_1, \dots, R_M \in \mathbb{R}^3$, and real charges $z_1, \dots, z_M \geq 1$. Put

$$Z = \sum_{j=1}^M z_j, \quad \nu = \sum_{j=1}^M z_j \delta_{R_j}, \quad K(x) = |x|^{-1}, \quad V = K * \nu.$$

For $n \geq 1$ let $\mathcal{H}_n = \bigwedge^n L^2(\mathbb{R}^3; \mathbb{C}^2)$, so that there are two electron spin states and antisymmetry exchanges spatial and spin variables together. The Hamiltonian is the self-adjoint operator associated with the closed, bounded-below Coulomb form on $\mathcal{Q}_n = \mathcal{H}_n \cap H^1((\mathbb{R}^3)^n; (\mathbb{C}^2)^{\otimes n})$:

$$H_n = \sum_{i=1}^n \left(-\frac{1}{2} \Delta_i - V(x_i) \right) + \sum_{1 \leq i < j \leq n} |x_i - x_j|^{-1}. \quad (1.1)$$

The nuclei are fixed; their mutual repulsion is a scalar independent of n and is omitted. Write $E_n = \inf \sigma(H_n)$ and $E_0 = 0$. We say that sector n *binds strictly* if $E_n < E_{n-1}$.

Theorem 1.1 (Uniform excess charge). *There is a universal finite constant C such that every nuclear system specified above and every integer $n \geq 1$ satisfy*

$$E_n < E_{n-1} \implies n \leq Z + CM.$$

Moreover, $E_n = E_{n-1}$ whenever $n > Z + CM$. For one nucleus this gives $n \leq Z + C$ under strict binding, for every real $Z \geq 1$. The same constant is independent of all charges and nuclear positions.

The positions may be arbitrarily close or far apart. The theorem resolves positively the excess-charge form of the ionization conjecture in this strict-binding formulation [7, 19]. Equality of adjacent energies does not by itself exclude a ground state at threshold; such threshold states are not needed here.

For the radius conclusion specialize to one nucleus at the origin with integer charge $Z \geq 1$. Write $H_{n,Z}$ and $E(n, Z)$ when the nuclear charge needs to be displayed. If Ψ_Z is a normalized ground state of $H_{Z,Z}$, its spin-summed density is

$$\rho_{\Psi_Z}(x) = Z \sum_{\sigma_1, \dots, \sigma_Z} \int |\Psi_Z(x, \sigma_1; x_2, \sigma_2; \dots; x_Z, \sigma_Z)|^2 dx_2 \cdots dx_Z,$$

where the integral is absent for $Z = 1$. Its mass is Z . Define

$$R(\Psi_Z) = \inf \left\{ r \geq 0 : \int_{|x| > r} \rho_{\Psi_Z}(x) dx \leq \frac{1}{2} \right\}. \quad (1.2)$$

Theorem 1.2 (Uniform outer half-electron radius). *There are universal constants $0 < c \leq C < \infty$ such that, for every integer $Z \geq 1$ and every normalized ground state of $H_{Z,Z}$,*

$$c \leq R(\Psi_Z) \leq C.$$

The existence of these ground states is included in Lemma 2.5. No uniqueness or spherical symmetry is assumed. The exterior mass in (1.2) stays fixed as Z grows; it describes outer electrons rather than a fixed fraction of the atom.

Corollary 1.3 (Two-sided first-ionization bound). *For the full nonrelativistic Coulomb atom specified above, with integer nuclear charge $Z \geq 1$, one-electron kinetic term $-\Delta/2$, and two electron spin states, write $E_Z(n) = E(n, Z)$. There are universal constants $0 < c \leq C < \infty$ such that the first ionization energy of the neutral atom satisfies*

$$c \leq I_1(Z) := E_Z(Z-1) - E_Z(Z) \leq C \quad \text{for every integer } Z \geq 1.$$

1.2 History and the scales of the problem

Simon’s atomic formulation asks whether $N_0(Z) - Z$ stays bounded, where $N_0(Z)$ is the first sector at which the energy equals every later sector energy [19, Problem 9]. Since the sector energies are nonincreasing and ultimately constant, this is the largest strictly binding sector used here. The corresponding molecular conjecture has the form $N_c \leq Z + CM$ [7, Section 2.1.2].

The early charge bounds already distinguished atoms from molecules. Lieb proved bounds linear in the total nuclear charge for fixed atoms and molecules, without requiring Fermi statistics [8]. For fermionic atoms, Lieb, Sigal, Simon and Thirring established asymptotic neutrality: the maximal strictly bound particle number $N_c(Z)$ satisfies $N_c(Z)/Z \rightarrow 1$ as $Z \rightarrow \infty$ [12, 13]. Fefferman and Seco, and Seco, Sigal and Solovej, subsequently obtained quantitative sublinear bounds [2, 18]. Ivrii obtained further semiclassical charge bounds for atoms and heavy molecules under the hypothesis that each nuclear charge is comparable to the electron number, with improved exponents for atoms and for suitably separated nuclei [5, Theorem 5.2.1]. For explicit finite-charge estimates, Nam proved $N_c(Z) < 1.22Z + 3Z^{1/3}$ [14]. More recently, Hundertmark, Pattakos and Schulz obtained $N_c(Z) < 1.1185Z + 4Z^{1/3}$ for $Z \geq 4$ [4]. These estimates supply useful quantitative control while leaving the uniform excess-charge assertion distinct.

For the full many-electron Schrödinger atom, Seco, Sigal and Solovej also obtained, for large Z , an upper bound for the neutral first ionization energy and a lower bound for an exterior-mass radius, both given by powers of the nuclear charge [18, Theorems 1 and 3]. Their radius leaves one expected electron outside, whereas (1.2) leaves one half. Their energy upper bound grows with Z , and their radius lower bound decays with Z . The upper bound in Corollary 1.3 establishes the usual boundedness assertion for $I_1(Z)$ [22, Section 2]; the uniform positive lower bound is an additional conclusion.

The statistical theories introduced by Thomas and Fermi [3, 23] predict a particularly rigid exterior screening profile. Lieb and Simon established the variational theory and its leading large-charge energy and density limits [9, 10]. Uniform ionization bounds were proved in Thomas–Fermi–von Weizsäcker theory by Benguria and Lieb [1], and in reduced and unrestricted Hartree–Fock theory by Solovej [20, 21]. Solovej’s unrestricted Hartree–Fock analysis compares screened potentials through successive spatial scales and also establishes uniform bounds and asymptotics for outer-electron radii [21, Definition 1.2, Theorem 1.5, and Sections 10–13]. Those conclusions concern their stated variational models; the states in (1.1) may have arbitrary many-body correlations.

The bulk density limit lives on the length scale $Z^{-1/3}$, whereas the normalized mass outside the radius in (1.2) is $1/(2Z)$. Its vanishing fraction makes bulk convergence insufficient to prove Theorem 1.2. The fixed exterior-mass viewpoint is also central to the generalized ionization conjecture [22, Section 3].

Companion papers use the estimates developed here, with additional arguments, to obtain Thomas–Fermi asymptotics for full two-spin Coulomb atoms. The energy result sends the number of removed electrons to infinity while it remains negligible relative to the nuclear charge [15, Theorem 1.1]; the neutral-radius result first takes upper and lower large-charge limits at fixed expected exterior electron mass, and then lets that mass tend to infinity [16, Theorem 1.1].

The analytic ingredients have separate roles. The Lieb–Thirring kinetic inequality provides a positive coarse density bound, for which we use Rumin’s spectral-splitting argument [11, 17]. The exact Thomas–Fermi coefficient comes instead from coherent-state filling and Neumann eigenvalue counting, with explicit remainders. These constructions have classical antecedents in the semiclassical analysis of Lieb and Simon and in the coherent-state comparison of Solovej [10, Theorems III.10–III.13] [21, Lemma 8.2]. Spatial square partitions are the usual IMS localization;

sector-valued localization for correlated states is systematically developed by Lewin [6, Proposition 3.1 and Example 3.3]. Our conditional core construction retains the removed positions and spins, so that the remaining correlated state can be compared with local trial electrons through its own screened field.

1.3 Obstacles and the common argument

Fixing the total electron number obstructs a local density comparison: a ball may contain too many or too few electrons, and its conditional core need not minimize any fixed local-particle-number problem. We therefore first work relative to the minimum $E = \inf_{n \geq 0} E_n$ over all sectors. A coarse binding bound guarantees that it is attained at a largest strictly bound sector. A normalized state with form energy at most $E + \delta$ will be called a δ -state. Keeping δ visible makes local estimates available again after an event is imposed, after particles are deleted, and after a shell weight changes the probability law.

Three steps are common to the molecular and atomic conclusions. First, a spatial cut records every removed electron and leaves an antisymmetric conditional core. In the hole, arbitrary expected trial mass is allowed because each actual trial has an integer sector with energy at least E . The coherent upper and Neumann lower comparisons have the same kinetic coefficient. Their comparison produces a local Thomas–Fermi minimizer ρ and its screened field W , related by

$$\rho = k(W_+)^{3/2}, \quad k = \frac{2^{3/2}}{3\pi^2}.$$

It also controls the Coulomb discrepancy from the actual fine density and the energy of that density in the negative part of W . A bootstrap for the positive field closes a bound on the second moment of local electron counts.

Next, smoothing and noisy observations produce a conditional density μ and a screened field. In the atomic case the field is $H = V - K * \mu$. On nucleus-free patches we transfer the local Thomas–Fermi law to these quantities, as one-sided comparisons between μ and $k(H_+)^{3/2}$ at positive, bounded rescaled heights. The molecular comparison retains the nearby nuclear potential explicitly: after splitting off its singular part, we compare the continuous conditional offset through an averaged Thomas–Fermi response.

To prove these comparisons, observe all positions with independent errors of width ℓ and forget the particle labels. An observation event of probability p has a symmetric square-root likelihood multiplier. Applied to a sector eigenstate, its energy cost is bounded by $C\ell^{-2} \log^5(e/p)$, without a particle-number factor. A hypothetical failure can therefore be tested by the local quantum estimates. The original observation posterior and the fresh patch posterior are different: two separate conditional-expectation identities compare their fields after averaging in the event law. The negative-field penalty controls the rare configurations needed to pass a lower field estimate to expectation.

Finally, a positive comparison field is continued outward. A local maximum chooses between a shifted nonlinear subsolution and a shifted observed field. The inverse comparisons let us replace the model reaction by the actual conditional density on one more layer. Conditional averaging then forgets one observation array, with convexity preserving the nonlinear inequality. The errors are summed through their expected integrals. At the terminal scale, comparison with a compact source potential and its spherical mean turns the positive field into a lower bound on the nuclear charge minus the enclosed electron mass, up to the accumulated error.

The geometry determines how this mechanism closes. For a molecule, the local length measures nearby nuclear charge. Empty nuclear annuli must be counted without a loss for their number of

scales. A decomposition by nested nuclear representatives gives a total cost proportional to M ; it bounds the electrons outside the terminal region and closes the excess-charge estimate.

For an atom, local balls away from the nucleus use their distance from it; this distinct scale reaches inside the bulk length. The continuation produces an interior electron deficit for sector ground states with $n \leq 3Z$ and bounded energy offset from E . We first use it at offset zero. Deleting exterior particles then proves $E_{Z-1} \leq E + C$, so both neutral and singly ionized ground states become eligible for the fixed-offset estimates. The neutral deficit gives the lower radius bound. A spatially averaged insertion into the singly ionized state gives a uniform positive gap $E_{Z-1} - E_Z \geq c$. Together with the offset bound this proves Corollary 1.3, after adjusting the constants at finitely many smaller charges. The gap also yields the expected tail bound C/R through a shell-weighted change of law. Here the shell weight is measurable in the revealed data, which is why this change of law preserves the surviving conditional field.

1.4 Organization and normalization

Section 2 provides the sector facts, and Sections 3–5 prove the shared localization, screening, and conditioning estimates. The two local geometries are specified in Section 4. From these shared tools, Sections 6–9 prove the molecular excess-charge bound. A reader interested in the neutral atomic results may instead proceed directly to Sections 10–14; these proofs do not use the molecular excess-charge theorem. The atomic comparison and intermediate subsolutions come first, followed by deletion, insertion, and the tail argument in the order described above. Section 15 collects the precise inputs for the companion asymptotic problems, including the classical Thomas–Fermi comparisons proved in Section 7.

All density integrals include spin summation. For a real number or function v , write $v_+ = \max(v, 0)$ and $v_- = \max(-v, 0)$. The constants $C, c > 0$ may change from line to line and are universal unless a dependence is stated. With the kinetic normalization in (1.1),

$$c_{\text{TF}} = \frac{3}{10}(3\pi^2)^{2/3}, \quad k = (5c_{\text{TF}}/3)^{-3/2} = \frac{2^{3/2}}{3\pi^2}, \quad c_F = 4\pi k. \quad (1.3)$$

The coarse kinetic lower bound does not assert that its unspecified constant equals c_{TF} .

2 A ground state at the minimum over all sectors

The local comparisons in the proof may replace electrons in a region by a different number of electrons. We therefore first reduce to a ground state whose energy is minimal among *all* particle-number sectors. A coarse number bound makes this reduction possible; no convexity of the sector energies is needed.

2.1 The Coulomb form and strict binding

Use the Hilbert spaces and form domains defined in the Introduction, writing n for a variable particle number. The quadratic form associated with (1.1) is

$$q_n[\psi] = \frac{1}{2} \sum_{i=1}^n \|\nabla_i \psi\|_2^2 - \sum_{i=1}^n \int V(x_i) |\psi|^2 + \sum_{1 \leq i < j \leq n} \int K(x_i - x_j) |\psi|^2. \quad (2.1)$$

We also evaluate this expression on the ambient spin-valued H^1 space when a localization preserves antisymmetry only within groups of variables. Sector-energy bounds will be applied only to the

groups that remain antisymmetric. Hardy's inequality in three dimensions, followed by Cauchy-Schwarz, shows that each Coulomb term is infinitesimally form bounded relative to the kinetic energy. For a pair term, apply the same argument to the relative coordinate $x_i - x_j$. Thus, for each fixed system and sector, the sum of the absolute Coulomb forms is bounded by

$$\eta \sum_i \|\nabla_i \psi\|_2^2 + C_\eta \|\psi\|_2^2 \quad (\eta > 0).$$

Consequently (2.1) is closed and bounded below on $\mathcal{Q}_n$. Its associated self-adjoint operator is the Hamiltonian H_n in the Introduction, and

$$E_n = \inf_{\substack{\psi \in \mathcal{Q}_n \\ \|\psi\|_2=1}} q_n[\psi] = \inf \sigma(H_n).$$

Smooth compactly supported antisymmetric functions form a form core: approximate in H^1 and apply the finite antisymmetrization projection. Set $\mathcal{H}_0 = \mathcal{Q}_0 = \mathbb{C}$, $q_0 = 0$, and $E_0 = 0$. The constants in these elementary domain facts may depend on the fixed system and on n .

Lemma 2.1 (Monotonicity and strict-binding compactness). *For every fixed nuclear system, $E_{n+1} \leq E_n$ for $n \geq 0$, and $E_1 < 0$. If $E_n < E_{n-1}$, there is a normalized $\Psi \in \mathcal{Q}_n$ with $q_n[\Psi] = E_n$.*

Proof. Approximate an n -electron trial state by one with compact support. Wedge it with a normalized spin orbital in a disjoint distant ball. Dilate the orbital to make its kinetic energy small and separate the supports far enough to make the added repulsion small. Disjoint spatial supports make the energy of the normalized wedge the sum of the group energies and their direct cross repulsion. Dropping the added nuclear attraction proves $E_{n+1} \leq E_n$. A normalized orbital dilated about one nucleus has kinetic energy $O(s^{-2})$ and attraction at most $-cz_m s^{-1}$, proving $E_1 < 0$.

Suppose $\Delta = E_{n-1} - E_n > 0$, and take a normalized minimizing sequence ψ_j . The form bound above gives a uniform H^1 bound. Choose nonnegative Lipschitz functions g_S, h_S with

$$g_S^2 + h_S^2 = 1, \quad g_S = 1 \text{ on } B(0, S), \quad g_S = 0 \text{ outside } B(0, 2S), \quad |\nabla g_S| + |\nabla h_S| \leq C/S.$$

They may be constructed as the cosine and sine of a cutoff angle. For each subset I of the particle labels, multiply ψ_j by $\prod_{i \in I} g_S(x_i) \prod_{i \notin I} h_S(x_i)$. The squared norms of these localized states sum to one. The product partition identity for the kinetic energy adds at most Cn/S^2 to the sum of their forms.

The all- g_S component has energy at least E_n times its squared norm. If $k < n$ coordinates carry g_S , slice in the remaining coordinates. Antisymmetry in the k retained coordinates gives their energy lower bound $E_k \geq E_{n-1}$. Drop the other kinetic terms and all pair terms involving an h_S coordinate. For $S > 2 \max_m |R_m|$, their total nuclear attraction is bounded below by $-2Zn/S$ times the component's squared norm. If $p_j(S)$ is the total squared mass of components other than all- g_S , it follows that

$$(\Delta - 2Zn/S)p_j(S) \leq q_n[\psi_j] - E_n + Cn/S^2.$$

The probability that any $|x_i| \geq 2S$ is at most $p_j(S)$, because the all- g_S product vanishes there. Taking first $j \rightarrow \infty$, then $S \rightarrow \infty$, proves tightness.

Local Rellich compactness and tightness give, along a subsequence, strong L^2 convergence to a normalized antisymmetric Ψ , weak H^1 convergence, and almost-everywhere convergence. The attraction converges: the infinitesimal form bound applied to $\psi_j - \Psi$, with its relative-bound parameter subsequently tending to zero, gives

$$\sum_i \int V(x_i) |\psi_j - \Psi|^2 \rightarrow 0.$$

Cauchy–Schwarz with this weight handles the difference of the attractive expectations. The kinetic energy is weakly lower semicontinuous, and the nonnegative repulsion is lower semicontinuous by Fatou’s lemma. Hence Ψ minimizes. Variation in the form domain gives the weak eigen-equation at E_n . $\square$

2.2 A coarse number bound

The following reciprocal-potential test is a weighted-electron argument of the type used in charge-linear ionization bounds [8]. We include the calculation, including the regularization needed for the form domain.

Lemma 2.2 (Reciprocal-potential bound). *If $E_n < E_{n-1}$ and every $z_m \geq 1$, then $n \leq 3Z$.*

Proof. Let Ψ be the minimizer from Lemma 2.1. For $l, \varepsilon > 0$ put

$$V_l(x) = \sum_m z_m (|x - R_m|^2 + l^2)^{-1/2}, \quad U = V_l + \varepsilon, \quad w = U^{-1}.$$

For these fixed parameters, w and its first derivatives are bounded. Test the weak eigen-equation with $(\sum_i w(x_i))\Psi$ and take real parts. This summed multiplier is symmetric; after taking this admissible test we may expand its individual summands in the ambient form.

In the i th summand, the Hamiltonian on the other $n-1$ coordinates, minus E_n , has nonnegative pairing after weighting by $w(x_i)$. This follows by slicing in x_i and its spin and using $E_{n-1} \geq E_n$. The remaining kinetic pairing is also nonnegative. Indeed, with $\phi = w(x_i)\Psi$,

$$\begin{aligned} \operatorname{Re} \int \nabla_i \Psi \cdot \overline{\nabla_i (w(x_i)\Psi)} &= \int U(x_i) |\nabla_i \phi|^2 - \frac{1}{2} \int (\Delta V_l)(x_i) |\phi|^2 \geq 0, \\ \Delta V_l(x) &= -3l^2 \sum_m z_m (|x - R_m|^2 + l^2)^{-5/2} \leq 0. \end{aligned}$$

The integration by parts is justified by large-radius cutoffs and H^1 approximation; all displayed coefficients and their needed derivatives are bounded for fixed l, ε . We obtain

$$\mathbb{E}_\Psi \sum_{i < j} \frac{w(x_i) + w(x_j)}{|x_i - x_j|} \leq \mathbb{E}_\Psi \sum_i w(x_i) V(x_i). \quad (2.2)$$

Let $l \downarrow 0$ at fixed ε . Dominated convergence applies, using V/ε for attraction and $2K/\varepsilon$ for a pair. These functions have finite expectations on the form domain. Hence (2.2) holds with $w = (V + \varepsilon)^{-1}$, and its right side is at most n .

For a configuration outside the null set of nuclear collisions, set

$$q_{im} = \frac{w(x_i) z_m}{|x_i - R_m|}, \quad s_i = \sum_m q_{im} = \frac{V(x_i)}{V(x_i) + \varepsilon} \leq 1.$$

The triangle inequality gives

$$\frac{w(x_i) + w(x_j)}{|x_i - x_j|} \geq \sum_m \frac{q_{im} q_{jm}}{z_m}.$$

In fact, multiplying the right side by $|x_i - x_j|$ bounds it above by $w(x_i)w(x_j)(V(x_i) + V(x_j)) \leq w(x_i) + w(x_j)$. Weighted Cauchy–Schwarz and $z_m \geq 1$ now imply

$$\begin{aligned} \sum_{i < j} \sum_m \frac{q_{im} q_{jm}}{z_m} &= \frac{1}{2} \sum_m \frac{(\sum_i q_{im})^2 - \sum_i q_{im}^2}{z_m} \\ &\geq \frac{(\sum_i s_i)^2}{2Z} - \frac{n}{2}. \end{aligned}$$

Here $\sum_m q_{im}^2/z_m \leq \sum_m q_{im}^2 \leq s_i^2 \leq 1$. As $\varepsilon \downarrow 0$, each s_i tends to one almost surely, and the squared sum is bounded by n^2 . Taking expectations in the last inequality and using (2.2) gives $n \geq n^2/(2Z) - n/2$, as asserted. $\square$

Lemma 2.3 (Reduction to a global sector minimum). *For every fixed nuclear system there is a largest integer N such that $E_N < E_{N-1}$. It satisfies $N \leq 3Z$, has a normalized minimizer Ψ , and*

$$E := E_N = \inf_{n \geq 0} E_n. \quad (2.3)$$

Any bound on this N bounds every strictly bound sector of the same system.

Proof. The set of strict sectors is nonempty since $E_1 < 0$, and finite by Lemma 2.2. After its largest member there are no strict drops. Monotonicity therefore gives $E_n = E_N$ for $n > N$, whereas $E_n \geq E_N$ for $n < N$. Existence of Ψ follows from Lemma 2.1. Every other strict sector has particle number at most N . $\square$

For each fixed nuclear system put $E = \inf_{n \geq 0} E_n$. A δ -state is any normalized $\psi \in \mathcal{Q}_n$ in a finite sector n with $q_n[\psi] \leq E + \delta$, where $\delta \geq 0$. Expectation in the squared law of a state includes the spin sum; when only positions are observed, particle labels are retained unless an unordered observation is explicitly specified.

Lemma 2.4 (Energy cost of a multiplier). *Let $\Psi \in \mathcal{Q}_n$ be a normalized eigenstate with eigenvalue e_Ψ . If F is a bounded real Lipschitz function of the spatial configuration, invariant under particle permutations, then*

$$q_n[F\Psi] - e_\Psi \|F\Psi\|_2^2 = \frac{1}{2} \mathbb{E}_\Psi |\nabla F|^2. \quad (2.4)$$

Here ∇ denotes the full spatial gradient in $\mathbb{R}^{3n}$.

Proof. Both $F\Psi$ and $F^2\Psi$ belong to $\mathcal{Q}_n$. Test the weak eigen-equation with $F^2\Psi$ and expand the gradients. Subtraction from $q_n[F\Psi]$ cancels all multiplication potentials and mixed-gradient terms, leaving (2.4). $\square$

2.3 Atomic sectors and neutral ground states

In the atomic specialization $M = 1$, $R_1 = 0$, and $z_1 = Z$, we write $E_n = E(n, Z)$. The compactness proof of Lemma 2.1 applies to every normalized minimizing sequence, so the set of normalized ground states of a strictly bound sector is compact in L^2 .

This is Zhislin's atomic binding condition $n < Z + 1$ [24, 25], specialized to integer Z . We give the proof needed here.

Lemma 2.5 (Atomic binding). *For integer $Z \geq 1$, $E_n < E_{n-1}$ for $1 \leq n \leq Z$. Each of these sectors has a ground state, and every normalized minimizing sequence has an L^2 convergent subsequence with normalized ground-state limit. The multiplier identity of Lemma 2.4 holds for every such ground state with $e_\Psi = E_n$.*

Proof. Strict binding follows inductively beginning with the vacuum. Suppose Φ is a ground state in sector $n - 1$. With the radial core cutoff g_S used in the compactness proof, put $F_S = \prod_i g_S(x_i)$. Its norm tends to one, and the multiplier identity gives normalized energy $E_{n-1} + O_Z(S^{-2})$. Wedge

this state with a normalized radial spatial orbital of fixed spin, supported in $3S < |x| < 4S$, with kinetic energy $O(S^{-2})$. The supports are separated. Newton's spherical mean identity is

$$\frac{1}{4\pi} \int_{\mathbb{S}^2} |s\omega - x|^{-1} d\omega = \frac{1}{\max(s, |x|)}. \quad (2.5)$$

It follows directly by integrating $(s^2 + |x|^2 - 2s|x|t)^{-1/2}/2$ over $-1 < t < 1$. Every core radius is at most $2S$, so the added electron's attraction plus direct repulsion is at most $-(Z - n + 1)/(4S)$. For $n \leq Z$ this strictly dominates the kinetic and cutoff errors when S is large. Lemma 2.1 supplies the next ground state. Its tightness argument gives the asserted compactness for every minimizing sequence, completing the induction. $\square$

Lemma 2.6 (A minimum over all sectors). *Every strict sector, namely every n with $E_n < E_{n-1}$, satisfies $n \leq 2Z + 1$. There is consequently a largest strict index N_* , and*

$$Z \leq N_* \leq 2Z + 1 \leq 3Z, \quad E := \inf_{n \geq 0} E_n = E_{N_*}. \quad (2.6)$$

The infimum E is attained by a normalized ground state in sector N_* .

Proof. Let Φ be a ground state in a strict sector, supplied by Lemma 2.1. For $l, \varepsilon > 0$ set

$$U(x) = (|x|^2 + l^2)^{-1/2} + \varepsilon, \quad w(x) = U(x)^{-1}.$$

For fixed parameters w and its first derivatives are bounded. Test the weak eigen-equation by $(\sum_i w(x_i))\Phi$ and take real parts. This is a symmetric admissible multiplier. In its i th summand, the form on the other $n - 1$ coordinates is at least $E_{n-1}\mathbb{E}_\Phi w(x_i)$, by slicing in x_i and its spin. The selected-coordinate kinetic pairing is nonnegative. Indeed, writing $\phi = w(x_i)\Phi$ and hence $\Phi = U(x_i)\phi$, integration by parts gives

$$\begin{aligned} \operatorname{Re} \int \nabla_i \Phi \cdot \overline{\nabla_i (w(x_i)\Phi)} &= \int U(x_i) |\nabla_i \phi|^2 - \frac{1}{2} \int \Delta U(x_i) |\phi|^2 \geq 0, \\ \Delta U(x) &= -3l^2(|x|^2 + l^2)^{-5/2} \leq 0. \end{aligned}$$

The identity follows first for smooth compactly supported functions; the bounded coefficients for fixed l, ε , approximation in H^1 , and large-radius cutoffs give it in the present domain. Since $E_{n-1} - E_n > 0$, discarding that additional nonnegative term leaves

$$\mathbb{E}_\Phi \sum_{i < j} \frac{w(x_i) + w(x_j)}{|x_i - x_j|} \leq \mathbb{E}_\Phi \sum_i w(x_i) \frac{Z}{|x_i|}. \quad (2.7)$$

Let $l \downarrow 0$ at fixed ε . Domination by $2K(x_i - x_j)/\varepsilon$ for pairs and $V(x_i)/\varepsilon$ for attraction, which are integrable on the form domain, justifies the limit. Then $w(x) = |x|/(1 + \varepsilon|x|)$. As $\varepsilon \downarrow 0$, both sides increase and the right side tends to nZ . The left side tends to the expectation of $\sum_{i < j} (|x_i| + |x_j|)/|x_i - x_j|$, which is at least $n(n - 1)/2$ by the triangle inequality. Thus $n \leq 2Z + 1$.

The strict indices form a nonempty finite set by Lemma 2.5. After its largest element N_* there can be no further strict drop; monotonicity implies $E_n = E_{N_*}$ for every $n > N_*$. For $n < N_*$ monotonicity gives $E_n \geq E_{N_*}$. This proves (2.6), without any assumption on possible plateaus before N_* . Attainment follows once more from Lemma 2.1. $\square$

3 Common localization and quantum comparisons

We next isolate the local variational tools. A spatial cut records the electrons removed from a region and leaves an antisymmetric conditional state for the others. Its conditional electrostatic field enters an exact energy identity. The state may belong to any finite particle-number sector. Because E is the minimum over all sectors, we may insert a variable number of trial electrons into the resulting hole. Upper and lower semiclassical comparisons will have the same kinetic coefficient

$$c_{\text{TF}} = \frac{3}{10}(3\pi^2)^{2/3}.$$

An additional kinetic bound, whose coefficient need not be sharp, controls localization errors.

3.1 Coulomb energy and a kinetic bound

Write $\dot{H}^1(\mathbb{R}^3)$ for the completion of $C_c^\infty(\mathbb{R}^3)$ in the gradient norm, represented in $L^6(\mathbb{R}^3)$ by Sobolev's inequality. For a compactly supported real $f \in L^{5/3}(\mathbb{R}^3)$, put

$$D(f) = \frac{1}{2} \int f(K * f).$$

This definition allows signed f .

Lemma 3.1 (Coulomb duality and radial smearing). *For compactly supported real $f \in L^{5/3}(\mathbb{R}^3)$, its Coulomb potential is continuous and belongs to $\dot{H}^1$, and*

$$D(f) = \frac{1}{8\pi} \|\nabla(K * f)\|_2^2, \quad \left| \int f v \right| \leq C \sqrt{D(f)} \|\nabla v\|_2 \quad (v \in \dot{H}^1). \quad (3.1)$$

The potential and energy assertions also hold for a nonnegative $f \in L^{5/3}$ whose distribution belongs to $(\dot{H}^1)^$, with $K * f$ defined by its nonnegative convolution. Its dual pairing with any $v \in \dot{H}^1$ is the absolutely convergent integral $\int f v$.*

*If η is a radial nonnegative probability density supported in $B(0, b)$, then $K * \eta \leq K$, with equality outside that ball. In particular, $D(\eta * \rho) \leq D(\rho)$ for every nonnegative compactly supported $\rho \in L^{5/3}$.*

Proof. For compactly supported $f \in L^{5/3}$ we also have $f \in L^{6/5}$. Riesz representation on $\dot{H}^1$ gives a unique solution u of $-\Delta u = 4\pi f$. The convolution $K * f$ is continuous: near the singularity use $K \in L_{\text{loc}}^{5/2}$ and continuity of translations in that space; away from it use dominated convergence. It is bounded locally and decays like $O(|x|^{-1})$ at infinity, hence belongs to L^6 . It has the same distributional Laplacian as u . A harmonic L^6 function is zero, by mollification and the mean-value property on arbitrarily large balls. Thus $u = K * f$. Testing its weak equation with u gives the first identity in (3.1); testing with v and using Cauchy–Schwarz gives the second.

For the noncompact nonnegative case, positivity first gives

$$\int f |\varphi| \leq \|f\|_{(\dot{H}^1)^*} \|\nabla \varphi\|_2 \quad (\varphi \in C_c^\infty).$$

To obtain this estimate, approximate the absolute value by smooth convex truncations with derivative bounded by one. Completion and an almost-everywhere convergent subsequence extend the estimate to $\dot{H}^1$ and identify its dual pairing with actual integration. Let u be the Riesz solution, and let $u_j = K * (\mathbf{1}_{B(0, j)} f)$. Testing the difference equation with $(u_j - u)_+$ gives $u_j \leq u$.

These positive-part tests belong to $\dot{H}^1$: an L^6 function with gradient in L^2 is in the completion by large-radius cutoff and mollification. For the cutoff error, Hölder's inequality bounds the gradient of the cutoff times the function by its L^6 norm on the escaping annulus. Now $u_j \uparrow K * f$ and $u_j \leq u \in L^6$, so dominated convergence in L^6 , the distributional equation, and harmonic uniqueness give $K * f = u$. Its energy identity follows by testing with u . The convolution is continuous also in this case. Choose a point x_0 where its value is finite. On any fixed bounded set of evaluation points, the sufficiently distant kernel tails are dominated by a constant times $K(x_0 - \cdot)f$, which is integrable. Dominated convergence handles that tail, and the local $L^{5/3}$ – $L^{5/2}$ argument handles the remaining compact part.

The spherical average of K at distance d from a sphere of radius s is $1/\max(d, s)$, by direct angular integration. Decomposing a radial probability into spherical averages proves the smearing assertion. The difference of two independent radial displacements is again radial; applying the same assertion to its distribution and integrating against $\rho(x)\rho(y)$ proves the energy inequality. $\square$

For a positive trace-class operator γ on $L^2(\mathbb{R}^3; \mathbb{C}^2)$, write ρ_γ for its spatial density, with spins summed. Its kinetic energy is $\text{Tr}(-\Delta/2)\gamma$, defined by the nonnegative form. We use the Lieb–Thirring kinetic inequality [11] in the following non-sharp form. The short proof by spectral projections also follows the distribution-energy approach of Rumin [17].

Lemma 3.2 (Occupation and kinetic energy). *If $0 \leq \gamma \leq 1$ has finite trace, then*

$$\text{Tr}(-\Delta/2)\gamma \geq c \int \rho_\gamma^{5/3} \quad (3.2)$$

for a universal $c > 0$. The one-body density matrix of an antisymmetric k -particle vector f is bounded above by $\|f\|_2^2$ times the identity. The same bound holds after integration in auxiliary variables, and such matrices may be summed whenever the corresponding squared norms sum to one.

Proof. For a unit spin orbital v , the number operator $\sum_{i=1}^k |v\rangle\langle v|_i$ is a projection on the antisymmetric space: in a Slater basis containing v , its eigenvalue is zero or one. Its expectation is at most $\|f\|_2^2$. This proves the occupation assertion and its versions with auxiliary integration and mixtures.

For the kinetic bound, take an eigen-expansion $\gamma = \sum_\alpha \lambda_\alpha |u_\alpha\rangle\langle u_\alpha|$, and let P_s be Fourier projection onto $|p|^2/2 \leq s$. The density of its low-frequency parts is at most $Cs^{3/2}$: Bessel's inequality bounds each spin evaluation against the corresponding Fourier vector, and there are two spins. The triangle inequality in the weighted orbital-index and spin norm gives, almost everywhere,

$$\sum_{\alpha, \sigma} \lambda_\alpha |((1 - P_s)u_\alpha)(x, \sigma)|^2 \geq (\sqrt{\rho_\gamma(x)} - Cs^{3/4})_+^2.$$

Integrating the left side in x and then in $s > 0$ gives the kinetic trace by Plancherel and the layer-cake formula. The integral of the right side in s is a positive constant times $\rho_\gamma(x)^{5/3}$. Truncation proves the assertion also when the kinetic trace is infinite. $\square$

Lemma 3.3 (Global atomic density at an energy offset). *For one nucleus of charge $Z \geq 1$, every δ -state in a sector $n \leq 3Z$ satisfies*

$$\int \rho_\psi^{5/3} \leq C(Z^{7/3} + \delta). \quad (3.3)$$

Proof. Drop the nonnegative pair repulsion, use $E \leq 0$, and split attraction at $R = Z^{-1/3}$. Outside that ball it is at most $Zn/R \leq 3Z^{7/3}$. Inside, Young's inequality gives

$$\int_{|x| < R} \frac{Z}{|x|} \rho_\psi \leq \frac{c}{2} \int \rho_\psi^{5/3} + C \int_{|x| < R} (Z/|x|)^{5/2} \leq \frac{c}{2} \int \rho_\psi^{5/3} + CZ^{7/3},$$

where c is the constant in Lemma 3.2. Combine this with that kinetic lower bound and the energy bound $q_n[\psi] \leq E + \delta \leq \delta$. $\square$

3.2 Cuts, conditional states, and the exact energy identity

The underlying tools are the IMS formula [21, Lemma 2.4] and localization into correlated particle-number sectors [6, Proposition 3.1 and Example 3.3]. Here we retain the removed coordinates as data, so that their conditional core field remains available in the exact energy identity.

Consider a finite sequence of deterministic nonnegative Lipschitz functions (s_h, c_h) on $\mathbb{R}^3$, with $s_h^2 + c_h^2 = 1$. At stage h the factor s_h removes particles from what remains of the core. Define the first-out factors and the final core factor by

$$p_h = s_h \prod_{l < h} c_l, \quad c = \prod_h c_h, \quad o = \left(\sum_h p_h^2 \right)^{1/2}. \quad (3.4)$$

Thus $o^2 + c^2 = 1$. For a normalized $\psi \in \mathcal{Q}_n$ and $I \subset \{1, \dots, n\}$, put

$$\psi_I = \prod_{i \in I} o(x_i) \prod_{i \notin I} c(x_i) \psi.$$

Under the squared laws of these functions, whose total mass is one, record I and the positions and spins of its out particles. Decorate each such particle with stage h with probabilities p_h^2/o^2 at its position. Denote the resulting data by X .

Given X , normalize the slice in the remaining core variables. It is an antisymmetric form-domain state almost surely. Write e_X for its energy, ρ_X^c for its one-body density, and

$$\Phi_X = V - K * \rho_X^c.$$

For the vacuum slice set $e_X = 0$ and $\rho_X^c = 0$.

Lemma 3.4 (Conditional localization identity). *With the preceding definitions, set*

$$T_o = \frac{1}{2} \sum_I \sum_{i \in I} \|\nabla_i \psi_I\|_2^2.$$

There is a nonnegative localization error satisfying

$$\text{IMS} \leq \frac{1}{2} \mathbb{E}_\psi \sum_i \sum_h (|\nabla s_h|^2 + |\nabla c_h|^2)(x_i) \quad (3.5)$$

such that

$$q_n[\psi] + \text{IMS} = \mathbb{E} e_X + T_o - \mathbb{E} \sum_{i \text{ out}} \Phi_X(x_i) + \mathbb{E} \sum_{i < j \text{ out}} K(x_i - x_j). \quad (3.6)$$

The out kinetic energy satisfies

$$T_o \geq c \int (o^2 \rho_\psi)^{5/3},$$

where ρ_ψ is the raw one-body density. Conditional quantities have versions jointly measurable in the data and the field evaluation point. Appending deterministic cuts admits a coupling in which these decorated data are nested, and hence conditional expectations of raw quantities satisfy the tower property.

Proof. Successive squared-gradient splitting gives

$$\sum_h |\nabla p_h|^2 + |\nabla c|^2 = \sum_h \left(\prod_{l < h} c_l^2 \right) (|\nabla s_h|^2 + |\nabla c_h|^2).$$

The norm-map inequality bounds $|\nabla o|^2$ by $\sum_h |\nabla p_h|^2$. Expanding the product partition in the kinetic form, the mixed derivatives cancel because $o\nabla o + c\nabla c = 0$. This proves the stated IMS bound. Multiplication potentials split exactly by the squared partition.

Fix the recorded subset I , and write u for its out positions and spins and ξ for the remaining positions and spins. Put

$$\kappa_I(u) = \int |\psi_I(u, \xi)|^2 d\xi, \quad \chi_X(\xi) = \frac{\psi_I(u, \xi)}{\sqrt{\kappa_I(u)}} \quad \text{when } \kappa_I(u) > 0,$$

where the integral includes the core spin sum. Thus χ_X is the normalized core vector, with its phase retained. The stage decoration depends only on u and does not change this normalized slice. Fubini gives its H^1 regularity almost surely; permutations of core variables preserve the multiplying factors and therefore its antisymmetry. The conditional core-out repulsion is exactly

$$\sum_{i \text{ out}} \int K(x_i - x) \rho_X^c(x) dx.$$

Grouping the core form, out kinetic form, and the remaining multiplication potentials proves (3.6). All absolute Coulomb terms and kinetic terms are integrable in this calculation by the form bounds and the finite partition. The slice ratios and their integrals give the stated measurable versions; assign arbitrary values on null slices.

For each I , apply the occupation assertion of Lemma 3.2 to the out group, integrating the other variables. Its one-body matrix γ_I^o obeys $\gamma_I^o \leq \|\psi_I\|_2^2$. Thus $\gamma_o = \sum_I \gamma_I^o \leq 1$, even though the out particle number varies with I . Its kinetic trace is T_o , and its density is $o^2 \rho_\psi$, by summing the squared partition with one label required to be out. Equation (3.2) proves the kinetic assertion.

Finally, the entire construction has a single sampling description. Conditional on the raw configuration, independently label each coordinate by its first-out stage or by final core, with probabilities p_h^2 and c^2 . An appended cut only splits the previous core probability. Forgetting its new-stage labels and their variables recovers the old observation. This is the required nesting and proves the tower assertion. $\square$

3.3 Insertion of trial electrons

We use the coherent-packet construction [21, Lemma 8.2]. The next lemma turns a nonnegative density into a genuine fermionic trial state. Its particle number is random, with integer values; the global sector minimum in (2.3) is what makes this permissible.

Fix once and for all a real smooth radial g supported in the unit ball, with $\int g^2 = 1$, and set $g_b(x) = b^{-3/2} g(x/b)$.

Lemma 3.5 (Trial particles in a hole). *Fix a conditional core slice with finite energy e_X and field Φ_X as above. Let $\rho \geq 0$ belong to $L^{5/3}$ and have compact support. Suppose its packet region $\text{supp } \rho + \overline{B(0, b)}$ has positive distance from the spatial support allowed to the core particles. Then*

$$E \leq e_X + c_{\text{TF}} \int \rho^{5/3} + Cb^{-2} \int \rho - \int \Phi_X(g_b^2 * \rho) + D(\rho). \quad (3.7)$$

The constant depends only on the fixed function g . The vacuum trial also gives $E \leq e_X$.

Proof. For each center y , momentum p , and spin σ , use the orbital

$$v_{y,p,\sigma}(x, \tau) = g_b(x - y)e^{ip \cdot x} \mathbf{1}_{\{\tau=\sigma\}}.$$

Integrate its projector over both spins and $|p| \leq (3\pi^2\rho(y))^{1/3}$ with measure $dy dp/(2\pi)^3$, obtaining an operator γ . The unrestricted packet resolution is the identity: apply Plancherel in p and then integrate g_b^2 in y . Restriction therefore gives $0 \leq \gamma \leq 1$. Momentum-ball integration gives

$$\begin{aligned} \text{Tr } \gamma &= \int \rho, & \rho_\gamma &= g_b^2 * \rho, \\ \text{Tr}(-\Delta/2)\gamma &= c_{\text{TF}} \int \rho^{5/3} + \frac{1}{2} \|\nabla g\|_2^2 b^{-2} \int \rho. \end{aligned} \quad (3.8)$$

The reality of g eliminates the kinetic cross term. These traces are finite. Every positive-eigenvalue orbital of γ belongs to H^1 and is supported in the packet region.

Take a finite eigen-truncation of γ . Occupy each retained orbital independently with Bernoulli probability equal to its eigenvalue. Each realization is a normalized Slater determinant, including the possible vacuum. The average one-body energy is the trace energy. Its average repulsion is the direct energy of the truncated density minus

$$\frac{1}{2} \sum_{\sigma, \tau} \iint K(x - y) |\gamma(x, \sigma; y, \tau)|^2 dx dy,$$

where here γ denotes the finite truncation. In particular the average repulsion is at most the direct energy. Equal-orbital direct and exchange contributions cancel, as required by the Bernoulli occupation rule.

Wedge each realization with the normalized core slice. Because their spatial supports are separated, the terms assigning the two groups to different sets of particle labels have disjoint supports. Normalized antisymmetrization consequently preserves the sum of the group energies and adds exactly their direct cross repulsion. Each resulting vector belongs to its own antisymmetric particle-number sector and has energy at least E . Averaging gives the desired inequality with the finite-truncation density and kinetic energy.

For completeness, passage to the full operator needs no selection of infinite Slater states. The truncated densities increase almost everywhere to $g_b^2 * \rho$, and their kinetic energies increase to (3.8). The full density is bounded and compactly supported, so nuclear integrals converge. The core potential is bounded on the packet region by support separation, so its integrals converge as well. Direct energies converge monotonically. Lemma 3.1 gives $D(g_b^2 * \rho) \leq D(\rho)$, proving (3.7). The same argument with no inserted orbitals proves $E \leq e_X$. $\square$

The inequality is pointwise for every admissible core slice outside a null set. Later we choose ρ as a measurable function of its conditional field. Only the displayed functional is then averaged; no measurable choice of Slater determinants is necessary.

3.4 A lower comparison with the same kinetic coefficient

An arbitrary out configuration also determines a bounded density, obtained by spreading each retained particle over a small radial kernel. To retain the coefficient c_{TF} in the lower bound, use Neumann eigenvalues in cubes and then average the grid. This is the Neumann-bracketing approach to the semiclassical kinetic coefficient [10, Theorems III.10–III.13]; the retained subset and its radial fine density are specified below.

For $b > 0$, define the radial probability kernel

$$\vartheta_b(x) = \int_{\text{SO}(3)} b^{-6} \prod_{j=1}^3 (b - |(Qx)_j|)_+ \, dQ, \quad (3.9)$$

where Haar measure has mass one. It is bounded by Cb^{-3} and supported in $\overline{B(0, \sqrt{3}b)}$.

Lemma 3.6 (Fine density and lower energy bounds). *In the setting of Lemma 3.4, let n_o be the out particle count. Choose, measurably, any subset of the out particles and set*

$$\sigma(x) = \sum_{i \text{ retained}} \vartheta_b(x - x_i).$$

Then

$$T_o \geq \mathbb{E} \left[c_{\text{TF}} \int \sigma^{5/3} - Cb^{-2}(n_o^{4/3} + n_o) \right], \quad (3.10)$$

$$\sum_{i < j \text{ out}} K(x_i - x_j) \geq D(\sigma) - Cn_o/b \quad \text{almost surely.} \quad (3.11)$$

Proof. On a cube of side b , the spin-two Neumann eigenvalues are $\pi^2|m|^2/(2b^2)$, for $m \in \mathbb{N}_0^3$, each with two spin states. The number of lattice points of radius at most s , with this multiplicity, is bounded by

$$\frac{\pi}{3}(s + \sqrt{3})^3 :$$

attach a disjoint unit cube to each octant lattice point and enclose their union in the octant ball of radius $s + \sqrt{3}$. Thus the j th ordered radius is at least $((3j/\pi)^{1/3} - \sqrt{3})_+$. Summing their squared radii and using $\sum_{j \leq n} j^{2/3} \geq (3/5)n^{5/3}$ gives the lower bound

$$b^{-2}(c_{\text{TF}}n^{5/3} - C(n^{4/3} + n)) \quad (3.12)$$

for the sum of the first n eigenvalues. The leading coefficient is exactly c_{TF} .

Partition the out coordinates into assignments to the cubes of a grid. Restriction to a cube is admissible for its Neumann form; no zero extension across its boundary is involved. Within each cube the corresponding variables remain antisymmetric. The occupation bound, now in the Neumann basis, gives (3.12) after slicing all auxiliary variables. If n_C denotes the out count in a cube, the leading term is

$$c_{\text{TF}} \int \left(\sum_C \frac{n_C}{b^3} \mathbf{1}_C(x) \right)^{5/3} dx.$$

The error is at most $Cb^{-2}(n_o^{4/3} + n_o)$, since $\sum_C n_C^{4/3} \leq n_o^{4/3}$.

Average the grid over a uniform translation in a fundamental cube and over rotations. At a fixed out configuration, its averaged cell density is $\sum_{i \text{ out}} \vartheta_b(x - x_i)$. Before rotation, the probability

that two points are in the same translated cube gives precisely the triangular product in (3.9). Jensen's inequality bounds the power integral from below by the power of this averaged density. Dropping any chosen particles can only decrease that density and its nonnegative power. This proves (3.10).

For the pair bound, first discard all pairs not both retained. Smearing each retained point by ϑ_b decreases every distinct pair interaction by Lemma 3.1, because the difference of two independent radial displacements is radial. Each self energy is bounded by C/b , using the kernel's support and L^∞ bound. Subtracting these self energies from $D(\sigma)$ proves (3.11). $\square$

We now have both sides of the local comparison. The upper trial uses an arbitrary nonnegative density in a hole; the lower bound uses the fine density of the electrons actually removed. Their matching $c_{\text{TF}} \int \rho^{5/3}$ terms will identify the same convex Thomas–Fermi functional, while Lemma 3.1 measures the difference of the two densities.

4 Local screening and electron counts

We now compare the quantum state in a ball with an unconstrained Thomas–Fermi minimizer in the same ball. The comparison has two outputs: a uniform second-moment bound for the number of nearby electrons, and a quantitative Coulomb-distance estimate for a finer local density. Both estimates hold for every δ -state. This stability under a small increase of energy will be essential when we later condition on observations. Throughout this section, ψ is a normalized δ -state in any finite sector n : $q_n[\psi] \leq E + \delta$, where $E = \inf_k E_k$. No eigenstate assumption or bound on n is used here. Expectations refer to its position and spin law, together with any localization labels that are specified.

4.1 Two local geometries

Fix $A = 40$. We use either of the following scale geometries; the choice is kept fixed throughout each application of this section. All balls in these definitions are open.

$$\begin{aligned} \text{nuclear scale:} \quad L &= 10^6, \quad \mathcal{Y} = \mathbb{R}^3, \\ a_y &= \sup\{s > 0 : \nu(B(y, Ls)) \leq s^{-3}\}; \\ \text{atomic distance scale:} \quad L &= 10^5, \quad \nu = Z\delta_0, \\ \mathcal{Y} &= \mathbb{R}^3 \setminus \{0\}, \quad a_y = |y|/L. \end{aligned} \tag{4.1}$$

Put $B_y = B(y, a_y)$ and $d_y = \min_m |y - R_m|$; in the atomic geometry $d_y = |y|$. The nuclear scale is used for molecular estimates, including balls containing nuclei. The atomic distance scale gives nucleus-free patches at every positive distance from the origin, also below $Z^{-1/3}$. These scales are distinct: for a single nucleus the first is $\max(Z^{-1/3}, |y|/10^6)$.

Lemma 4.1 (Properties of the nuclear scale). *In the nuclear-scale geometry, $a : \mathbb{R}^3 \rightarrow (0, \infty)$ is L^{-1} -Lipschitz and satisfies*

$$Z^{-1/3} \leq a_y < \infty, \quad \nu(B(y, La_y)) \leq a_y^{-3}, \quad \nu(B(y, 2La_y)) > (2a_y)^{-3}, \quad d_y \leq La_y. \tag{4.2}$$

If $d_y/L > 1$, then $a_y = d_y/L$. For every fixed $c < L/2$, $B(y, ca_y)$ is covered by a bounded number of balls B_z with $a_z \asymp a_y$. The covering bound depends only on c and L .

Proof. The set of admissible s in (4.1) is downward closed. It contains $Z^{-1/3}$ and is bounded above, since every sufficiently large ball contains all nuclei. Increasing open balls show that the condition also holds at the supremum. The radius $2a_y$ is not admissible, proving the third inequality in (4.2).

If $s = a_x - |x - y|/L > 0$, then $B(y, Ls) \subset B(x, La_x)$, and hence $\nu(B(y, Ls)) \leq a_x^{-3} \leq s^{-3}$. Thus $a_y \geq s$. Interchanging x and y proves the Lipschitz bound. If $d_y > La_y$, a slightly larger ball in the defining family would still contain no nucleus, a contradiction. If $d_y/L > 1$, every radius below d_y/L is admissible, whereas every strictly larger radius contains a charge at least one and has $s^{-3} < 1$. This proves the asserted equality. Finally, the Lipschitz bound makes all scales in $B(y, ca_y)$ comparable to a_y ; a mesh of spacing a sufficiently small multiple of a_y gives the stated cover. $\square$

In the atomic geometry the scale is also L^{-1} -Lipschitz, and $B(y, La_y)$ contains no nucleus. Thus in both geometries

$$\nu(B(y, La_y)) \leq a_y^{-3}.$$

For every fixed $c < L/2$, the ball $B(y, ca_y)$ stays in the target domain $\mathcal{Y}$ and has a bounded cover by cells B_z of scales comparable to a_y . In the atomic case this follows from $|y| = La_y$ and the same mesh argument. The proofs below use these common local properties. The lower scale bound $a_y \geq Z^{-1/3}$ is not assumed in the atomic case.

For the present δ -state define

$$\begin{aligned} n_y &= \#\{i : x_i \in B_y\}, & m_{0,y} &= \max(a_y^{-3}, 1), & m_y &= m_{0,y} + \sqrt{\delta a_y}, & e_y &= \frac{m_y^2}{a_y}, \\ P &= \max\left(1, \sup_{y \in \mathcal{Y}} \frac{\mathbb{E} n_y^2}{m_y^2}\right). \end{aligned} \tag{4.3}$$

Initially P is merely finite, since $n_y \leq n$ and $m_y \geq 1$. We shall prove that it is bounded by a universal constant. Notice that

$$m_y \geq 1, \quad a_y m_y \geq 1, \quad a_y m_y^{2/3} \geq 1, \quad e_y \asymp \max(a_y^{-7}, a_y^{-1}, \delta). \tag{4.4}$$

These facts hold also when $a_y > 1$. Define the raw far field

$$J_y = \int_{|v-y| \geq Aa_y} \frac{d\nu(v)}{|y-v|} - \sum_{|x_i-y| \geq Aa_y} \frac{1}{|y-x_i|}. \tag{4.5}$$

The exclusion removes its singularities and, for each fixed system, $|J_y| \leq (Z+n)/(Aa_y)$. In the atomic geometry the nuclear term is exactly $V(y) = Z/|y|$, since $A < L$.

4.2 A positive-field estimate

A deterministic cut and its recorded out data X have the meaning of Lemma 3.4. The next estimate permits a finite initial sequence of cuts; the proof appends further deterministic cuts only to rule out a large positive field.

Lemma 4.2 (Conditional positive fields). *Fix $C_0 \geq 1$. Suppose the initial sequence is absent, or its total out support is covered by at most C_0 balls B_z and the sum of its cuts' squared gradients is bounded by*

$$\sum_z D_z^2 a_z^{-2} \mathbf{1}_{B_z}, \quad \frac{D_z^2}{a_z m_z} \leq C_0 \sqrt{P}.$$

If X denotes its data (trivial for an absent sequence), and e_* is the maximum of e_y and the energies e_z of these covering cells, then

$$\|(\mathbb{E}[J_y | X])_+\|_2 \leq C(C_0)\sqrt{\frac{Pe_*}{a_y}}. \quad (4.6)$$

Proof. We first give the upper trial at a target y . Append a cut that is full out on $B(y, Aa_y)$, supported in $B(y, 2Aa_y)$, and has gradients bounded by C/a_y . Write X' for the enlarged data, $a = a_y$, and $h = \mathbb{E}[J_y | X']$. The sliced core vanishes on $B(y, Aa)$. Take uniform packet centers of total mass $M_{\text{tr}} \geq 0$ in $B(y, a)$, with packet width a . The packet support lies in $B(y, 2a)$ and is separated from the core. The radial mean of the core field and of each far nuclear field is its value at y . Near nuclei contribute positively, and J_y subtracts far out electrons as well as the core. Therefore Lemma 3.5 gives

$$E \leq e_{X'} - M_{\text{tr}}h + C \left(\frac{M_{\text{tr}}^{5/3}}{a^2} + \frac{M_{\text{tr}}}{a^2} + \frac{M_{\text{tr}}^2}{a} \right). \quad (4.7)$$

Here M_{tr} is the expected occupation of the packet trial; it need not be an integer. This use of an arbitrary trial mass is justified by $E = \inf_k E_k$.

For $ah > C_1 m_{0,y}$ choose $M_{\text{tr}} = \lambda ah$. The three positive terms in (4.7), divided by $M_{\text{tr}}h$, are at most

$$\frac{C\lambda^{2/3}}{a(ah)^{1/3}}, \quad \frac{C}{a(ah)}, \quad C\lambda.$$

The inequalities $am_{0,y}^{1/3} \geq 1$ and $am_{0,y} \geq 1$ allow us to choose $\lambda > 0$ small and then C_1 large so that their sum is less than a fixed number below one. In the other case use the vacuum trial. For all data this proves

$$E \leq e_{X'} - cah_+^2 + C \frac{m_{0,y}^2}{a}. \quad (4.8)$$

The insertion estimate makes a large positive field expensive. We bound the energy of the removed electrons in terms of positive fields at nearby centers. If the present field exceeded a universal multiple of its proposed bound, this comparison would force a larger normalized field after one further deterministic cut. Repetition gives doubly exponential growth, whereas the elementary nuclear bound and the Lipschitz scale permit only exponential growth. We now quantify this argument, retaining the original e_* throughout.

Start with target $v_0 = y$ and the given data $X_0 = X$. At stage j , append the cut just constructed at the current deterministic target v_j ; denote the enlarged data by X_{j+1} . The upper estimate (4.8) applies with $y = v_j$ and $X' = X_{j+1}$. Cover the supports of all cuts, including this new cut, by at most $C(j+1)$ local cells, and let $\mathcal{U}_j$ be the union of their doubled balls $2B_z$. These are the possible next targets: we estimate the removed energy using their fields under the enlarged data, writing $X' = X_{j+1}$ in this calculation. Each newly added scale is comparable to a preceding covering scale by the common local scale properties above; the original target and initial covering scales need not be comparable to one another. Let $a_{\min} > 0$ be the minimum scale of the original target and initial covering cells. The Lipschitz property and the bounded local meshes give, for every later target or covering cell at iteration j ,

$$a_v \geq C^{-(j+1)} a_{\min}, \quad e_v \leq C^{j+1} e_*. \quad (4.9)$$

Indeed each new scale is comparable to a preceding scale; the quantities $m(a) = \max(a^{-3}, 1) + \sqrt{\delta a}$ and $e(a) = m(a)^2/a$ change by a bounded factor when a does, uniformly in δ . The total out count obeys

$$\|n_o\|_2 \leq \sum_{\text{covering cells } z} \|n_z\|_2 \leq \sqrt{P} \sum_z m_z. \quad (4.10)$$

The localization error is at most $C^{j+1}Pe_*$: for an initial cell this follows from

$$D_z^2 a_z^{-2} \mathbb{E} n_z \leq \sqrt{P} \frac{D_z^2}{a_z m_z} e_z \leq C_0 P e_z,$$

and for a later cell it follows from $a_z m_z \geq 1$ and $\mathbb{E} n_z \leq \sqrt{P} m_z$.

For a covering cell B_z , let V_z be the potential of nuclei in $B(z, 4a_z)$. Let $h_z(v)$ be the conditional raw signed field, given the present X' , of all sources at distance at least $4a_z$ from z . It has a jointly measurable harmonic version on $3B_z$: all sources are separated from compact subballs, so their kernel derivatives are bounded for the fixed system and may be integrated against the conditional law. On B_z the field of the conditional core satisfies

$$\Phi_{X'} \leq V_z + h_z + \text{the far out-electron potential}. \quad (4.11)$$

For $v \in 2B_z$, the exclusion in J_v contains $B(z, 4a_z)$, since $A = 40$ and nearby scales are comparable. Extra electrons in h_z can only lower the field. Extra nuclei have total charge at most a_z^{-3} and distance at least $2a_z$ from v . Hence

$$h_z(v) \leq \mathbb{E}[J_v \mid X'] + C a_z^{-4}. \quad (4.12)$$

Set

$$G(v, X') = \sqrt{\frac{a_v}{e_*}} \|(\mathbb{E}[J_v \mid X'])_+\|_2.$$

Positive parts of harmonic functions are subharmonic. Their ball submean inequality, a fixed covering of B_z , and Minkowski's inequality give

$$\begin{aligned} \left\| \sup_{B_z} (h_z)_+ \right\|_2 &\leq C a_z^{-3} \int_{2B_z} \|(h_z(v))_+\|_2 dv \\ &\leq C^{j+1} \sqrt{\frac{e_*}{a_z}} \left(1 + \sup_v G(v, X') \right), \end{aligned} \quad (4.13)$$

where the supremum is over $v \in \mathcal{U}_j$. The term a_z^{-4} was absorbed using $a_z^{-7} \leq e_z \leq C^{j+1}e_*$.

The out addback in (4.11) is at most

$$C \sum_l \frac{n_l}{\max(a_l, a_z)}.$$

Indeed the far exclusion gives distance at least $3a_z$; if $a_l \gg a_z$, Lipschitzness of the scale forces every point of B_l to be a distance at least a fixed multiple of a_l from B_z . Using $\|n_l\|_2 \leq \sqrt{Pe_l a_l}$ and $\sqrt{a_l}/\max(a_l, a_z) \leq a_z^{-1/2}$ bounds this addback in L^2 by $C^{j+1} \sqrt{Pe_*/a_z}$.

For completeness, the nuclear singularities have the required energy scale. If $Q_z = \nu(B(z, 4a_z)) \leq a_z^{-3}$, weighted convexity gives

$$\int_{B_z} V_z^{5/2} \leq Q_z^{3/2} \sum_m z_m \int_{B_z} |v - R_m|^{-5/2} dv \leq C Q_z^{5/2} a_z^{1/2} \leq C a_z^{-7}. \quad (4.14)$$

In the sum over cells, the power of the number of cells arising from convexity is bounded by C^{j+1} . Lemma 3.2 and Young's inequality therefore absorb this attraction at cost $C^{j+1}e_*$. Multiplying the other bounds by $\|n_z\|_2 \leq \sqrt{Pe_z a_z}$ and summing cells, the conditional localization identity, with out-pair repulsion dropped, yields

$$q_n[\psi] + \text{IMS} \geq \mathbb{E}e_{X'} - C^{j+1}e_* \left(P + \sqrt{P} \sup_v G(v, X') \right). \quad (4.15)$$

This calculation used the actual list of cuts; it did not assume the conclusion of the lemma for that list.

Combine (4.8) and (4.15), use $\delta \leq e_*$, and apply conditional Jensen from the enlarged data back to the preceding data. With $G(v_j, X_j)$ defined by the same normalization,

$$G(v_j, X_j)^2 \leq C^{j+1} \left(P + \sqrt{P} \sup_{v \in \mathcal{U}_j} G(v, X_{j+1}) \right). \quad (4.16)$$

Write $g_j = G(v_j, X_j)/\sqrt{P}$. Whenever $g_j^2 \geq 2C^{j+1}$, the last inequality permits a deterministic next target $v_{j+1} \in \mathcal{U}_j$ with

$$g_{j+1} \geq \frac{g_j^2}{4C^{j+1}}, \quad \log g_{j+1} \geq 2 \log g_j - (j+1) \log C - \log 4.$$

The series

$$S = \sum_{j \geq 0} 2^{-j-1} ((j+1) \log C + \log 4)$$

converges. Choose $M_0 > 0$ so that $\exp(2^{j+1}M_0) \geq 2C^{j+1}$ for every j . If $\log g_0 > S + M_0$, induction gives $\log g_j \geq 2^j M_0$; thus the condition for selecting each next target always holds. All supremums here are of numerical norms for deterministic cuts; choosing a target close to one of them never chooses a target from sampled data. On the other hand, $J_v \leq Z/(Aa_v)$ pointwise, and (4.9) gives

$$G(v, X') \leq \sqrt{\frac{a_v}{e_*}} \frac{Z}{Aa_v} \leq C^{j+1} \frac{Z}{A\sqrt{e_* a_{\min}}}.$$

For the fixed system and initial cuts the last factor is finite. Its logarithm grows at most linearly in j , contradicting the positive multiple of 2^j forced above. This argument works in both geometries and requires no global lower bound on the scale. It proves (4.6). $\square$

4.3 Comparison in a ball

We now use the positive-field estimate to control the full local quantum energy. Besides bounding counts, the comparison retains its Coulomb error and the cost of placing electrons where the comparison field is negative. We use the constants

$$p = \frac{3}{2}, \quad k = \left(\frac{5}{3} c_{\text{TF}} \right)^{-3/2}, \quad c_{\text{F}} = 4\pi k.$$

Lemma 4.3 (Conditional comparison in an interior ball). *Fix $y \in \mathcal{Y}$, write $a = a_y$, $m = m_y$, $e = e_y$, and choose $0 < \beta < 1/100$ such that*

$$\frac{\beta^{-2}}{am} \leq \sqrt{P}. \quad (4.17)$$

Set $b = \beta a$. There is a deterministic $t_0 \in [5a, 6a]$ and a cut full out on $B(y, t_0)$, supported in $B(y, t_0 + b)$, with gradients at most C/b . Let X be its out data, ρ_X^c the conditional core density, $\Phi_X = V - K * \rho_X^c$, and $\Omega = B(y, t_0 - 4b)$. There is a unique minimizer $\rho = \rho_X \geq 0$, extended by zero outside Ω , of

$$\mathcal{T}_X(\rho) = c_{\text{TF}} \int \rho^{5/3} - \int \Phi_X \rho + D(\rho). \quad (4.18)$$

It is measurable in X and, with $W = \Phi_X - K * \rho$, satisfies

$$\rho = kW_+^{3/2} \quad \text{in } \Omega. \quad (4.19)$$

Retain the out particles with $|x_i - y| < t_0 - 7b$ and put $\sigma = \sum_{i \text{ retained}} \vartheta_b(\cdot - x_i)$, using the radial fine kernel from Lemma 3.6. Write

$$\mathcal{L} = c_{\text{TF}} \int \left(\sigma^{5/3} - \rho^{5/3} - \frac{5}{3} \rho^{2/3} (\sigma - \rho) \right). \quad (4.20)$$

For every sufficiently small $\epsilon > 0$,

$$\begin{aligned} & \mathbb{E} \left[D(\sigma - \rho) + \mathcal{L} + \int W_- \sigma \right] \\ & \leq C \left[\delta + (\epsilon + \sqrt{\beta}) P e + \epsilon^{-3/2} \beta^{1/2} a^{-7} + b^{-1} \sqrt{P} m + b^{-2} \left(P^{2/3} m^{4/3} + \sqrt{P} m \right) \right]. \end{aligned} \quad (4.21)$$

Moreover, $\mathbb{E} \int \sigma^{5/3}$ is bounded by $C P e$ plus a constant times this right-hand side, and the out kinetic energy obeys

$$T_o \leq C \left(P e + b^{-2} \sqrt{P} m \right). \quad (4.22)$$

Finally, independently of the data,

$$W \leq V_{\text{loc}} + C a^{-4} \quad \text{on } B(y, 3a), \quad \int_{B(y, 2a)} \rho \leq C a^{-3}, \quad (4.23)$$

where V_{loc} is the potential of nuclei in $B(y, 20a)$. If this local nuclear measure is zero, as it always is in the atomic distance geometry, the same construction has the sharper bound

$$\mathbb{E} \left[D(\sigma - \rho) + \mathcal{L} + \int W_- \sigma \right] \leq C \left[\delta + \sqrt{\beta} P e + b^{-1} \sqrt{P} m + b^{-2} \left(P^{2/3} m^{4/3} + \sqrt{P} m \right) \right], \quad (4.24)$$

and its interior bounds include

$$W \leq C a^{-4}, \quad \rho \leq C a^{-6} \quad \text{on } B(y, 3a), \quad \int_{B(y, 2a)} \rho \leq C a^{-3}. \quad (4.25)$$

The Coulomb error $D(\sigma - \rho)$ controls smooth tests of $\sigma - \rho$ in mean, leaving exceptional cut data possible. The penalty $\int W_- \sigma$ integrates the negative field against the fine density of retained electrons. Together with the later oscillation estimates and a lower bound on this density's local mass, it provides the additional control needed to pass a lower field estimate to expectation.

Proof. We divide the proof into the choice of cut, the two energy comparisons, and the deterministic interior bound.

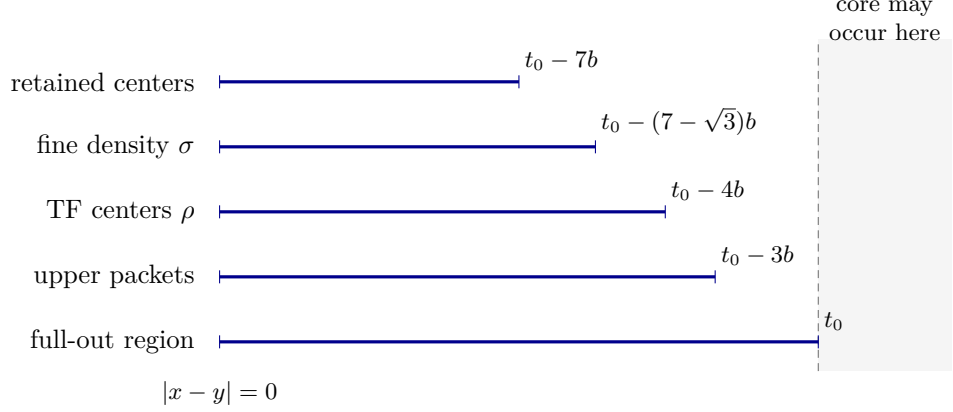


Figure 1: Radial supports in the interior comparison (schematic, not to scale). Fine smears of retained particles lie inside the TF domain, and the upper packets remain at distance at least $3b$ from the conditional core. This particle-support geometry is common to both applications. The atomic distance scale makes $B(y, 20a)$ free of nuclei; the molecular scale may leave nuclei there, and their packet loss is retained in (4.21).

Choice of cut and the conditional field. For a candidate $t \in [5a, 6a]$, let the cut transition in a shell of width b , using sine and cosine of an angle. Let n_{del} count the out particles not retained. Such a particle lies in $t - 7b \leq |x - y| \leq t + b$. For each raw pair of particles the set of t for which both lie in this shell has length at most $8b$. Averaging over t therefore bounds its expected squared count by $C\beta Pm^2$, since $B(y, 7a)$ has a bounded comparable-scale cover. The same averaging, applied to the nuclear integral, uses (4.14) to give a bound $C\beta a^{-7}$. Averaging the sum of the two normalized nonnegative quantities selects one t_0 for which both bounds hold:

$$\begin{aligned} \mathbb{E}n_{\text{del}}^2 &\leq C\beta Pm^2, & \mathbb{E}n_o^2 &\leq CPm^2, & \text{IMS} &\leq Cb^{-2}\sqrt{P}m, \\ \int_{\{t_0-7b \leq |x-y| \leq t_0+b\}} V_{\text{loc}}^{5/2} &\leq C\beta a^{-7}. \end{aligned} \quad (4.26)$$

The full out-count and IMS bounds in the same display follow from the comparable-scale cover and $\mathbb{E}n_z \leq \sqrt{P}m_z$.

On the cut support and on Ω we have

$$\Phi_X \leq V_{\text{loc}} + F, \quad F \geq 0, \quad \|F\|_2 \leq C \frac{\sqrt{P}m}{a}. \quad (4.27)$$

To see this, the raw sources outside $B(y, 20a)$ give a harmonic conditional field on $B(y, 10a)$; these electrons are all core particles. Its value at $v \in B(y, 10a)$ is at most $\mathbb{E}[J_v \mid X] + Ca^{-4}$, by the same exclusion comparison as (4.12). Take F to be the positive supremum on $B(y, 7a)$. Harmonic means and Lemma 4.2 prove (4.27). The cells of this initial cut have scale comparable to a and $D_z \leq C\beta^{-1}$, so (4.17) verifies that lemma's hypothesis.

The unconstrained minimizer. The core is at distance at least $4b$ from Ω , so its potential is bounded and smooth there. Hence $\Phi_X \in L^{5/2}(\Omega)$. Young's inequality makes (4.18) coercive in $L^{5/3}$. The positive Coulomb form is weakly lower semicontinuous: by Lemma 3.1 it is a squared Hilbert norm, or one may express it as the supremum of its affine dual forms. The direct method on

the closed positive cone gives a minimizer, and strict convexity of the kinetic term gives uniqueness. Nonnegative variations, followed by signed compact multiplicative variations, give

$$\frac{5}{3}c_{\text{TF}}\rho^{2/3} - \Phi_X + K * \rho \geq 0, \quad \left(\frac{5}{3}c_{\text{TF}}\rho^{2/3} - \Phi_X + K * \rho \right) \rho = 0,$$

which is (4.19).

This minimizer depends continuously in $L^{5/3}$ on its field in $L^{5/2}$. Indeed convergent fields give uniformly bounded minimizers; weak lower semicontinuity and comparison with the limit minimizer identify each weak limit and give convergence of the minimum values. Lower semicontinuity of D then forces convergence of the kinetic norms. Uniform convexity of $L^{5/3}$ gives strong convergence. The conditional field, being jointly measurable and taking values in the separable space $L^{5/2}(\Omega)$, is measurable as a function with values in that space. The continuity just proved establishes the stated measurability of ρ .

The diameter of Ω is at most $12a$, so $D(\rho) \geq (\int \rho)^2/(24a)$. The minimum is nonpositive because zero is admissible. Absorbing V_{loc} with Young's inequality in (4.27) gives

$$\frac{(\int \rho)^2}{Ca} \leq Ca^{-7} + F \int \rho.$$

It follows that

$$\int \rho \leq C(aF + a^{-3}), \quad \int \rho^{5/3} \leq C(aF^2 + a^{-7}). \quad (4.28)$$

When $V_{\text{loc}} = 0$, the same nonpositive-minimum argument uses only $\Phi_X \leq F$ and gives the sharper estimates

$$\int \rho \leq CaF, \quad \int \rho^{5/3} \leq CaF^2. \quad (4.29)$$

In particular all trial errors below are integrable in X .

Upper quantum comparison. The centers ρ lie in $\Omega = B(y, t_0 - 4b)$; packets of width b remain at distance at least $3b$ from the core. Thus Lemma 3.5 applies, and radial smearing leaves the core potential unchanged. The nuclear loss is bounded at centers by

$$V_b^{\text{loss}}(x) = \sum_{R_m \in B(y, 20a)} \frac{z_m}{|x - R_m|} \mathbf{1}_{\{|x - R_m| \leq 2b\}}.$$

Weighted convexity, the total local charge bound, and integration of $|x|^{-5/2}$ over a ball of radius $2b$ give

$$\int (V_b^{\text{loss}})^{5/2} \leq C\beta^{1/2}a^{-7}.$$

Nuclei not in the indicated local ball are harmonic on all packets. Young's inequality now gives, pathwise,

$$E \leq e_X + \mathcal{T}_X(\rho) + Cb^{-2} \int \rho + C\epsilon \int \rho^{5/3} + C\epsilon^{-3/2}\beta^{1/2}a^{-7}. \quad (4.30)$$

Lower quantum comparison. Reserve a fraction ϵT_o of the out kinetic energy. The coarse kinetic inequality absorbs both V_b^{loss} and the nuclear attraction of deleted particles, at cost $C\epsilon^{-3/2}\beta^{1/2}a^{-7}$, using (4.26). The deleted F contribution costs at most

$$\|F\|_2 \|n_{\text{del}}\|_2 \leq C\sqrt{\beta}Pe.$$

The retained fine density σ is supported in Ω , because its centers have radius less than $t_0 - 7b$ and its kernel radius is $\sqrt{3}b$. The core potential is harmonic on these kernels, and the loss of nuclear attraction is again covered by V_b^{loss} at the actual particle positions. Applying the remaining fraction of Lemma 3.6 and its pair-repulsion bound in the localization identity gives

$$\begin{aligned} q_n[\psi] + \text{IMS} &\geq \mathbb{E} \left[e_X + \mathcal{T}_X(\sigma) - \epsilon c_{\text{TF}} \int \sigma^{5/3} \right] - C\mathcal{R}, \\ \mathcal{R} &= \epsilon^{-3/2} \beta^{1/2} a^{-7} + \sqrt{\beta} P e + b^{-1} \sqrt{P} m + b^{-2} \left(P^{2/3} m^{4/3} + \sqrt{P} m \right). \end{aligned} \quad (4.31)$$

Here $\mathbb{E} n_o^{4/3} \leq (\mathbb{E} n_o^2)^{2/3}$; no higher out-count moment is required.

The Euler equation gives the exact expansion

$$\mathcal{T}_X(\sigma) - \mathcal{T}_X(\rho) = D(\sigma - \rho) + \mathcal{L} + \int W_- \sigma. \quad (4.32)$$

Besides being nonnegative, the convex remainder satisfies

$$\mathcal{L} \geq c \int \sigma^{5/3} - C \int \rho^{5/3},$$

by Young's inequality applied to the tangent term. For small ϵ this absorbs $\epsilon c_{\text{TF}} \int \sigma^{5/3}$ into half of $\mathcal{L}$, with error $C\epsilon \int \rho^{5/3}$. Average (4.30), compare it with (4.31), and use (4.28), (4.27), and the IMS bound. The terms $\epsilon \mathbb{E} \int \rho^{5/3}$ cost $C\epsilon P e$; the packet kinetic term costs $C b^{-2} (\sqrt{P} m + a^{-3}) \leq C b^{-2} \sqrt{P} m$. This proves (4.21), after multiplying its constant to restore the full nonnegative remainder, and also the asserted bound for $\mathbb{E} \int \sigma^{5/3}$.

The source-free improvement. If $\nu(B(y, 20a)) = 0$, every nucleus and the conditional core are outside the packet supports. The field Φ_X is harmonic on those supports, so radial averaging is exact and the upper comparison is

$$E \leq e_X + \mathcal{T}_X(\rho) + C b^{-2} \int \rho. \quad (4.33)$$

For the lower comparison, use all the out kinetic energy. Harmonicity and the deleted-count bound give

$$\sum_{i \text{ retained}} \Phi_X(x_i) = \int \Phi_X \sigma, \quad \mathbb{E}(F n_{\text{del}}) \leq C \sqrt{\beta} P e.$$

The full sharp kinetic and pair bounds of Lemma 3.6 therefore give

$$q_n[\psi] + \text{IMS} \geq \mathbb{E}[e_X + \mathcal{T}_X(\sigma)] - C \left[\sqrt{\beta} P e + b^{-1} \sqrt{P} m + b^{-2} (P^{2/3} m^{4/3} + \sqrt{P} m) \right]. \quad (4.34)$$

Subtract the average upper inequality and use the exact expansion (4.32), the mass estimate (4.29), and the IMS bound. This proves (4.24), including its full convex remainder, without the nuclear loss or an ϵ error.

There is a useful separate kinetic bound. In the localization identity use only $e_X \geq E$, drop pair repulsion, and use (4.27). Half the coarse kinetic inequality absorbs V_{loc} at cost $C a^{-7}$; also $\mathbb{E}(F n_o) \leq C P e$. The result is (4.22).

Interior field bound. The core vanishes in $B(y, 4a)$ and this ball lies in Ω . There

$$\Delta W = -4\pi\nu + c_F W_+^{3/2}.$$

Put $s = 4a$ and

$$U(x) = V_{\text{loc}}(x) + \frac{Cs^4}{(s^2 - |x - y|^2)^4}.$$

The Laplacian of the bump is at most $80Cs^6/(s^2 - |x - y|^2)^6$. For a sufficiently large universal C this is at most c_F times its $3/2$ power. Thus U is a supersolution for the displayed equation. The difference $W - V_{\text{loc}}$ is bounded above: the far nuclear field is bounded on this ball, the core potential is nonnegative, and $K * \rho \geq 0$. Consequently $(W - U)_+$ has compact support inside the ball. Nuclear singularities cancel in $W - U$, which is locally H^1 ; testing the difference inequality by its positive part yields

$$-\int |\nabla(W - U)_+|^2 \geq c_F \int_{\{W > U\}} (W_+^{3/2} - U^{3/2})(W - U) \geq 0.$$

The test is valid by compact H^1 approximation: the densities are locally in $L^{5/3}$, and the nuclear point terms have already canceled. Hence $W \leq U$. On $B(y, 3a)$ the bump is Ca^{-4} , proving the first part of (4.23). If $V_{\text{loc}} = 0$, this yields $W \leq Ca^{-4}$; the Euler equation then gives $\rho \leq Ca^{-6}$ and proves (4.25). In this source-free case,

$$\Delta W = c_F W_+^{3/2} = 4\pi\rho \quad \text{on } B(y, 4a). \quad (4.35)$$

Finally, weighted powers and $\nu(B(y, 20a)) \leq a^{-3}$ give

$$\int_{B(y, 2a)} \rho \leq C \int_{B(y, 2a)} (V_{\text{loc}}^{3/2} + a^{-6}) \leq C(a^{-9/2}a^{3/2} + a^{-3}) \leq Ca^{-3}.$$

□

4.4 Closing the count estimate

The comparison in a hole bounds its electron count through a smooth Coulomb test. The resulting inequality improves any sufficiently large value of the provisional constant P , and therefore makes that constant universal.

Lemma 4.4 (Local second moments). *For every normalized δ -state in any finite sector and every $y \in \mathcal{Y}$,*

$$\mathbb{E}n_y^2 \leq C \left(\max(a_y^{-6}, 1) + \delta a_y \right). \quad (4.36)$$

In particular the number P in (4.3) is at most a universal constant.

Proof. Fix small $\epsilon > 0$ and $0 < \beta < 1/100$, to be chosen in that order. We need only consider $P \geq \beta^{-4}$, since β will be fixed universally. Then $\beta^{-2}/(am) \leq \sqrt{P}$ at every target, and the patch of Lemma 4.3 is available. Every raw particle in B_y is retained, and its fine smear is contained in $B(y, 1.1a)$ because $\beta < 1/100$. Choose a nonnegative smooth χ , equal to one there, supported in $B(y, 2a)$, with $\|\nabla\chi\|_2 \leq C\sqrt{a}$. Lemma 3.1 and (4.23) give

$$n_y \leq \int \chi\sigma \leq Ca^{-3} + C\sqrt{aD(\sigma - \rho)}, \quad \mathbb{E}n_y^2 \leq Ca^{-6} + Ca \mathbb{E}D(\sigma - \rho).$$

Divide by Pm^2 and use (4.21). Since $\delta a \leq m^2$, the result is at most

$$C \left[P^{-1} + \epsilon + \sqrt{\beta} + \frac{\epsilon^{-3/2} \beta^{1/2}}{P} + \frac{\beta^{-1}}{\sqrt{P}m} + \frac{\beta^{-2}}{P^{1/3} a m^{2/3}} + \frac{\beta^{-2}}{\sqrt{P} a m} \right]. \quad (4.37)$$

In the atomic distance geometry, using (4.24) instead gives the sharper intermediate inequality

$$\frac{\mathbb{E} n_y^2}{P m_y^2} \leq C \left[P^{-1} + \sqrt{\beta} + \frac{\beta^{-1}}{\sqrt{P} m_y} + \frac{\beta^{-2}}{P^{1/3} a_y m_y^{2/3}} + \frac{\beta^{-2}}{\sqrt{P} a_y m_y} \right]. \quad (4.38)$$

For the common bound it suffices to use (4.37) in both geometries. First choose ϵ small, then β small, and finally a universal threshold P_0 large. The inequalities (4.4) show that the expression is at most $1/2$ whenever $P > P_0$. Enlarge P_0 so that $P_0 \geq \beta^{-4}$; this verifies (4.17) at every target. If $P > P_0$, we would obtain $\mathbb{E} n_y^2 / m_y^2 \leq P/2$ for every y , contradicting the definition of P . Thus $P \leq P_0$. Since $m_y^2 \leq 2 \max(a_y^{-6}, 1) + 2\delta a_y$, this proves (4.36). $\square$

Proposition 4.5 (Uniform local counts and conditional fields). *In either geometry, every normalized δ -state in a finite sector satisfies $P \leq C$. Fix $y \in \mathcal{Y}$. Suppose the initial finite sequence of cuts has total out support covered by at most C_0 cells B_z , with $C_0^{-1} \leq a_z / a_y \leq C_0$, and satisfies the gradient hypothesis of Lemma 4.2. The absent-cut case is allowed. Then*

$$P \leq C, \quad \|(\mathbb{E}[J_y | X])_+\|_2 \leq C(C_0) \frac{m_y}{a_y}. \quad (4.39)$$

Neither constant depends on the nuclear system, particle number, state, target, or δ .

Proof. The count assertion is Lemma 4.4. Comparable scales have comparable m and $e = m^2/a$, uniformly in δ , so $e_* \leq C(C_0)e_y$ in Lemma 4.2. Substitute $P \leq C$ into that lemma to obtain the field assertion. $\square$

Lemma 4.6 (Fine comparison and local density). *There is a universal $a_0 > 0$ with the following property. If $a = a_y < a_0$ and ψ is a δ -state with $\delta \leq a^{-7+0.02}$, choose the patch of Lemma 4.3 with $b = a^{1+0.2}$. Its densities obey*

$$\mathbb{E} \left[D(\sigma - \rho) + \mathcal{L} + \int W_- \sigma \right] \leq C a^{-7+0.02}, \quad \mathbb{E} \int \sigma^{5/3} \leq C a^{-7}. \quad (4.40)$$

The raw one-body density satisfies

$$\int_{B(y, 4a)} \rho_\psi^{5/3} \leq C a^{-7}. \quad (4.41)$$

In particular, the two estimates needed in the atomic application are

$$\mathbb{E} \left[D(\sigma - \rho) + \int W_- \sigma \right] \leq C a^{-7+0.02}, \quad \int_{B(y, 4a)} \rho_\psi^{5/3} \leq C a^{-7}. \quad (4.42)$$

Proof. Take $\beta = a^{0.2}$ and $\epsilon = a^{0.02}$. The energy assumption gives $m \leq C a^{-3}$, while $\beta^{-2}/(a m) \leq a^{1.6}$, so the patch hypothesis holds for small a . By Lemma 4.4, $P \leq C$. The terms on the right of (4.21) are bounded respectively by constant multiples of

$$a^{-6.98}, \quad a^{-6.98} + a^{-6.9}, \quad a^{-6.93}, \quad a^{-4.2}, \quad a^{-6.4} + a^{-5.4}.$$

All are at most $C a^{-7+0.02}$ for $a < 1$, proving (4.40). Equation (4.22) gives $T_o \leq C a^{-7}$. Since the cut is full out on $B(y, 4a)$, the out density equals ρ_ψ there. Applying Lemma 3.2 proves (4.41). $\square$

Corollary 4.7 (Raw near-Coulomb bound). *Under the hypotheses of Lemma 4.6, for $0 < u \leq 4a$,*

$$\mathbb{E} \sum_{|x_i - y| < u} \frac{1}{|x_i - y|} \leq Ca^{-4}(u/a)^{1/5}. \quad (4.43)$$

Proof. The expectation is $\int_{B(y,u)} |x - y|^{-1} \rho_\psi(x) dx$. Hölder's inequality, the raw density bound, and direct radial integration give

$$\int_{B(y,u)} \frac{\rho_\psi(x)}{|x - y|} dx \leq \left(\int_{B(y,4a)} \rho_\psi^{5/3} \right)^{3/5} \left(\int_{|z| < u} |z|^{-5/2} dz \right)^{2/5} \leq Ca^{-21/5} u^{1/5}.$$

This is the stated bound in either geometry. $\square$

4.5 Source-free oscillation and atomic annuli

The next estimate controls also a very negative field. It applies when the patch contains no local nuclear source; molecular patches containing nuclei instead use the offset oscillation estimate proved with the nuclear splitting below.

Lemma 4.8 (Interior oscillation). *For every patch in Lemma 4.3 with $\nu(B(y, 20a)) = 0$, and every $0 < s' \leq a/2$, almost surely in its data,*

$$\sup_{x \in B(y, s')} |W(x) - W(y)| \leq C \frac{s'}{a} (a^{-4} + W(y)_-). \quad (4.44)$$

Proof. By (4.35) and (4.25), $0 \leq \Delta W \leq Ca^{-6}$ on $B(y, 2a)$. Write

$$Q(x) = \frac{1}{4\pi} \int_{B(y, 2a)} \frac{\Delta W(z)}{|x - z|} dz.$$

For $x \in B(y, 2a)$, direct integration of $|x - z|^{-1}$ and $|x - z|^{-2}$ gives

$$0 \leq Q(x) \leq Ca^{-4}, \quad |\nabla Q(x)| \leq Ca^{-5}.$$

Since $\Delta Q = -\Delta W$ inside that ball, $H = W + Q$ is harmonic there. By (4.25), $H \leq C_1 a^{-4}$. Choose $C_2 > C_1$ universally and put $h = C_2 a^{-4} - H$, a positive harmonic function on $B(y, 2a)$. The Poisson formula on a sphere of radius $3a/2$ gives

$$\sup_{B(y, a)} |\nabla h| \leq Ca^{-1} h(y).$$

Indeed the differentiated Poisson kernel on $B(y, a)$ is bounded by C/a times its value at the center; integrating the nonnegative boundary values gives precisely this inequality. Also $h(y) \leq Ca^{-4} + W(y)_-$, since $Q(y) \geq 0$. Integrate the gradient along segments in $B(y, s')$, and add the bound for ∇Q . This proves (4.44). $\square$

Corollary 4.9 (Annular counts). *In the atomic distance geometry, for every $0 < \lambda < \Lambda < \infty$ and $u > 0$, every normalized δ -state of finite particle number satisfies*

$$\|\#\{i : \lambda u < |x_i| < \Lambda u\}\|_2 \leq C_{\lambda, \Lambda} (\max(u^{-3}, 1) + \sqrt{\delta u}). \quad (4.45)$$

All constants are independent of Z and of the particle number.

Proof. The compact annulus $\{\lambda \leq |x| \leq \Lambda\}$ has a finite cover by cells B_z whose radii are comparable to one, with constants allowed to depend on λ, Λ, L . Dilate this cover by u . Because $a_{uz} = ua_z$, the dilated balls are again cells, now with radii comparable to u . The annular count is bounded by their sum of counts. Minkowski's inequality and (4.39) bound its L^2 norm by $C \sum_z m_z$, which is the right side of (4.45). Boundary spheres carry zero raw probability, since the one-electron density is absolutely continuous. Thus the same estimate holds with any choice of open or closed endpoints. $\square$

5 Common observation and change-of-law estimates

The two geometries use different master widths and nuclear fields, but share the observation likelihood, the cost of imposing an event, and the transfer from the observed field to a fresh patch field. We prove these statements here. In particular, the last transfer compares expectations under a specified law; it does not identify the two conditional fields on individual configurations.

5.1 Likelihoods and conditional versions

Let Ψ be a normalized antisymmetric state in a finite n -electron sector. Under a probability law P , raw positions and spins (x, σ) have density $|\Psi|^2$. Fix finitely many widths $\ell_0, \dots, \ell_{m-1} > 0$. Independently of the raw data, take independent scalar noises $U_{ji,b}$ with density

$$\zeta(u) = c_\zeta \exp\left(-\frac{1}{1-u^2}\right) \mathbf{1}_{(-1,1)}(u).$$

At scale j , form the labeled observations $\hat{Y}_{ji,b} = x_{i,b} + \ell_j U_{ji,b}$ and then forget the particle labels separately in each array. Represent the result by its lexicographically sorted tuple Y_j in the strict sorted chamber $\mathcal{C} \subset (\mathbb{R}^3)^n$. Ties have probability zero. Set

$$\mathcal{F}_j = \sigma(Y_j, \dots, Y_{m-1}), \quad \mathcal{F}_m = \{\emptyset, \Omega\}.$$

Thus the sigma-fields decrease as j increases.

Let t_z be a deterministic measurable width with $t_z \geq t_{\min} > 0$ for the fixed system, and use the smooth radial packet of Section 3 to define $\mathcal{K}_z(y) = g_{t_z}(y - z)^2$. Write $V = V_s + V_c$, where V_s is the deterministic nuclear potential retained in the observed field. In the molecular application it is the smoothed nuclear potential. In the atomic application $V_s = V$ and $V_c = 0$. Where relevant put $\nu_s = -(4\pi)^{-1} \Delta V_s$ distributionally; it is the smoothed nuclear measure in the first case and the original point measure in the second.

Lemma 5.1 (Conditional densities and fields). *There are jointly Borel versions of*

$$\mu_j(y) = \mathbb{E}_P \left[\sum_{i=1}^n \mathcal{K}_{x_i}(y) \mid \mathcal{F}_j \right], \quad H_j(y) = V_s(y) - (K * \mu_j)(y),$$

*with the following properties. At every positive-density data tuple, $\mu_j \geq 0$, $\int \mu_j = n$, and μ_j is smooth in its spatial variable. Every spatial derivative has a deterministic bound for the fixed system. The potential $K * \mu_j$ is C^1 , with bounded value and gradient, and*

$$\Delta H_j = -4\pi\nu_s + 4\pi\mu_j \tag{5.1}$$

in distributions. In particular, the always valid regularity statement is that $H_j - V_s$ is C^1 and bounded with bounded gradient. If V_s and its gradient are bounded and continuous, these conclusions hold for H_j itself. Conditional averaging satisfies

$$\mathbb{E}_P[\mu_j(y) \mid \mathcal{F}_{j+1}] = \mu_{j+1}(y), \quad \mathbb{E}_P[H_j(y) \mid \mathcal{F}_{j+1}] = H_{j+1}(y) \quad (5.2)$$

where $V_s(y)$ is finite. The same towers hold for nonnegative spatial integrals, and for integrable signed observables. These statements include the trivial data at $j = m$. None of the fixed-system regularity bounds is asserted to be uniform in the nuclear system or in $t_{\min}$.

Proof. On $\mathcal{C}$ the exact likelihood of an array is

$$L_j(x, Y_j) = \sum_{\pi \in S_n} \prod_{i=1}^n \prod_{b=1}^3 \ell_j^{-1} \zeta \left(\frac{Y_{j, \pi(i), b} - x_{i, b}}{\ell_j} \right). \quad (5.3)$$

Set it equal to zero outside the chamber. Integration of the permutation sum over $\mathcal{C}$ equals integration of the labeled product over all arrays, and is therefore one. Define

$$\Lambda_{\geq j}(x, Y) = \prod_{h=j}^{m-1} L_h(x, Y_h), \quad p_{\geq j}(Y) = \sum_{\sigma} \int |\Psi(x, \sigma)|^2 \Lambda_{\geq j}(x, Y) dx,$$

with the empty product and $p_{\geq m}$ both equal to one. At a tuple with $p_{\geq j} > 0$, use the probability density $|\Psi|^2 \Lambda_{\geq j} / p_{\geq j}$ as the conditional raw law. Equivalently, the conditional mean of a nonnegative Borel raw observable G is the ratio

$$\frac{\sum_{\sigma} \int G(x, \sigma) |\Psi(x, \sigma)|^2 \Lambda_{\geq j}(x, Y) dx}{p_{\geq j}(Y)}.$$

On zero-density tuples use the unconditional raw law. All these versions are jointly Borel, including their dependence on an additional spatial variable. At every parent tuple with $p_{\geq j+1} > 0$, the conditional density of the next array is $p_{\geq j} / p_{\geq j+1}$. Tonelli's theorem and $\int L_j dY_j = 1$ show that it integrates to one. Substitution of the ratios gives the tower property, also after nonnegative spatial integration; positive and negative parts give the integrable signed case.

For each multi-index α , $|\partial_y^\alpha \mathcal{K}_z(y)| \leq C_\alpha t_{\min}^{-3-|\alpha|}$ uniformly in z, y . Differentiation under the conditional integral therefore gives all claimed derivatives and bounds. Tonelli gives the mass n . For the potential and its gradient, split the kernels $|u|^{-1}$ and $|u|^{-2}$ at unit distance. The near parts are integrable and are controlled by $\|\mu_j\|_\infty$; the far parts are controlled by n . Splitting again at a small radius, then using dominated convergence on its complement, proves continuity of the potential and gradient. Finally $-\Delta K = 4\pi\delta_0$ gives (5.1). In the atomic case this argument does not remove the nuclear singularity of V_s . $\square$

5.2 Dimension-free energy cost

Lemma 5.2 (Energy cost of a data event). *Suppose Ψ is an eigenstate of its sector Hamiltonian with energy e_Ψ , where $0 \leq e_\Psi - E \leq D_0$. In particular, $e_\Psi = E_n$ for a sector ground state. Suppose further that*

$$\sum_{h=j}^{m-1} \ell_h^{-2} \leq C_\ell \ell_j^{-2}$$

with a fixed numerical C_ℓ . This holds for the dyadic scales $\ell_h = r_h^{1.01}$, $r_{h+1} = 2r_h$, with $C_\ell = (1 - 2^{-2.02})^{-1}$. If $A \in \mathcal{F}_j$ has probability $p = P(A) > 0$, put $p_A(x) = P(A \mid x)$ and

$$\Psi_A = \sqrt{p_A(x)/p} \Psi.$$

Then Ψ_A is a normalized antisymmetric form-domain vector, its raw position-and-spin law is the raw marginal of P conditioned on A , and

$$q_n[\Psi_A] \leq e_\Psi + C\ell_j^{-2} \log^5(e/p) \leq E + D_0 + C\ell_j^{-2} \log^5(e/p). \quad (5.4)$$

The constant depends only on C_ℓ and the chosen noise law, not on n , the number of arrays, or the nuclear system. The same assertion holds for an event of one unordered observation at any width ℓ , using ℓ in place of ℓ_j . No upper bound on the particle number in terms of the nuclear charge is needed.

Proof. In labeled observation coordinates place the sorting map inside the indicator of A . The translated product density is continuously differentiable in L^1 , since ζ is smooth and compactly supported. Hence p_A is continuously differentiable even for a Borel event. The scalar translation score is

$$S(u) = -\frac{\zeta'(u)}{\zeta(u)} = \frac{2u}{(1-u^2)^2}, \quad |u| < 1.$$

Its law is symmetric and centered. It satisfies $\|S(U)\|_q \leq Cq^2$ for $q \geq 2$: indeed, $T = (1 - U^2)^{-1}$ has a finite exponential moment of every order less than one, so integration of its exponential tail gives $\|T\|_q \leq Cq$, while $|S(U)| \leq 2T^2$.

For a unit vector $v \in \mathbb{R}^{3n}$ the directional derivative of the likelihood pairs the event indicator with

$$Q_v = \sum_{h=j}^{m-1} \sum_{i=1}^n \sum_{b=1}^3 \ell_h^{-1} v_{i,b} S(U_{hi,b}).$$

Multiplication by independent random signs preserves the joint law of the independent symmetric scores. For deterministic coefficients,

$$\prod_{\alpha} \cosh(sc_{\alpha}) \leq \exp\left(\frac{s^2}{2} \sum_{\alpha} c_{\alpha}^2\right)$$

gives the Gaussian tail estimate and consequently $\|\sum_{\alpha} \epsilon_{\alpha} c_{\alpha}\|_q \leq C\sqrt{q}(\sum_{\alpha} c_{\alpha}^2)^{1/2}$. Conditioning on the scores and then using the triangle inequality in $L^{q/2}$ yields

$$\|Q_v\|_q \leq Cq^{5/2} \left(\sum_{h,i,b} \ell_h^{-2} v_{i,b}^2 \right)^{1/2} \leq Cq^{5/2} \ell_j^{-1}.$$

The identity $\sum_{i,b} v_{i,b}^2 = 1$ and the geometric square-sum bound are precisely what avoid factors depending on dimension or on the number of observations.

Hölder's inequality at fixed x gives $|D_v p_A(x)| \leq Cq^{5/2} \ell_j^{-1} p_A(x)^{1-1/q}$. For $p_A(x) > 0$ choose $q = \max\{2, \log(e/p_A(x))\}$ and then take the supremum over unit vectors. At a zero the gradient of the nonnegative differentiable function p_A vanishes. Therefore

$$|\nabla p_A| \leq C\ell_j^{-1} p_A \log^{5/2}(e/p_A), \quad (5.5)$$

with right side zero at $p_A = 0$.

Regularize the square root by $\sqrt{p_A + \eta}$. Since $\sqrt{u} \log^{5/2}(e/u)$ is bounded on $[0, 1]$, these functions have a common Lipschitz bound. Their uniform limit $\sqrt{p_A}$ is Lipschitz. Its gradient vanishes almost everywhere on its zero set, and the ordinary chain rule applies elsewhere. Thus

$$\left| \nabla \sqrt{p_A/p} \right|^2 \leq C \ell_j^{-2} \frac{p_A}{p} \log^5(e/p_A).$$

The permutation sum in (5.3) shows that p_A is invariant under raw particle permutations and is spin independent. The bounded Lipschitz multiplier therefore preserves antisymmetry and the form domain. Its squared norm and its raw law are as claimed by the definition of p_A .

Lemma 2.4, applied with eigenvalue e_Ψ , now gives

$$q_n[\Psi_A] - e_\Psi \leq C \ell_j^{-2} \mathbb{E}_P[\log^5(e/p_A(x)) \mid A].$$

Under this conditional law,

$$P\{p_A(x) < e^{-u} \mid A\} = p^{-1} \mathbb{E}_P[p_A \mathbf{1}_{\{p_A < e^{-u}\}}] \leq \min\{1, e^{-u}/p\} \quad (u \geq 0).$$

Integration of this tail, split at $\log(e/p)$, bounds the fifth moment by $C \log^5(e/p)$ and proves (5.4). The proof for a single array is identical with the square-sum constant equal to one. $\square$

5.3 Transfer to a fresh patch

The following quantitative formulation isolates exactly the local screening estimates needed by both applications. It allows either a patch containing singular nuclei or a patch containing none. The fine density below is the actual sum of packets around retained raw particles, not its conditional average.

Lemma 5.3 (Averaged field transfer). *Fix a point y where $V_s(y)$ is finite and a length $a > 0$. Let $A \in \mathcal{F}_j$ have positive P -probability, and let Q be P conditioned on A , extended by fresh localization labels. Its raw marginal is the law of Ψ_A from Lemma 5.2. Let X denote the fresh patch data, and let ρ_X^c be the conditional density of core particles under this tilted localization law. Partition the remaining raw particles into retained and deleted particles. Let*

$$\sigma = \sum_{i \text{ retained}} \vartheta_b(\cdot - x_i), \quad h_X = V_s - K * \rho_X^c - K * \rho_X, \quad W_X = V_c + h_X,$$

where ϑ_b is the radial probability kernel of Lemma 3.6, supported in $B(0, \sqrt{3}b)$, and $\rho_X \geq 0$ is a measurably chosen patch comparison density. Assume ρ_X and σ belong to $L^{5/3}$ and have compact support for almost every patch datum. Put $t = t_y$, $\tau = t/a$, and $\beta = b/a$, and suppose $0 < t, b \leq a/10$. Assume the following estimates, with fixed constants:

1. The master width $z \mapsto t_z$ is Lipschitz with constant at most $1/2$. The raw one-particle density ρ_{Ψ_A} satisfies

$$\int_{B(y, 4a)} \rho_{\Psi_A}^{5/3} \leq C a^{-7}.$$

2. Deleted particles have $|x_i - y| \geq ca$, and their count obeys $\mathbb{E}_Q n_{\text{del}} \leq C a^{-3} \sqrt{\beta}$.

3. For some $\varepsilon_a \geq 0$,

$$\mathbb{E}_Q D(\rho_X - \sigma) \leq C a^{-7} \varepsilon_a^2.$$

4. On $B(y, t)$ either the source-free bound $\rho_X \leq Ca^{-6}$ holds, or

$$\rho_X = k(W_X)_+^{3/2}, \quad W_X \leq V_{\text{loc}} + Ca^{-4}, \quad V_{\text{loc}}(z) = \sum_{\alpha} \frac{q_{\alpha}}{|z - R_{\alpha}|}, \quad \sum_{\alpha} q_{\alpha} \leq Ca^{-3},$$

with $q_{\alpha} \geq 0$. The latter alternative permits arbitrary locations of the local nuclei.

Then the two expectation towers are

$$\mathbb{E}_Q(K * \mu_j)(y) = \mathbb{E}_Q P^{\text{ms}}(y), \quad \mathbb{E}_Q(K * \rho_X^c)(y) = \mathbb{E}_Q P^c(y), \quad (5.6)$$

where P^{ms} is the potential of the raw master packets and P^c is the raw core potential. Moreover,

$$|\mathbb{E}_Q(H_j - h_X)(y)| \leq Ca^{-4}[\tau^{1/5} + \beta^{1/5} + \sqrt{\beta} + \varepsilon_a \tau^{-1/2} + R(\tau) + (\tau + \sqrt{3}\beta)^{1/5}], \quad (5.7)$$

where $R(\tau) = \tau^2$ in the source-free alternative and $R(\tau) = \tau^2 + \tau^{1/2}$ in the nuclear alternative. Thus the common upper bound $R(\tau) \leq C\tau^{1/2}$ is always available. All constants are independent of the number of particles; they depend only on the constants in the displayed hypotheses.

Proof. Write

$$P^{\text{raw}}(y) = \sum_i K(y - x_i), \quad P^{\text{ms}}(y) = \sum_i (K * \mathcal{K}_{x_i})(y), \quad P^{\text{raw}} = P^c + P^{\text{ret}} + P^{\text{del}}.$$

The first identity in (5.6) follows by multiplying the original P -conditional expectation defining μ_j by $\mathbf{1}_A$, using $A \in \mathcal{F}_j$, and dividing by $P(A)$. The fresh labels do not alter this expectation. For the second identity, condition the raw-plus-label marginal of Q on X . That marginal is exactly the localization law of Ψ_A ; the original observations can be integrated out. Tonelli justifies both identities initially for nonnegative potentials. In particular the first tower is under the original observation law, while the second uses the fresh tilted patch law.

Subtracting the definitions of H_j and h_X , and then using these towers, gives the exact algebraic identity

$$\begin{aligned} \mathbb{E}_Q(H_j - h_X)(y) &= \mathbb{E}_Q[(P^{\text{raw}} - P^{\text{ms}}) - P^{\text{del}} \\ &\quad - (P^{\text{ret}} - K * \sigma) + K * (\rho_X - \sigma)](y). \end{aligned} \quad (5.8)$$

The estimates below also establish finiteness of all terms, so this subtraction involves no difference of infinite expectations.

For $0 < u \leq 4a$, the local density bound and Hölder's inequality give

$$\begin{aligned} \mathbb{E}_Q \sum_{|x_i - y| < u} K(y - x_i) &\leq \left(\int_{B(y, 4a)} \rho_{\Psi_A}^{5/3} \right)^{3/5} \left(\int_{|z| < u} |z|^{-5/2} dz \right)^{2/5} \\ &\leq Ca^{-4}(u/a)^{1/5}. \end{aligned} \quad (5.9)$$

The complementary raw potential is bounded by $n/(4a)$. By radial smearing, $P^{\text{raw}} - P^{\text{ms}} \geq 0$ and only centers with $|x_i - y| \leq t_{x_i}$ can contribute. Lipschitz continuity of the width implies $|x_i - y| \leq t_y + |x_i - y|/2$, hence these centers lie in $B(y, 2t)$. Equation (5.9) bounds its mean by $Ca^{-4}\tau^{1/5}$. The same Newton comparison for each fine packet gives

$$0 \leq \mathbb{E}_Q(P^{\text{ret}} - K * \sigma)(y) \leq Ca^{-4}\beta^{1/5}.$$

The distance and count assumptions for the deleted particles give $\mathbb{E}_Q P^{\text{del}}(y) \leq Ca^{-4}\sqrt{\beta}$.

For the signed density difference take $v_t(z) = \min\{|y - z|^{-1}, t^{-1}\}$. This is in $\dot{H}^1$ and

$$\|\nabla v_t\|_2^2 = 4\pi \int_t^\infty r^{-2} dr = 4\pi/t.$$

Lemma 3.1 and Cauchy–Schwarz in probability give

$$\mathbb{E}_Q \left| \int v_t(\rho_X - \sigma) \right| \leq Ct^{-1/2} \sqrt{\mathbb{E}_Q D(\rho_X - \sigma)} \leq Ca^{-4} \varepsilon_a \tau^{-1/2}.$$

It remains to control the two nonnegative remainders of the truncation inside $B(y, t)$. In the source-free case,

$$\int_{B(y, t)} K(y - z) \rho_X(z) dz \leq Ca^{-6} t^2 = Ca^{-4} \tau^2.$$

For the nuclear alternative, weighted convexity gives, with $Q_{\text{loc}} = \sum_\alpha q_\alpha$,

$$V_{\text{loc}}^{3/2} \leq Q_{\text{loc}}^{1/2} \sum_\alpha q_\alpha |z - R_\alpha|^{-3/2}.$$

The assertion is read as zero if $Q_{\text{loc}} = 0$. Uniformly in $R \in \mathbb{R}^3$,

$$\int_{B(y, t)} \frac{dz}{|y - z| |z - R|^{3/2}} \leq Ct^{1/2}.$$

For completeness, apply Hölder to the second factor with any exponent $q \in (3/2, 2)$ and to the first with conjugate exponent $q' \in (2, 3)$. Integrals of these radial decreasing powers over a ball are no larger than their centered-ball integrals; the powers of t combine to $t^{1/2}$. Since $\rho_X \leq C(a^{-6} + V_{\text{loc}}^{3/2})$, its remainder is at most

$$C(a^{-6} t^2 + Q_{\text{loc}}^{3/2} t^{1/2}) \leq Ca^{-4}(\tau^2 + \tau^{1/2}).$$

This proves the asserted nuclear bound even when a nucleus lies at y ; only $V_s(y)$, not the unsmoothed $V(y)$, was required finite.

For σ , only packets centered within $t + \sqrt{3}b$ of y meet the truncation ball. Extend each such packet's kernel integral to all space and use radial smearing to bound it by its raw point potential. Equation (5.9) bounds the mean remainder by $Ca^{-4}(\tau + \sqrt{3}\beta)^{1/5}$. The truncated σ potential is integrable, being at most n/t . Together with Coulomb duality and the just obtained remainder bound, this also proves integrability of the full ρ_X potential. The raw and core potentials were already integrable by (5.9). Hence all the preceding subtractions are valid. Adding the four estimates in (5.8) proves (5.7). $\square$

In both later applications the screening estimates supply $\varepsilon_a = a^{.01}$ and $\beta = a^2$. The master geometry supplies $\tau \geq ca^w$, with $w = 10^{-5}$, and an upper bound on τ tending to zero with the outer small parameter. Consequently

$$a^{.01} \tau^{-1/2} \leq Ca^{.01 - w/2}, \quad .01 - w/2 = .009995 > 0,$$

and every term in the bracket in (5.7) tends to zero uniformly in the participating bands. Thus the transfer is $o(a^{-4})$. The specific lower probability thresholds for the events are used in those applications to verify the screening hypotheses through (5.4); no probability threshold is needed for the transfer identity itself.

5.4 Maxima before conditional averaging

The next local rule joins fields on actual open neighborhoods. It will be used in both propagation arguments after their common singularities have been canceled.

Lemma 5.4 (Maximum of weak subsolutions). *Let $\mathcal{O} \subset \mathbb{R}^3$ be open, let g be locally bounded, and let $F(y, t)$ be measurable in y , continuous in t , and bounded on compact spatial sets and bounded ranges of t . If finitely many functions $v_i \in C(\mathcal{O}) \cap H_{\text{loc}}^1(\mathcal{O})$ satisfy*

$$\Delta v_i \geq g + F(y, v_i)$$

in distributions, their maximum satisfies the same inequality with its own value in the reaction term. The assertion also holds for functions $v_i = V + a_i$ with continuous locally H^1 offsets a_i , when all inequalities have the common singular term $-4\pi\nu$ and the reaction vanishes in a neighborhood of $\text{supp } \nu$.

Proof. It suffices to consider two functions. For a nonnegative compactly supported smooth test φ , choose an increasing Lipschitz function θ_η that is zero on $(-\infty, 0]$ and one on $[\eta, \infty)$. Test the first inequality with $\theta_\eta(v_1 - v_2)\varphi$ and the second with $(1 - \theta_\eta(v_1 - v_2))\varphi$. These nonnegative compactly supported H^1 tests are legitimate by mollification, since the right sides are bounded on their support. The sum of the left sides is

$$\begin{aligned} & - \int [\theta_\eta(v_1 - v_2)\nabla v_1 + (1 - \theta_\eta(v_1 - v_2))\nabla v_2] \cdot \nabla \varphi \\ & - \int \theta'_\eta(v_1 - v_2) |\nabla(v_1 - v_2)|^2 \varphi. \end{aligned}$$

The second term is nonpositive. Deleting it and letting $\eta \downarrow 0$ gives the gradient of $\max(v_1, v_2)$ by the Sobolev positive-part formula. Dominated convergence gives the reaction at the maximum; the reactions agree at ties. Induction proves the finite version.

For the last assertion subtract V in each inequality. The singular measures cancel, and the reaction for the offsets becomes $F(y, V(y) + a_i(y))$. It is locally bounded: it vanishes near every nuclear point, and V is smooth off $\text{supp } \nu$. Apply the first assertion to the offsets and then add V back. $\square$

6 Electron counts in large nuclear voids

The local count bound of Lemma 4.4 controls the electrons in each annulus whose radius is comparable to its distance from the nuclei. Summing that bound over arbitrarily many annuli would lose a factor equal to the number of scales. We first remove this loss in a nuclear void. A geometric decomposition will then reduce the exterior region of an arbitrary molecule to only $O(M)$ such voids.

Throughout this section Ψ is the normalized eigenstate in the largest strictly bound sector, and $E = E_N = \inf_{j \geq 0} E_j$, as in Lemma 2.3. Expectations without an additional subscript refer to its position law, with spins summed. Constants may depend on the fixed numbers $A = 40$ and $L = 10^6$, but on no feature of the molecule.

Lemma 6.1 (A large nuclear void). *There are universal constants $t_0, C > 0$ with the following property. Let $O \in \mathbb{R}^3$, $t \geq t_0$ and $T = 2^J t$, where $J \geq 1$ is an integer. Suppose that at least one nucleus lies in $B(O, t/100)$ and that every nucleus satisfies*

$$|R_m - O| < t/100 \quad \text{or} \quad |R_m - O| > 100T.$$

Then

$$\|\#\{i : t \leq |x_i - O| < T\}\|_{L^2(\Psi)} \leq C. \quad (6.1)$$

Proof. Translate O to the origin. Put $u_i = |x_i|$, $s_k = 2^k t$ and

$$n_k = \#\{i : s_k \leq u_i < 2s_k\}, \quad L_{\text{ann}} = \sum_{k=0}^{J-1} n_k.$$

For $s_k/4 \leq |v| \leq 8s_k$, the nuclear assumptions give $cs_k \leq d(v) \leq Cs_k$, where $d(v)$ is the distance to the nearest nucleus. Indeed, the inner nucleus supplies the upper bound; the inner and outer alternatives supply the lower bound. Choose t_0 so large that $d(v)/L > 1$ on all these annuli. Lemma 4.1 then gives $a_v = d(v)/L \asymp s_k/L$.

Each enlarged annulus is covered by a universally bounded number of local balls $B(v, a_v)$ with comparable scales. For example, take a maximal set of centers in the annulus separated by cs_k/L , with c small; the disjoint balls of half that radius give the bound on their number. Lemma 4.4, applied to any δ -state ω in the fixed sector N , consequently yields

$$\|n_k\|_{L^2(\omega)} + \|n_{\text{neigh},k}\|_{L^2(\omega)} \leq C(1 + \sqrt{\delta s_k}), \quad n_{\text{neigh},k} = \#\{i : s_k/2 \leq u_i \leq 4s_k\}. \quad (6.2)$$

This proves the result when J is bounded. We may therefore assume that J exceeds a fixed sufficiently large integer.

A weighted eigenstate inequality. Choose $0 \leq \chi \leq 1$ supported in $(32t, T/32)$, equal to one on $[64t, T/64]$, and satisfying $|\chi'(u)| \leq C/u$, $|\chi''(u)| \leq C/u^2$. Such a function is obtained by multiplying a fixed cutoff at scale t by a fixed cutoff at scale T . Set $w(u) = u\chi(u)^2$. It is a bounded smooth multiplier that vanishes near the origin. For each weighted index, integration by parts gives

$$\begin{aligned} \text{Re}\langle w(u_i)\Psi, -\tfrac{1}{2}\Delta_i\Psi \rangle &= \tfrac{1}{2} \int w(u_i) |\nabla_i\Psi|^2 - \tfrac{1}{4} \int \Delta_i w(u_i) |\Psi|^2 \\ &\geq -C\mathbb{E} \frac{\mathbf{1}_{\{t \leq u_i < T\}}}{u_i}. \end{aligned}$$

These identities hold first after form-domain approximation and then by continuity, since w and its derivatives are bounded for fixed t, T . Summing the last bound costs at most $C \sum_k s_k^{-1} \mathbb{E} n_k \leq C/t \leq C$ by (6.2) with $\delta = 0$. After fixing x_i and its spin, the remaining Hamiltonian has energy at least $E_{N-1} \geq E$. Its contribution against the nonnegative weight $w(u_i)$ is therefore nonnegative. Testing the eigen-equation with $\sum_i w(u_i)\Psi$ shows that weighted repulsion minus attraction is at most C .

For an electron in a shell meeting the support of χ , keep only its repulsion against electrons at radius $< t$, at radius $\geq T$, and in shells $l \leq k-5$. Dropping the other repulsions preserves the upper bound. Write

$$P_k = \sum_{\substack{0 \leq l < J \\ l \leq k-5}} n_l, \quad q = \nu(B(0, t)) - \#\{i : u_i < t\},$$

and define the exterior signed potential at the origin by

$$B = \int_{|R| > 100T} \frac{d\nu(R)}{|R|} - \sum_{u_i \geq T} \frac{1}{u_i}.$$

For each such k , use a cut that is full out on $[s_k/2, 4s_k]$, supported in $[s_k/4, 8s_k]$, and has gradients bounded by C/s_k . Let X_k be its out data as in Lemma 3.4; a bar will mean conditional expectation given X_k in this paragraph and below. Set

$$\varphi(v) = V(v) - \sum_{u_i < t} \frac{1}{|v - x_i|} - \sum_{u_i \geq T} \frac{1}{|v - x_i|}, \quad \psi_k(v) = \overline{\varphi(v) - \sum_{l \leq k-5} \sum_{i \in l} \frac{1}{|v - x_i|}}.$$

Every supported shell satisfies $32t \leq s_k \leq T/64$. All sources defining ψ_k are outside $\{s_k/4 < |v| < 8s_k\}$: prefix electrons have radius $< s_k/16$, inner electrons have radius $< t \leq s_k/32$, and exterior electrons have radius $\geq T \geq 64s_k$. Thus ψ_k has a continuous harmonic version on this annulus. Newton's sphere-average formula gives

$$\langle \psi_k \rangle_u = \frac{\bar{q} - \bar{P}_k}{u} + \bar{B}, \quad s_k/4 < u < 8s_k, \quad (6.3)$$

where $\langle F \rangle_u = (4\pi u^2)^{-1} \int_{|v|=u} F(v) dS(v)$. The shell- k positions are all recorded in X_k . We may therefore condition the selected interaction terms while retaining their weights. The preceding eigenstate inequality becomes

$$-\mathbb{E} \sum_k \sum_{i \in k} u_i \chi(u_i)^2 \psi_k(x_i) \leq C, \quad (6.4)$$

where k ranges over the supported shells. Conditional harmonic versions and sphere averages are justified directly by the positive distance to these sources and their fixed finite total masses.

A size-biased law for one shell. Fix a supported k . Choose a smooth function b_k , equal to one on $[s_k, 2s_k]$, supported in $(s_k/2, 4s_k)$, with $0 \leq b_k \leq 1$ and $|b'_k| \leq C/s_k$. Put

$$S_k = \sum_i b_k(u_i)^2, \quad m_k = \mathbb{E} S_k.$$

Then $n_k \leq S_k \leq n_{\text{neigh},k}$. If $m_k = 0$, every shell- k contribution vanishes and can be omitted. Otherwise define

$$\Psi_k = \sqrt{S_k/m_k} \Psi,$$

and denote its expectation and L^2 norm by $\mathbb{E}_k$ and $\|\cdot\|_{2,k}$. The square root of a sum of squares is Lipschitz, and, off its zero set,

$$|\nabla \sqrt{S_k}|^2 = \frac{\sum_i b_k(u_i)^2 |\nabla b_k(u_i)|^2}{S_k} \leq C s_k^{-2} \mathbf{1}_{\{n_{\text{neigh},k} > 0\}} \leq C s_k^{-2} n_{\text{neigh},k}.$$

Its gradient is zero almost everywhere on its zero set. The multiplier identity in Lemma 2.4 shows that Ψ_k is a δ_k -state, with

$$\delta_k \leq \frac{C \mathbb{E} n_{\text{neigh},k}}{s_k^2 m_k}. \quad (6.5)$$

The random variable S_k is X_k -measurable, because its support is inside the full-cut zone. The joint position and cut-label law changes by the factor S_k/m_k . Its conditional position law given X_k is therefore unchanged on $S_k > 0$. In particular all the barred fields and counts above have the same conditional versions under this new law. This permits us to use the local estimates for Ψ_k without changing the fields in (6.4).

A positive bound and a signed harmonic comparison. On the broad annulus of harmonicity, the base and prefix sources lie outside the exclusion defining J_v . Its radius Aa_v is a fixed small fraction of $|v|$, since A/L is small. The additional electrons subtracted by J_v give the upper bound

$$\psi_k(v) \leq \overline{J_v} + C \sum_{l \geq k-4} \frac{\bar{n}_l}{\max(s_l, s_k)}. \quad (6.6)$$

For comparable shells the exclusion supplies the denominator cs_k ; for sufficiently larger shells radial separation supplies cs_l . These are precisely the electrons absent from the definition of ψ_k .

The cut defining X_k is covered by a universally bounded number of cells at scales comparable to s_k/L . Its gradients are C/s_k and $a_v \geq 1$ there, so it meets the initial-cut hypothesis of Lemma 4.2. For targets in the same broad annulus, that lemma and (6.2), now applied under Ψ_k , give

$$\|(\psi_k(v))_+\|_{2,k} \leq \frac{C(1 + \sqrt{\delta_k s_k})}{s_k}.$$

Conditional Jensen's inequality bounds $\|\bar{n}_l\|_{2,k}$ by $\|n_l\|_{2,k}$. Thus the added count terms are bounded by the convergent series

$$C \sum_{l \geq k-4} \frac{1 + \sqrt{\delta_k s_l}}{s_l} \leq \frac{C(1 + \sqrt{\delta_k s_k})}{s_k}.$$

Positive parts of harmonic functions are subharmonic. Covering the compact annulus $\{s_k/2 \leq |v| \leq 3s_k\}$ by a fixed number of interior balls in the broad annulus, and using the ball submean inequality and Minkowski's inequality, consequently gives

$$A_k = s_k \sup_{s_k/2 \leq |v| \leq 3s_k} (\psi_k(v))_+, \quad \|A_k\|_{2,k} \leq C(1 + \sqrt{\delta_k s_k}). \quad (6.7)$$

The function $A_k/s_k - \psi_k$ is nonnegative and harmonic in the interior of this latter annulus. Its value at any point of a sphere $s_k \leq u \leq 2s_k$ is at least c times its sphere mean, for one universal $c > 0$. Indeed, after scaling, a bounded chain of interior balls joins any two points of these spheres; on a ball, this comparison follows by comparing its Poisson kernels on a smaller concentric ball. Applying this comparison and (6.3) gives

$$u\psi_k(v) \leq CA_k + c(\bar{q} + u\bar{B} - \bar{P}_k), \quad |v| = u \in [s_k, 2s_k]. \quad (6.8)$$

The negative prefix term retains a fixed positive coefficient.

The endpoint fields. Put $s_- = 2t$ and $s_+ = T/2$. For a supported shell and $u \in [s_k, 2s_k]$,

$$\bar{q} + u\bar{B} = (1 - \vartheta(u))(\bar{q} + s_-\bar{B}) + \vartheta(u)(\bar{q} + s_+\bar{B}), \quad \vartheta(u) = \frac{u - s_-}{s_+ - s_-}, \quad 0 \leq \vartheta(u) \leq Cs_k/T.$$

At either endpoint all base sources are outside the J_v exclusion. Sphere averaging $\bar{\varphi}$ and adding back the annular electrons therefore yields

$$\bar{q} + s\bar{B} \leq s\langle (\overline{J_v})_+ \rangle_s + Cs \sum_l \frac{\bar{n}_l}{\max(s_l, s)}, \quad s \in \{s_-, s_+\}. \quad (6.9)$$

Indeed, the omitted contribution of an annular electron is nonnegative and at most its full Coulomb kernel. The latter has sphere average $1/\max(s, u_i) \leq 1/\max(s, s_l)$ for an electron in shell l . For

these targets $a_v \asymp s/L$. The initial cut is still at scale s_k/L , so the energy parameter in Lemma 4.2 satisfies $e_* \leq C(\min(s, s_k)^{-1} + \delta_k)$. The precise consequence is

$$\left\| s \langle (\bar{J}_v)_+ \rangle_s \right\|_{2,k} \leq C \sqrt{\frac{s}{\min(s, s_k)}} + \delta_k s. \quad (6.10)$$

At $s = s_-$ this is at most $C(1 + \sqrt{\delta_k s_k})$. At $s = s_+$ the interpolation weight gives instead

$$C \frac{s_k}{T} \sqrt{\frac{T}{s_k} + \delta_k T} \leq C \left(\sqrt{\frac{s_k}{T}} + \sqrt{\delta_k s_k} \sqrt{\frac{s_k}{T}} \right) \leq C(1 + \sqrt{\delta_k s_k}).$$

The inner endpoint count term in (6.9) has norm at most

$$C \sum_l 2^{-l} (1 + \sqrt{\delta_k 2^l t}) \leq C(1 + \sqrt{\delta_k t}).$$

For the outer endpoint, $s_+/\max(s_l, s_+) \leq 1$; we retain its contribution as $C(s_k/T)\bar{L}_{\text{ann}}$. Thus (6.8) supplies a nonnegative, X_k -measurable random variable D_k , the same for every electron in shell k , such that

$$u_i \psi_k(x_i) \leq D_k + C \frac{s_k}{T} \bar{L}_{\text{ann}} - c \bar{P}_k, \quad \|D_k\|_{2,k} \leq C(1 + \sqrt{\delta_k s_k}). \quad (6.11)$$

Summation over shells. Changing to the size-biased law and using (6.5), $s_k \geq 1$, and $m_k \leq \mathbb{E}n_{\text{neigh},k}$, gives

$$\begin{aligned} \mathbb{E}[n_k D_k] &\leq m_k \mathbb{E}_k D_k \leq C m_k (1 + \sqrt{\delta_k s_k}) \\ &\leq C \left(m_k + \sqrt{m_k \mathbb{E}n_{\text{neigh},k}/s_k} \right) \leq C \mathbb{E}n_{\text{neigh},k}. \end{aligned} \quad (6.12)$$

The neighbor annuli for supported k lie inside $\{t \leq |x| < T\}$ and have bounded overlap, so these costs sum to $C\mathbb{E}L_{\text{ann}}$. For the outer term, n_k is X_k -measurable and the tower property gives

$$\begin{aligned} \sum_k \frac{s_k}{T} \mathbb{E}[n_k \bar{L}_{\text{ann}}] &= \sum_k \frac{s_k}{T} \mathbb{E}[n_k L_{\text{ann}}] \\ &\leq \|L_{\text{ann}}\|_2 \sum_k \frac{s_k}{T} \|n_k\|_2 \leq C \|L_{\text{ann}}\|_2. \end{aligned}$$

Here the unconditioned norms use (6.2) with $\delta = 0$, and $\sum_k s_k/T \leq 1$. Insert (6.11) into (6.4). The weights $W_k = \sum_{i \in k} \chi(u_i)^2$ are X_k -measurable and satisfy $0 \leq W_k \leq n_k$. Removing the bar on P_k by the tower property, we obtain

$$\mathbb{E} \sum_k W_k P_k \leq C(1 + \|L_{\text{ann}}\|_2). \quad (6.13)$$

Let I_{mid} consist of the shells entirely contained in $[64t, T/64]$, and set $L_{\text{mid}} = \sum_{k \in I_{\text{mid}}} n_k$. On these shells $W_k = n_k$. The sum on the left of (6.13) therefore includes all products $n_k n_l$ with $k, l \in I_{\text{mid}}$ and $l \leq k - 5$. For every omitted close pair $|k - l| < 5$, including $k = l$, use the tilt at k and (6.2) to obtain

$$\mathbb{E} n_k n_l \leq m_k \mathbb{E}_k n_l \leq C m_k (1 + \sqrt{\delta_k s_l}) \leq C \mathbb{E}n_{\text{neigh},k}.$$

The last step uses $s_l \asymp s_k$ and the same calculation as (6.12). Each k has only a bounded number of such neighbors. Thus the close-pair costs sum to $C\mathbb{E}L_{\text{ann}}$, rather than a cost proportional to J . Expanding L_{mid}^2 now gives

$$\mathbb{E}L_{\text{mid}}^2 \leq C(1 + \|L_{\text{ann}}\|_2).$$

Only a fixed number of endpoint shells were omitted, so $\|L_{\text{ann}} - L_{\text{mid}}\|_2 \leq C$ by (6.2). Consequently $X = \|L_{\text{ann}}\|_2$ satisfies $X^2 \leq C(1 + X)$, proving the lemma. $\square$

Lemma 6.2 (Exterior count). *There is a universal constant C such that, for $0 < s \leq 1$,*

$$\mathbb{E}\#\{i : a_{x_i} \geq s\} \leq CMs^{-3}. \quad (6.14)$$

The constant is independent of the charges and the distances between nuclei.

Proof. Write $\mathcal{R} = \{R_1, \dots, R_M\}$ and $d(x) = \text{dist}(x, \mathcal{R})$. We use $d(x) \leq La_x$ from Lemma 4.1. For a Borel set B , write $n(B) = \#\{i : x_i \in B\}$. The local count bound (4.36), applied with $\delta = 0$, gives

$$\mathbb{E}n(B(y, a_y)) \leq C \max\{a_y^{-3}, 1\}. \quad (6.15)$$

We first prove that

$$\mathbb{E}n(\{d \geq U\}) \leq CM \quad (6.16)$$

for a sufficiently large universal constant U . A direct summation of local bounds over all large distance scales would give an infinite series. We instead group scales into a bounded number of annuli to which Lemma 6.1 applies.

Nested sets of nuclear representatives. Fix

$$h = 12, \quad \varepsilon = 2^{1-h} = \frac{1}{2048}, \quad D_0 = 1000, \quad q = 10.$$

Thus

$$\varepsilon < \frac{1}{200}, \quad D_0 - \varepsilon > 400, \quad D_0 > 3\varepsilon, \quad 2^q > D_0.$$

Choose an integer j_0 such that $2^{j_0}/L \geq 1$ and 2^{j_0-1} exceeds the lower threshold for the inner radius in Lemma 6.1. Set $U = 2^{j_0}$. Since the nuclei are distinct, there is an integer $l_0 < j_0 - h$ such that $\mathcal{R}$ is 2^{l_0} -separated. Put $P_{l_0} = \mathcal{R}$. For each $l > l_0$, choose a maximal 2^l -separated subset $P_l \subset P_{l-1}$, where separated means that distinct points have distance at least 2^l .

For each $p \in P_{l-1}$ choose a parent $\pi_l(p) \in P_l$, keeping $\pi_l(p) = p$ for $p \in P_l$. Maximality permits this choice with $|p - \pi_l(p)| < 2^l$ whenever p is removed. Composing these maps gives an ancestor $\alpha_l(R) \in P_l$ for every original nucleus R , and

$$|R - \alpha_l(R)| < 2^{l+1}. \quad (6.17)$$

Indeed, the accumulated displacement is less than $\sum_{k=l_0+1}^l 2^k < 2^{l+1}$; at $l = l_0$ it is zero. Every representative is an original nucleus, and a removed nucleus never reappears in the nested sets.

For $j \geq j_0$, put $u_j = 2^j$. Choose a nearest nucleus $R(x)$ measurably, breaking ties by the least index. Assign a point of the dyad $u_j \leq d(x) < 2u_j$ to $c = \alpha_{j-h}(R(x)) \in P_{j-h}$, and denote its assigned set by $A_{j,c}$. These are Borel sets, disjoint at each scale, and

$$|R(x) - c| < \varepsilon u_j, \quad A_{j,c} \subset B(c, (2 + \varepsilon)u_j). \quad (6.18)$$

Only finitely many crowded pairs. Call (j, c) crowded if there is a distinct $w \in P_{j-h}$ with $|w - c| \leq D_0 u_j$; otherwise call it isolated. We claim that the number of crowded pairs is at most CM . For each crowded pair choose one witness w . The points c and w cannot both survive to P_{j+q} , since that set is 2^{j+q} -separated and $2^{j+q} > D_0 u_j$. Let r be the first level at which either endpoint is removed, and charge (j, c) to one such removal event (r, p) , where $p \in P_{r-1} \setminus P_r$. Then

$$j - h < r \leq j + q, \quad \text{or equivalently} \quad r - q \leq j \leq r + h - 1. \quad (6.19)$$

A fixed removal event can therefore receive charges from at most $q + h$ levels j . At any one of these levels, a center c charging (r, p) is either p itself or has p as its witness. In either case, $c \in P_{j-h} \cap \overline{B}(p, D_0 u_j)$. Such centers are 2^{j-h} -separated. The disjoint balls of radius 2^{j-h-1} around them lie in the ball of radius $D_0 u_j + 2^{j-h-1}$ around p , so their number is at most $(1 + 2D_0 2^h)^3$. This is a fixed constant. There are at most $M - 1$ removal events, since the nested sets eventually contain one nucleus. Consequently

$$\#\{(j, c) : j \geq j_0, (j, c) \text{ is crowded}\} \leq CM. \quad (6.20)$$

Simultaneous removals cause no difficulty: the charge uses just one of the two removed endpoints.

Each crowded assigned set has bounded expected particle count. To see this, choose a maximal $u_j/(2L)$ -separated subset of $A_{j,c}$. By (6.18), a volume comparison bounds its cardinality by CL^3 . Its balls of radius $u_j/(2L)$ cover $A_{j,c}$. At every center y we have $a_y \geq d(y)/L \geq u_j/L \geq 1$, so these covering balls are contained in $B(y, a_y)$. Equation (6.15) therefore gives

$$\mathbb{E}n(A_{j,c}) \leq C. \quad (6.21)$$

Together, (6.20) and (6.21) show that all crowded assigned sets cost at most CM expected particles.

Isolated runs give empty annuli. For a fixed nucleus c , the levels $j \geq j_0$ at which $c \in P_{j-h}$ form an initial integer interval, possibly empty or infinite. Removing its crowded levels leaves consecutive runs of isolated levels. The number of these runs is at most one plus the number of crowded levels for c . Summing over c and using (6.20), the total number of isolated runs is at most CM .

At an isolated level j for c , every original nucleus R satisfies one of the two alternatives

$$|R - c| < \varepsilon u_j \quad \text{or} \quad |R - c| > (D_0 - \varepsilon) u_j. \quad (6.22)$$

Indeed, if $\alpha_{j-h}(R) = c$, the first alternative follows from (6.17). Otherwise isolation gives $|\alpha_{j-h}(R) - c| > D_0 u_j$, and the ancestor offset gives the second alternative.

The alternative for a fixed nucleus cannot change between consecutive isolated levels j and $j + 1$ for the same center c . A nucleus near c at level j remains too close to satisfy the far alternative at $j + 1$. Conversely, a nucleus far at j could be near at $j + 1$ only if its fixed distance from c were both greater than $(D_0 - \varepsilon) u_j$ and less than $2\varepsilon u_j$, which is impossible because $D_0 > 3\varepsilon$. Thus the set of near nuclei is constant throughout each isolated run. This argument permits ancestor maps to change and uses only the gap in (6.22).

Consider first a finite run $j = a, \dots, b$ and put

$$O = c, \quad t = u_a/2, \quad T = 4u_b.$$

The near nuclei have distance less than $\varepsilon u_a < t/100$ from c . All other nuclei have distance greater than $(D_0 - \varepsilon) u_b > 100T$. There is a nucleus at c , and $T/t = 2^{b-a+3}$ is a power of two. By the

choice of j_0 , Lemma 6.1 applies. Every assigned point from this run belongs to its annulus, since c is itself a nucleus and

$$t \leq u_a \leq d(x) \leq |x - c| < (2 + \varepsilon)u_b < T.$$

It follows that the union of the assigned sets over this run has expected particle count at most C , independently of its length. For an infinite run, apply the same argument to each finite initial segment and then use monotone convergence for their increasing assigned unions. There are at most CM runs, so the total contribution of all isolated assigned sets is at most CM . Combined with the crowded sets, this proves (6.16).

The remaining distances. It remains to count particles with $a_x \geq s$ and $d(x) < U$. For the part with $d(x) < s$, choose a maximal $s/2$ -separated subset of $\{d < s, a \geq s\}$. Its centers lie in $\bigcup_{m=1}^M B(R_m, s)$. Disjoint balls of radius $s/4$ around them show, by volume comparison, that there are at most CM centers. Their balls of radius $s/2$ cover the set and are contained in the corresponding local balls $B(y, a_y)$, since $a_y \geq s$. Equation (6.15) bounds this contribution by CMs^{-3} .

For a distance dyad $u \leq d(x) < 2u$, choose a maximal $u/(2L)$ -separated subset. Its centers lie in $\bigcup_m B(R_m, 2u)$, so there are at most CL^3M of them. Their local balls cover the dyad, because $a_y \geq d(y)/L \geq u/L$. Thus (6.15) bounds its expected count by

$$CM \max\{(u/L)^{-3}, 1\}.$$

Take $u = 2^k s$ for $0 \leq k \leq K$, where K is the least nonnegative integer such that $2^K s \geq U$. These dyads cover $s \leq d(x) < U$. The sum of the inverse-cube terms is bounded by Cs^{-3} . For the constant terms with $u < 1$, use $1 \leq u^{-3}$; the number of remaining dyads, with $1 \leq u \leq 2U$, is bounded by a universal constant. Their total contribution is therefore at most CMs^{-3} . Adding (6.16) proves (6.14). $\square$

7 Deterministic comparison fields

We now construct positive fields that can be compared across all the local scales. The construction uses only the nuclear measure. Its output is a family of continuous offsets satisfying a nonlinear subsolution inequality, together with a strictly positive gap when the scale doubles. The gap will allow the quantum field to replace a comparison field in the next stage of the proof.

Fix a nuclear system and a small parameter $r_* > 0$. Its value in terms of the excess charge, together with the dyadic scales used for the observations, will be chosen in Section 8. Every smallness threshold for r_* in the present deterministic construction is universal and independent of the nuclear system.

Throughout this section, $p = 3/2$, $k = ((5/3)c_{\text{TF}})^{-3/2}$, and $c_F = 4\pi k$. Recall that $K(x) = |x|^{-1}$ and that a_x is the nuclear scale of Lemma 4.1. All constants in this section are independent of the nuclear configuration and charges. They may depend on the fixed localization constants and the smooth radial function g used for wave packets.

7.1 Global Thomas–Fermi fields

The variational and comparison framework for point and smoothed sources is classical [10, Theorems II.14, II.17–II.18, and IV.7–IV.10]. We give a proof on the Coulomb-space cone used here, including the decay needed for comparison on an unbounded domain. If ζ is a finite sum of positive

point charges and nonnegative smooth compactly supported radial charges, write $V_\zeta = K * \zeta$. A radial charge is specified by its center, profile, and total mass. Set

$$\mathcal{X}_+ = \{\rho \geq 0 : \rho \in L^{5/3}(\mathbb{R}^3) \cap (\dot{H}^1(\mathbb{R}^3))^*\}.$$

By Lemma 3.1, the Coulomb energy of an element of this cone is the squared norm $D(\rho) = \|\nabla K * \rho\|_2^2 / (8\pi)$.

Lemma 7.1 (Global comparison problem). *Let ζ be as above, with total mass $Q > 0$. The functional*

$$\mathcal{T}_\zeta(\rho) = c_{\text{TF}} \int \rho^{5/3} - \int V_\zeta \rho + D(\rho), \quad \rho \in \mathcal{X}_+,$$

*has a unique minimizer. Its field $W = V_\zeta - K * \rho$ satisfies*

$$\rho = kW_+^p, \quad \Delta W = -4\pi\zeta + c_F W_+^p, \quad W \geq 0, \quad \int \rho \leq 2Q. \quad (7.1)$$

*The screening potential $K * \rho$ is continuous and tends to zero at infinity. Consequently W tends to zero at infinity and is continuous away from point charges.*

Proof. Smear the point charges at any fixed positive widths and leave the smooth charges unchanged. This decomposes $V_\zeta = V_s + V_c$ with $V_s \in \dot{H}^1$ and compactly supported $V_c \in L^{5/2}$. Thus

$$\left| \int V_s \rho \right| \leq C \|\nabla V_s\|_2 \sqrt{D(\rho)}, \quad \left| \int V_c \rho \right| \leq \|V_c\|_{5/2} \|\rho\|_{5/3}.$$

For the fixed source and smearing widths, both displayed potential norms are finite. Young's inequality proves coercivity in both spaces defining $\mathcal{X}_+$. A bounded minimizing sequence has weakly convergent subsequences in these two reflexive spaces. Their limits agree as distributions, since both converge against compact smooth tests; nonnegativity passes to this limit. The two linear terms are weakly continuous in their respective spaces, and both positive terms are weakly lower semicontinuous. This proves existence. The strictly convex kinetic term proves uniqueness.

Nonnegative smooth compact variations give

$$d := \frac{5}{3} c_{\text{TF}} \rho^{2/3} - V_\zeta + K * \rho \geq 0 \quad \text{almost everywhere.}$$

For bounded smooth compactly supported φ , the variations $(1 + t\varphi)\rho$ are admissible for sufficiently small positive and negative t . Indeed $|\varphi\rho| \leq \|\varphi\|_\infty \rho$, and positivity of the Coulomb kernel bounds its Coulomb energy. The nonnegative pairing property in Lemma 3.1 therefore justifies differentiating these variations and gives $\int d\varphi\rho = 0$. Hence $d\rho = 0$, proving the Euler equation in (7.1) and its distributional Laplacian.

To establish finite mass without assuming it, let the source components have centers R_m and masses Q_m , and put

$$U(x) = \sum_m \frac{Q_m}{|x - R_m|}, \quad w(x) = U(x)^{-1}.$$

Newton's theorem gives $V_\zeta \leq U$. Where $\rho > 0$, the Euler equation gives $K * \rho \leq V_\zeta$. For any ball B , multiply this inequality by $1_B w\rho$ and restrict the screening integral below to B . Write $q_m(x) = w(x)Q_m/|x - R_m|$, so that $\sum_m q_m = 1$. The triangle inequality gives

$$\frac{w(x) + w(y)}{|x - y|} \geq \sum_m \frac{q_m(x)q_m(y)}{Q_m}.$$

The resulting symmetrization and Cauchy–Schwarz yield, with $m_B = \int_B \rho$,

$$m_B \geq \frac{1}{2} \sum_m \frac{(\int_B q_m \rho)^2}{Q_m} \geq \frac{m_B^2}{2Q}.$$

All integrals in this calculation are local and finite; in particular w is bounded on B , and the local Coulomb energy is finite. Increasing B proves $\int \rho \leq 2Q$.

For completeness, $L^1 \cap L^{5/3}$ implies continuity and decay of the screening potential. The kernel restricted to a fixed ball belongs to $L^{5/2}$, so its pairing with translates of ρ is continuous. The complementary kernel contributes at most $\|\rho\|_1/R$ when the cutoff radius is R . For fixed R , the local $L^{5/3}$ norm of ρ on a ball whose center tends to infinity tends to zero. First letting the center tend to infinity and then $R \rightarrow \infty$ proves decay.

Finally, $(W + \varepsilon)_-$ is compactly supported away from point charges for every $\varepsilon > 0$: $W \rightarrow 0$ at infinity, and its positive Coulomb singularities dominate the continuous screening potential. On the support of this test $\rho = 0$, and hence $\Delta W \leq 0$. Testing the equation gives $\int |\nabla(W + \varepsilon)_-|^2 \leq 0$. The test is justified by compact H^1 approximation away from the singularities. Letting $\varepsilon \downarrow 0$ proves $W \geq 0$. $\square$

Lemma 7.2 (Point fields and mixtures). *For one point charge $Q > 0$ at R , the field in Lemma 7.1 satisfies*

$$W(x) \geq c \min \left(\frac{Q}{|x - R|}, |x - R|^{-4} \right). \quad (7.2)$$

For a single point charge or radial smooth charge of mass Q , its screening potential is bounded by $CQ^{4/3}$.

More generally, select source components of masses Q_i , truncate them to masses $q_i \leq Q_i$, and put $q = \sum_i q_i > 0$. Let W_i be the single-source field of mass q at the i th center with that source's normalized profile. If W is the full-source field, then

$$\sum_i \frac{q_i}{q} W_i \leq W. \quad (7.3)$$

Proof. For a single source, Newton's theorem and the Euler equation give $\rho(z) \leq k(Q/|z - R|)^{3/2}$. For every x, R and $s > 0$,

$$\int_{B(x,s)} \frac{dz}{|x - z| |z - R|^{3/2}} \leq Cs^{1/2}. \quad (7.4)$$

To see the uniformity in R , apply Hölder with exponent $q_1 \in (3/2, 2)$ to the second factor and its conjugate $q_1' \in (2, 3)$ to the first. The integrals of both radial powers over $B(x, s)$ are bounded by their integrals over centered balls of the same radius. Their powers of s combine to $s^{1/2}$. Split $K * \rho(x)$ at $s = Q^{-1/3}$. The near part is at most $CQ^{3/2}s^{1/2}$ by (7.4), and the far part at most $2Q/s$ by Lemma 7.1. Both are $CQ^{4/3}$.

For a point charge this gives $W \geq Q/(2d)$ for $d = |x - R| \leq cQ^{-1/3}$. Outside that sphere set $v = \lambda d^{-4}$. For sufficiently small universal $\lambda > 0$,

$$\Delta v = 12\lambda d^{-6} \geq c_F v^{3/2},$$

and its boundary value is below the preceding lower bound. On the exterior domain, testing the equation for $v - W$ with $(v - W - \varepsilon)_+$ gives a nonpositive gradient square bounded below by zero. This test has compact support since both fields decay. Thus $v \leq W$. Together with the inner estimate this proves (7.2).

For the mixture $F = \sum_i (q_i/q) W_i$, convexity gives

$$\Delta F \geq -4\pi\zeta_{\text{truncated}} + c_F F^p \geq -4\pi\zeta + c_F F^p.$$

Compare with the equation for W and test with $(F - W - \varepsilon)_+$. The support is compact. Near a point charge whose mass was strictly decreased, $F - W$ tends to $-\infty$, so the test vanishes there. If the masses agree, the singular terms cancel and the difference is locally H^1 . The remaining density terms are locally in $L^{5/3}$, so compact H^1 approximation is valid. Monotonicity of the power gives $F \leq W$. $\square$

7.2 Fields adapted to the nuclear scale

Fix a sufficiently small constant $c_1 \in (0, 1/100)$. Smear the m th nucleus with the radial probability $g_{c_1 a_{R_m}}^2$ and denote the resulting density by ν_s . Put

$$V_s = K * \nu_s, \quad V_c = V - V_s \geq 0. \quad (7.5)$$

The subscript c records the part confined near the nuclei. A nucleus contributing to $V_c(x)$ lies within $2c_1 a_x$ of x . Its scale is comparable to a_x , and the total contributing charge is at most a_x^{-3} . Therefore

$$\nu_s(x) \leq C a_x^{-6}, \quad V_c \in L_c^{5/2}(\mathbb{R}^3). \quad (7.6)$$

The smooth potentials of nuclei within any of the fixed local multiples of a_x used below sum to at most $C a_x^{-4}$. Indeed each such width is comparable to a_x , each unit-mass smooth potential is at most C/a_x , and the total local charge is at most a_x^{-3} .

Let W_0, ρ_0 be the field and density for the original nuclear measure ν . Its continuous offset is

$$h_0 = W_0 - V_c = V_s - K * \rho_0.$$

For a fixed positive θ , to be chosen below, let W_s be the field for the smooth source $\theta \nu_s$.

Lemma 7.3 (Two reference fields). *There are fixed constants $c_0 > 0$, $\theta > 0$, $c_\theta > 0$, and $C < \infty$ such that, off the nuclei,*

$$W_0(x) \geq c_0 a_x^{-4}, \quad |h_0(x)| \leq C a_x^{-4}, \quad (7.7)$$

and everywhere

$$c_\theta a_x^{-4} \leq W_s(x) \leq C a_x^{-4}, \quad |\nabla W_s(x)| \leq C a_x^{-5}, \quad W_s(x) \leq 1.01 W_0(x). \quad (7.8)$$

Here h_0 is continuous, and W_s is locally C^1 . Both fields decay at infinity.

Proof. By the scale definition, $\nu(B(x, 2La_x)) > (2a_x)^{-3}$. Truncate charges in this ball to total $q = (2a_x)^{-3}$ and apply Lemma 7.2. Their distances from x are at most $2La_x$, so (7.2) gives the first bound in (7.7) with a fixed $c_0 > 0$.

If some charges contribute to $V_c(x)$, denote their total by $q \leq a_x^{-3}$. The screening estimate for each single field of charge q , followed by the mixture comparison, gives

$$W_0(x) \geq \sum_{\text{contributors}} \frac{z_m}{|x - R_m|} - C q^{4/3} \geq V_c(x) - C a_x^{-4}.$$

If there are no contributors, the same conclusion follows from $W_0 \geq 0$.

For the upper bound, set $a = a_x$, $s = 4a$, and let V_{loc} be the potential of nuclei within $20a$ of x . On $B(x, s)$ the function

$$V_{\text{loc}}(z) + \frac{Cs^4}{(s^2 - |z - x|^2)^4}$$

is a supersolution of the field equation when C is large: the Laplacian of the second term is at most $80Cs^6/(s^2 - |z - x|^2)^6$, whereas its p th power has the same spatial dependence and coefficient $C^{3/2}$. The boundary blowup dominates $W_0 - V_{\text{loc}}$. The positive-difference test is legitimate because the point singularities cancel. Thus $W_0(x) \leq V_{\text{loc}}(x) + Ca^{-4}$. Subtracting $V_c(x)$ and using the smooth local potential bound proves the upper offset estimate. The expression defining h_0 proves its continuity.

For W_s , the same bump without V_{loc} is a supersolution after its coefficient is enlarged to absorb the locally bounded source $4\pi\theta\nu_s$. This proves $W_s \leq Ca^{-4}$. Its Laplacian is bounded by Ca^{-6} on a local ball. Subtract the Newton potential of this source restricted to that ball. Its gradient is bounded by Ca^{-5} , since the integral of $|z|^{-2}$ on a ball of radius a is Ca . The remaining harmonic function is bounded by Ca^{-4} on a smaller ball; its interior gradient bound proves the stated estimate and local C^1 regularity.

For the lower bound, select and truncate the same charges as in the first paragraph, before multiplying them by θ . Their smearing widths are at most $3c_1a$, because their centers are within $2La$. A single such blob of charge $\theta(2a)^{-3}$ has bare potential at x at least $c_L\theta a^{-4}$: every source point is within $(2L + 3c_1)a$ of x . Its screening potential is at most $C\theta^{4/3}a^{-4}$. Choose θ sufficiently small and apply the mixture comparison to obtain $W_s \geq c_\theta a^{-4}$.

Decrease θ further if necessary. The source bound and the lower bound for W_0 imply

$$4\pi\theta\nu_s \leq c_F(1.01^{3/2} - 1.01)W_0^{3/2}.$$

Consequently $1.01W_0$ is a supersolution for the field of $\theta\nu_s$; its extra negative point sources have the required sign. The positive-difference test for $W_s - 1.01W_0$, supported away from those point singularities, proves the final comparison. Decay follows from Lemma 7.1. $\square$

The reference fields now have positive lower bounds at every scale. To compare them with the local quantum constructions, we also need to control variation over a much smaller ball. The offset is the right quantity for this purpose: subtracting V_c removes every nuclear singularity from the function being compared.

Lemma 7.4 (Oscillation of offsets). *Fix y and write $a = a_y$. Let $h = h_0$, or let $h = W - V_c$ for an interior Thomas–Fermi patch centered at y satisfying the hypotheses of Lemma 4.3. Then h has a continuous representative. For $0 < s \leq a/10$ and $|z - y| \leq s$,*

$$|h(z) - h(y)| \leq Ca^{-4}(s/a)^{1/2} + C(s/a)h(y)_-. \quad (7.9)$$

For $h = h_0$ the final term can be omitted.

Proof. In either case the local upper field estimate in (4.23) or Lemma 7.3 gives $h \leq Ca^{-4}$ on a fixed ball about y . The patch core is absent there. The Euler equation, the bound on ν_s , and the weighted power inequality give

$$|\Delta h(z)| \leq Ca^{-6} + Ca^{-3/2} \sum_{\text{loc}} z_m |z - R_m|^{-3/2}. \quad (7.10)$$

Here all centers in the sum belong to one fixed local ball and their total charge is at most a^{-3} . The same estimate holds for h_0 by Lemma 7.3.

Take the Newton particular solution for this Laplacian restricted to a ball of radius a . Equation (7.4) bounds it by Ca^{-4} on a smaller ball. Its variation over distance s is at most $Ca^{-4}(s/a)^{1/2}$. In detail, the part within distance Cs of either evaluation point is at most

$$C(a^{-6}s^2 + a^{-9/2}s^{1/2}).$$

On a dyadic ring of radius $l \geq Cs$, the difference of the two Coulomb kernels is at most Cs/l^2 , and the source mass in that ring is at most $C(a^{-6}l^3 + a^{-9/2}l^{3/2})$. Summing the rings up to radius a gives the asserted variation.

The harmonic remainder is bounded above by Ca^{-4} . Harnack's inequality and the harmonic gradient estimate, applied to this upper constant minus the remainder, bound its variation by $C(s/a)(a^{-4} + h(y)_-)$. Combining the estimates gives (7.9), as well as continuity. For h_0 , Lemma 7.3 bounds its negative part by Ca^{-4} , which is absorbed by the first term. $\square$

7.3 Averaging at a smaller radius

Fix $w = 10^{-5}$, and use the small parameter r_* fixed above. Define the smearing widths, center kernels, and averaging operator by

$$t_z = c_1 a_z \min(a_z, r_*)^w, \quad \mathcal{K}_z(y) = g_{t_z}^2(y - z), \quad Rf(y) = \int \mathcal{K}_z(y) f(z) dz, \quad \tau_y = t_y/a_y. \quad (7.11)$$

Thus R smooths a mass at its own center-dependent width. It preserves total mass, although it need not preserve the constant function exactly.

Lemma 7.5 (Kernel and potential estimates). *Uniformly as $r_* \downarrow 0$, a center contributing to $Rf(y)$ satisfies $|z - y| \leq 2t_y$ and $t_z/t_y = 1 + O(r_*^w)$. Moreover,*

$$R1(y) = 1 + O(r_*^w), \quad \|\nabla_z \mathcal{K}_z(y)\|_\infty \leq Ct_y^{-4}, \quad \|\nabla_z \mathcal{K}_z(y)\|_2 \leq Ct_y^{-5/2}. \quad (7.12)$$

The norms in these gradient estimates are taken in the center variable z . The operator R is bounded on $L^{5/3}$ and preserves integrals. The correction

$$q_0 = K * (\rho_0 - R\rho_0) \quad (7.13)$$

is nonnegative, continuous, belongs to $\dot{H}^1$, and tends to zero at infinity. It satisfies

$$0 \leq q_0(y) \leq Ca_y^{-4} \tau_y^{1/2}. \quad (7.14)$$

Proof. The scale function is $1/L$ -Lipschitz. The function $a \mapsto c_1 a \min(a, r_*)^w$ has Lipschitz constant $O(r_*^w)$. Thus $|t_z - t_y| \leq O(r_*^w)|z - y|$. If $|z - y| \leq t_z$, this proves the support and comparability assertions. Replacing t_z by t_y changes the kernel by at most $Cr_*^w t_y^{-3}$ on a ball of radius $2t_y$, giving the first assertion of (7.12). The chain rule for a Lipschitz width gives the gradient bounds, first pointwise almost everywhere and then in L^2 by the support bound. Since $\int \mathcal{K}_z(y) dy = 1$, Fubini proves preservation of integrals. Weighted Jensen followed by Fubini, using the bound on $R1$, proves the $L^{5/3}$ estimate.

Newton's theorem gives the nonnegative representation

$$q_0(y) = \int \rho_0(z) [K(y - z) - (K * g_{t_z}^2)(y - z)] dz.$$

Only $|z - y| \leq 2t_y$ contributes. The local field bound gives

$$\rho_0(z) \leq Ca_y^{-6} + Ca_y^{-3/2} \sum_{\text{loc}} z_m |z - R_m|^{-3/2}.$$

Equation (7.4) therefore bounds the displayed integral by $C(a_y^{-6}t_y^2 + a_y^{-9/2}t_y^{1/2})$, proving (7.14). Both ρ_0 and $R\rho_0$ belong to $L^1 \cap L^{5/3}$, hence also to $L^{6/5}$. Their potentials are continuous and decay by the proof of Lemma 7.1; the Sobolev inequality and the Coulomb energy identity put their difference in $\dot{H}^1$. $\square$

Define the nonlinearity acting on an offset $h \in \mathbb{R}$ by

$$f_y(h) = c_F \int \mathcal{K}_z(y)(V_c(z) + h)_+^p dz. \quad (7.15)$$

For each y , this is finite, convex, continuous, and nondecreasing in h . These properties follow from those of the positive power and the local integrability of V_c^p . We will use this averaged law in place of the unsmeared Thomas–Fermi equation.

7.4 Barriers with a strict gap between scales

We have established the reference fields and the averaging estimates. The remaining task is to lower the reference field just enough that its averaged density pays for the Laplacian of the lowering term. The strict inequality between the coefficients below leaves room for all errors from averaging.

Lemma 7.6 (Comparison barriers). *There are fixed positive constants $\alpha, \epsilon_0, C'_0, c_b, c'$ with the following properties for all sufficiently small $r_* > 0$. For $0 < r \leq r_*$ define*

$$\Gamma_r = \epsilon_0 W_s + C'_0 r^{4\alpha} W_s^{1+\alpha}, \quad B_r = W_0 + q_0 - \Gamma_r, \quad b_r = B_r - V_c, \quad h_b = h_0 - c_b a^{-4}. \quad (7.16)$$

The offsets b_r are continuous and locally H^1 and decay at infinity. On $\{a \geq r\}$ their Laplacian densities satisfy

$$\Delta b_r \geq -4\pi\nu_s + f_y(b_r). \quad (7.17)$$

More precisely, this inequality holds almost everywhere there between the locally integrable densities representing the two sides; on every open set contained in this region it is also a distributional inequality. On the same region,

$$B_r \geq 0.9W_0 > 0, \quad h_b + c'a^{-4} \leq b_r \leq Ca^{-4}. \quad (7.18)$$

For every contributing center z in (7.15),

$$V_c(z) + h_b(y) \geq ca_y^{-4} \quad \text{almost everywhere in } z. \quad (7.19)$$

All these constants are independent of r, r_ and the nuclear system.*

Proof. First fix θ as in Lemma 7.3. That lemma implies

$$D_* := \sup_x \frac{|\nabla W_s(x)|^2}{W_s(x)^{5/2}} < \infty$$

with a fixed universal bound. Since $\Delta W_s \leq c_F W_s^{3/2}$, the chain rule gives

$$\Delta(W_s^{1+\alpha}) \leq (1 + \alpha)(c_F + \alpha D_*) W_s^{\alpha+3/2}.$$

Choose $\alpha > 0$ so small that $(1 + \alpha)(1 + \alpha D_*/c_F) \leq 1.1$. Consequently

$$\Delta \Gamma_r \leq 1.1 c_F \sqrt{W_s} \Gamma_r. \quad (7.20)$$

All chain rules are local: W_s is positive and locally C^1 with locally bounded Laplacian, so multiplication in the weak equation justifies them.

Choose $c_b > 0$ smaller than a sufficiently small fixed fraction of c_0 . On $a \geq r$, $r^{4\alpha} W_s^\alpha \leq C^\alpha$. We may therefore choose positive ϵ_0, C'_0 so small that, throughout this region, Γ_r is a tiny fixed fraction of W_s , is at most $0.1 c_0 a^{-4}$, and is at most $(c_b/2) a^{-4}$. At the same time,

$$\Gamma_r \geq \epsilon_0 c_\theta a^{-4}. \quad (7.21)$$

These choices precede every smallness requirement on r_* .

The equations for h_0 and q_0 give the exact cancellation

$$\Delta(h_0 + q_0) = -4\pi\nu_s + c_F R(W_0^p). \quad (7.22)$$

Fix y with $a_y \geq r$. For a contributing center z , the offset oscillation estimate and (7.14) imply

$$\begin{aligned} V_c(z) + b_r(y) &= W_0(z) + h_0(y) - h_0(z) + q_0(y) - \Gamma_r(y) \\ &\leq W_0(z) - \Gamma_r(y) + C a_y^{-4} \tau_y^{1/2}. \end{aligned}$$

By (7.21), the final error is $o(\Gamma_r(y))$ uniformly as $r_* \rightarrow 0$. Furthermore,

$$W_0(z) \geq \frac{W_s(z)}{1.01} \geq \frac{1 - o(1)}{1.01} W_s(y),$$

using the gradient and lower bounds for W_s and $|z - y| \leq 2t_y$. The chosen small ratio Γ_r/W_s and the derivative of the power therefore give

$$W_0(z)^p - (V_c(z) + b_r(y))_+^p \geq 1.35 \sqrt{W_s(y)} \Gamma_r(y)$$

for sufficiently small damping fractions and then sufficiently small r_* . Indeed the limiting coefficient is $p/\sqrt{1.01} > 1.49$. After integration in z , the factor $R1(y) = 1 + o(1)$ leaves the lower bound $1.3\sqrt{W_s(y)}\Gamma_r(y)$. Combine this with (7.22) and (7.20) to prove (7.17). All terms represent locally integrable densities, so the assertion includes points of a level set of a up to a Lebesgue-null set; no regularity of those level sets is required.

Because $q_0 \geq 0$ and $\Gamma_r \leq 0.1W_0$, we have $B_r \geq 0.9W_0$. The upper offset bound follows from $b_r \leq h_0 + q_0$. Also

$$b_r - h_b = q_0 - \Gamma_r + c_b a^{-4} \geq (c_b/2) a^{-4},$$

so one may take $c' = c_b/2$. On the kernel support,

$$V_c(z) + h_b(y) = W_0(z) + h_0(y) - h_0(z) - c_b a_y^{-4} \geq c a_y^{-4},$$

by the lower bound for W_0 , scale comparability, and the offset oscillation estimate, after decreasing r_* further. This proves (7.19).

Finally, h_0, q_0 are continuous and locally H^1 , and the same is true of Γ_r . Their decay proves the remaining assertions. $\square$

Lemma 7.7 (Gap under doubling). *Fix $\kappa \in (0, 1/10)$. The barriers in Lemma 7.6 have a fixed constant $\gamma > 0$ such that, whenever $R_+ = 2r \leq r_*$,*

$$b_r - b_{R_+} \geq \gamma R_+^{-4} \quad \text{on } \{R_+ \leq a \leq (1 + \kappa)R_+\}. \quad (7.23)$$

Proof. The difference is exactly

$$b_r - b_{R_+} = C'_0(1 - 2^{-4\alpha})R_+^{4\alpha}W_s^{1+\alpha}.$$

The lower bound for W_s gives the assertion with

$$\gamma = C'_0(1 - 2^{-4\alpha})c_\theta^{1+\alpha}(1 + \kappa)^{-4(1+\alpha)} > 0.$$

□

The lower margin in (7.19) also controls inversion of the averaged law. If $\delta_0 > 0$ is a sufficiently small fixed constant, differentiation under the integral gives

$$\frac{d}{dh} \frac{f_y(h)}{4\pi} \geq ca_y^{-2} \quad \text{for } h \geq h_b(y) - \delta_0 a_y^{-4}. \quad (7.24)$$

Indeed $(V_c(z) + h)_+^{p-1} \geq ca_y^{-2}$ on the kernel support, and $R1(y)$ is bounded below. This derivative bound and the gap (7.23) are the two margins used in the conditional field comparisons and the subsequent passage from r to $2r$.

8 Noisy observations and conditional screening

We compare the conditional electron density with the nonlinear reaction law of Section 7. The observation noise makes a rare data event inexpensive to impose on the eigenstate. A local quantum comparison in the resulting state will then rule out each failure of the desired inverse relation. We use the master kernels and deterministic barriers of Section 7, with the small parameter now chosen from the proposed excess charge.

Suppose, towards a contradiction, that the asserted uniform excess-charge bound fails. By Lemma 2.3, choose systems at their largest strictly bound sectors such that

$$K_{\text{exc}} := N - Z > 0, \quad K_{\text{exc}}/M \longrightarrow \infty.$$

Fix a universal constant $B_0 > 2$, to be enlarged in the final argument, and define

$$r_* = (B_0 M / K_{\text{exc}})^{1/3} \longrightarrow 0, \quad r_0 = 2^{-n} r_*, \quad r_j = 2^j r_0 \quad (0 \leq j \leq n), \quad (8.1)$$

where the finite integer $n \geq 1$ is chosen so that $r_0 < Z^{-1/3}$. The crude bound $N \leq 3Z$ and $M \geq 1$ imply $r_* > Z^{-1/3}$. All subsequent smallness assertions concern this sequence of systems; their thresholds will be universal and independent of n .

8.1 The common observation construction in molecular geometry

Set $\ell_j = r_j^{1.01}$ for $j < n$. Use the unordered arrays and likelihood versions of Section 5, with N particles, indices $j, \dots, n-1$, and $\mathcal{F}_n$ trivial. The master kernels are those of (7.11). Define

$$\mu_j(y) = \mathbb{E} \left[\sum_{i=1}^N \mathcal{K}_{x_i}(y) \mid \mathcal{F}_j \right], \quad H_j = V_s - K * \mu_j.$$

Lemma 5.1 gives the joint versions, fixed-system regularity, Poisson equation, and the tower property. Here the ground state attains E , so Lemma 5.2 gives offset $C\ell_j^{-2} \log^5(e/P_A)$ for every event of probability $P_A > 0$.

8.2 Upper tails and a spatial cap

The observations have two uses. First, their conditional electron density is a continuous field to which elliptic comparisons apply. Second, an event of these observations can be imposed at the small energy cost in Lemma 5.2. We first use that cost to exclude large positive fields and excessive local observed counts.

For this subsection fix a numerical constant $L_1 \geq 1$. Its value will be chosen after the universal field caps have been obtained. At scale $r = r_j$, put $R_+ = 2r$. Constants in estimates of small probabilities may depend on L_1 ; the field thresholds below do not.

Lemma 8.1 (Positive conditional-field tail). *If $r/2 \leq a_y \leq 5L_1r$, then, for a universal C and every $s > C$,*

$$\mathbb{P}\{H_j(y) > sa_y^{-4}\} \leq e \exp\{-c_{L_1}(s - C)^{2/5}a_y^{-1+.004}\}. \quad (8.2)$$

All estimates hold for sufficiently small r_ , uniformly in $j < n$.*

Proof. Write $a = a_y$ and tilt the event in question, if it has positive probability P_A . By the tower property, its average of $K * \mu_j$ is the tilted raw average of the potential of the master smears. Nuclei within a fixed multiple of a contribute at most Ca^{-4} to V_s . More distant electron smears have exactly their bare potential at y : a contributing smear near y has center within $2t_y \ll a$. Dropping the remaining negative electron terms therefore gives

$$\mathbb{E}[H_j(y) \mid A] \leq Ca^{-4} + \mathbb{E}_{\psi_A} J_y.$$

Lemmas 4.2 and 4.4, with no initial cut, bound the last expression by $Ca^{-4} + C\sqrt{\delta/a}$. Indeed, for small a , their scale quantity is bounded by $C(a^{-7} + \delta)$. Lemma 5.2 and $\ell_j \geq c_{L_1}a^{1+.01}$ imply

$$s - C \leq Ca^{7/2}\ell_j^{-1} \log^{5/2}(e/P_A) \leq C_{L_1}a^{2.49} \log^{5/2}(e/P_A).$$

Solving for P_A proves the assertion. The deterministic threshold comes solely from the universal local field estimates. $\square$

Call $\{r \leq a \leq 2L_1R_+\}$ the participating band. Use a grid of closed cubes Q of side $r/100$ meeting this band. Let Q' be concentric enlargements, chosen with bounded overlap to contain every ball of radius r centered in Q , with an additional margin. The Lipschitz bound for a ensures that $a_Q := \inf_{Q'} a \geq r/2$ and that all points in Q' fall in the range of Lemma 8.1. Choose smooth $0 \leq \chi_Q \leq 1$ equal to one on Q , supported in Q' , with $|\Delta\chi_Q| \leq Cr^{-2}$.

Lemma 8.2 (Spatial cap and its exceptional cost). *There are universal thresholds $C'_{\text{cap}}, C_{\text{cap}}$ with the following properties. Define*

$$\begin{aligned} Q \text{ is bad} &\iff \sup_Q H_j > C'_{\text{cap}}a_Q^{-4}, \\ B_Q &= \mathbf{1}_{\{Q \text{ bad}\}} \sup_Q (H_j)_+, \quad \Pi = \sum_Q B_Q \chi_Q, \quad \mathcal{E} = \bigcup_{Q \text{ bad}} Q'. \end{aligned}$$

Then $H_j - \Pi \leq C_{\text{cap}}a^{-4}$ on the participating band. For each fixed $m \geq 1$ and $A' > 0$,

$$\mathbb{P}(Q \text{ bad}) + a_Q^{4m} \mathbb{E} B_Q^m \leq C_{L_1, m, A'} r^{A'}. \quad (8.3)$$

At each deterministic point, $\mathbb{P}(y \in \mathcal{E})$ has the same arbitrarily high polynomial bound. The cube families are finite, and

$$\sum_Q |Q'| \leq C_{L_1} M r^3. \quad (8.4)$$

All these random objects are jointly measurable in the data and the spatial variable. For the fixed system and scale, their sizes and supports have deterministic bounds.

Proof. Since $\Delta H_j \geq -Ca_Q^{-6}$ on Q' , adding a quadratic and applying the ball submean inequality gives, for any fixed large C' and a universal C'' ,

$$\sup_Q H_j \leq C''a_Q^{-4} + Cr^{-3} \int_{Q'} (H_j - C'a_Q^{-4})_+.$$

The same inequality follows for distributional Laplacians by mollification. Choose C' above the threshold in Lemma 8.1. Because a_z/a_Q is bounded and $a_z \leq 5L_1r$ on Q' , integration of that tail yields

$$\mathbb{E}(H_j(z) - C'a_Q^{-4})_+^m \leq Ca_Q^{-4m} r^{A'}$$

for arbitrary fixed m, A' . Jensen's inequality for the spatial average gives the same bound for the integral term. Taking $C'_{\text{cap}} > C'' + 1$, Markov's inequality gives the bad-cube probability. On a bad cube its supremum is bounded by the integral term plus $C''a_Q^{-4}$; the probability just proved controls the latter contribution to every moment. This proves (8.3).

If a point lies in a bad cube, the corresponding term of Π dominates $(H_j)_+$ there. Otherwise the good-cube bound gives the asserted cap, since a_Q and the local scale are comparable by a universal factor. At a fixed point only a bounded number of enlarged cubes occur, proving the estimate for $\mathcal{E}$. Every participating or enlarged cube lies within distance $C_{L_1}r$ of some nucleus, by $d_y \leq La_y$. Counting grid cubes in this union proves finiteness and (8.4). Cube suprema are measurable through a countable dense subset, because H_j is continuous. The bounds from Lemma 5.1 also give the last assertion. $\square$

8.3 The inverse comparison and the order of constants

We now fix the constants needed for propagation. Choose C_{inv} larger than the barrier upper bound in a^{-4} units and than C_{cap} . Choose $C_{\text{top}} > C_{\text{inv}}$ sufficiently large and put $h_{\text{top}}(y) = C_{\text{top}}a_y^{-4}$; this also exceeds h_b with a fixed margin. If γ is the constant in Lemma 7.7, choose $0 < \lambda_2 < \lambda_1 < \gamma/4$. Choose L_1 sufficiently large that, when $a > L_1R_+$, subtracting $\lambda_2R_+^{-4}$ from either upper cap places it strictly below b_{R_+} . This is possible since $|b_{R_+}| \leq Ca^{-4}$ there. The preceding probability estimates now apply with this fixed L_1 .

Choose a safety constant $e_0 > 0$ so small that $\eta = e_0R_+^{-4}$ is smaller than the barrier margin above h_b throughout $R_+ \leq a \leq 2L_1R_+$ and than the shift gap. Finally choose $\delta_0 > 0$ small compared with that margin, λ_2 , and $\lambda_1 - \lambda_2$, so that

$$16\delta_0 + 2e_0 < \lambda_1 - \lambda_2, \quad \delta_0 < \lambda_2, \quad 2e_0 < c'(2L_1)^{-4}. \quad (8.5)$$

These choices precede the final smallness requirement on r_* .

Define

$$T_y(h) = \frac{f_y(h)}{4\pi}, \quad I_y(m) = \begin{cases} h_b(y), & m \leq T_y(h_b(y)), \\ T_y^{-1}(m), & T_y(h_b(y)) < m < T_y(h_{\text{top}}(y)), \\ h_{\text{top}}(y), & m \geq T_y(h_{\text{top}}(y)). \end{cases}$$

The positivity on the master kernel established with the barriers implies

$$T'_y(h) \geq ca_y^{-2} \quad (h \geq h_b(y) - \delta_0a_y^{-4}). \quad (8.6)$$

Indeed, differentiating the power $3/2$ leaves a positive square root bounded below by ca_y^{-2} on a kernel of integral $1 + o(1)$. Thus the inverse on the indicated interval exists and is continuous. The clipped inverse satisfies

$$\begin{aligned} m \geq T_y(h_b) &\implies T_y(I_y(m)) \leq m, \\ m \leq T_y(h_{\text{top}}) &\implies T_y(I_y(m)) \geq m. \end{aligned} \tag{8.7}$$

Lemma 8.3 (One-sided inverse comparisons). *Fix $j < n$, and put $r = r_j$ and $R_+ = 2r$. For each deterministic y with $r \leq a_y \leq 2L_1R_+$, the failure probability of each implication below is less than r^{40} :*

$$\begin{aligned} \mu_j(y) \geq T_y(h_b(y)) &\implies H_j(y) \geq I_y(\mu_j(y)) - \delta_0 a_y^{-4}, \\ \mu_j(y) \leq T_y(h_{\text{top}}(y)) &\implies H_j(y) \leq I_y(\mu_j(y)) + \delta_0 a_y^{-4}. \end{aligned} \tag{8.8}$$

The constants and the smallness threshold for r_* are independent of the system and the number of scales.

Proof. The argument has three steps. Observation matching first relates the conditional density to the particles retained by a fresh patch. The patch comparison then determines the local offset except for a small event, whose negative-field cost must also be controlled. Finally, two separate tower identities transfer the offset comparison back to the master field.

A count-good event and the fresh patch. Write $a = a_y$, $t = t_y$, $\tau = t/a$. Recall that $Rf(y) = \int \mathcal{K}_z(y)f(z)dz$: the averaging and the kernel gradient estimates below concern the center variable z . There is a universal K_0 such that

$$\mathbb{P}\{\#\{i : Y_{ji} \in B(y, 3a)\} > K_0 a^{-3}\} < r^{42}. \tag{8.9}$$

Otherwise tilt this event. Lemma 5.2 gives $\delta \leq Cr^{-2.02} \log^5(e/r^{42}) \leq a^{-7+.02}$ for small r_* . The last inequality is uniform: after multiplication by $a^{6.98}$ its left side is at most $C_{L_1} r^{4.96} \log^5(e/r^{42})$. Every compatible raw configuration has the same excessive count in $B(y, 4a)$, because $\sqrt{3}\ell_j = o(a)$. A fixed cell cover and Lemma 4.4 contradict this for large enough K_0 .

Call the complement of this excessive-count event count-good. On such an observation path, match its entries to any compatible raw configuration. The kernel Lipschitz bound gives

$$\left| \sum_i \mathcal{K}_{x_i}(y) - \sum_i \mathcal{K}_{Y_{ji}}(y) \right| \leq Ca^{-3} \ell_j t^{-4}. \tag{8.10}$$

Only observations in $B(y, 3a)$ can contribute to this difference. The conditional ratios in Lemma 5.1 are supported on compatible configurations. Averaging (8.10) therefore gives the same bound with $\mu_j(y)$ replacing the raw sum.

If an implication in (8.8) fails with probability at least r^{40} , intersect its failure event with count-good data and call the result A . Then $P_A := \mathbb{P}(A) \geq r^{41}$ for small r . Write P for the original joint law of raw variables and all noisy arrays, and $Q = P(\cdot | A)$. Its raw marginal is the law of $\psi_A = \sqrt{p_A/P_A} \Psi$, with $\delta \leq a^{-7+.02}$. Apply Lemma 4.3 to this state, choosing $b = a^{1+.2}$ and $\beta = b/a = a^{.2}$. Once selected by shell averaging, the patch radius is deterministic. Extend Q by sampling the fresh cut labels conditionally on the raw variables with the cut's squared partition probabilities. The raw-label marginal is exactly the patch construction for ψ_A . Write X for its cut data, ρ_X^c for its conditional core density, ρ for its local Thomas–Fermi minimizer, and σ for its retained fine density. These are conditional objects under this tilted patch law. In contrast, μ_j remains the conditional density defined under P .

Law	Construction and conditional quantities
$P = \mathbb{P}$	The raw state Ψ with independent noisy arrays. The fields μ_j, H_j retain their definitions using $\mathcal{F}_j$ under this law.
Q	$P(\cdot \mid A)$, extended by the fresh cut labels. The quantities ρ_X^c, ρ, σ use its raw-label marginal and the cut data X .

The fine smearing moves retained centers by at most $\sqrt{3}b$. Every particle relevant to the master kernel is in the retained central region. Thus (8.10) implies, Q -almost surely,

$$|R\sigma(y) - \mu_j(y)| \leq Ca^{-3}(\ell_j + b)t^{-4} = o(a^{-6}). \quad (8.11)$$

Let $G = \{D(\sigma - \rho) \leq a^{-7+.01}\}$. Lemma 4.6 gives

$$\mathbb{E}_Q \left[D(\sigma - \rho) + \int W_- \sigma \right] \leq Ca^{-7+.02}, \quad \mathbb{E}_Q \int \sigma^{5/3} \leq Ca^{-7}. \quad (8.12)$$

In particular $Q(G^c) \leq Ca^{.01}$. On G , Lemma 3.1 gives

$$|R\rho(y) - R\sigma(y)| \leq Ct^{-5/2}a^{-3.5+.005} = o(a^{-6}).$$

For clarity, after division by a^{-6} these two errors are bounded by

$$C\{a^{.01}\tau^{-4} + a^{.2}\tau^{-4} + a^{.005}\tau^{-5/2}\} = o(1). \quad (8.13)$$

The lower bound $\tau \geq c_{L_1}a^w$, $w = 10^{-5}$, makes every power positive.

The local inverse comparison and rare negative fields. Put $h_X = W - V_c$. On G , in the high-density case one must have

$$h_X(y) \geq I_y(\mu_j(y)) - \frac{1}{4}\delta_0a^{-4}. \quad (8.14)$$

Indeed, the opposite inequality and Lemma 7.4 imply on the master kernel $h_X(z) \leq I_y(\mu_j(y)) - \delta_0a^{-4}/8$. To cover arbitrarily negative $h_X(y)$, note that $h \mapsto h + C\tau h_-$ is increasing for small τ ; evaluate it at the proposed upper threshold, which is bounded in a^{-4} units. The oscillation error there is $o(a^{-4})$. The Thomas–Fermi equation, (8.6), and the first part of (8.7) then yield $R\rho(y) \leq \mu_j(y) - c\delta_0a^{-6}$, contradicting (8.13). In the low-density case the lower oscillation estimate and the increasing function $h \mapsto h - C\tau h_-$ similarly give, on G ,

$$h_X(y) \leq I_y(\mu_j(y)) + \frac{1}{4}\delta_0a^{-4}. \quad (8.15)$$

Here the second clipping inequality is used.

The deterministic upper bound $h_X(y) \leq Ca^{-4}$ controls the positive error on G^c in the low-density case. The high-density case requires a bound on negative fields there. By kernel positivity, $\mu_j \geq T_y(h_b) \geq ca^{-6}$. Equation (8.11) and $\mathcal{K}_z(y) \leq Ct^{-3}$ force, on every configuration,

$$m_\sigma := \int_{B(y, 2t)} \sigma \geq ca^{-6}t^3 = ca^{-3}\tau^3.$$

The upper oscillation bound implies on this ball

$$W_- \geq (1 - C\tau)h_X(y)_- - V_c - Ca^{-4}\tau^{1/2}.$$

Integrate against σ , divide by m_σ , and average over G^c . The patch bound controls the $W_- \sigma$ numerator by $Ca^{-7+.02}$. For the nuclear term, weighted convexity and the uniform integral of $|z - R_m|^{-5/2}$ on a radius- $2t$ ball give

$$\int_{B(y, 2t)} V_c^{5/2} \leq Ca^{-7} \tau^{1/2}.$$

Hölder's inequality on probability times space therefore gives

$$\begin{aligned} \mathbb{E}_Q \mathbf{1}_{G^c} \int_{B(y, 2t)} V_c \sigma &\leq \left(\mathbb{E}_Q \int \sigma^{5/3} \right)^{3/5} \left(Q(G^c) \int_{B(y, 2t)} V_c^{5/2} \right)^{2/5} \\ &\leq Ca^{-7+.004} \tau^{.2}. \end{aligned}$$

We conclude that

$$\mathbb{E}_Q \mathbf{1}_{G^c} h_X(y)_- \leq Ca^{-4} \{a^{.02} \tau^{-3} + a^{.004} \tau^{-2.8} + Q(G^c) \tau^{1/2}\} = o(a^{-4}). \quad (8.16)$$

Since $|I_y| \leq Ca^{-4}$, the bounds on G now imply the corresponding one-sided comparisons in expectation, with error $\delta_0 a^{-4}/4 + o(a^{-4})$.

Transfer of averaged fields. The remaining step compares the expectation of this fresh patch offset with that of the master offset under the event law Q . Apply Lemma 5.3 with the nuclear splitting $V = V_s + V_c$, master width $t = t_y$, and fine width $b = a^{1.2}$. Lemma 4.6 gives the raw local density and Coulomb error bounds; the deleted shell estimate is (4.26). The pointwise density bound required for the nuclear remainder follows from $\rho = kW_+^{3/2}$ and (4.23); its potential on a radius- t ball is $Ca^{-4}(t/a)^{1/2}$ by (7.4). Thus the explicit transfer estimate is

$$|\mathbb{E}_Q(H_j - h_X)(y)| \leq Ca^{-4} \{\tau^{1/5} + \beta^{1/5} + \beta^{1/2} + a^{.01} \tau^{-1/2} + \tau^{1/2}\} = o(a^{-4}). \quad (8.17)$$

Here $\beta = a^2$, $c_{L_1} a^w \leq \tau \leq c_1 r_*^w$, and $.01 - w/2 > 0$. The two expectations in the common lemma use different conditional laws; this verification identifies neither posterior pathwise.

On the high failure event, every path has $H_j(y) < I_y(\mu_j(y)) - \delta_0 a^{-4}$. Equations (8.14) and (8.16), together with (8.17), instead give

$$\mathbb{E}_Q H_j(y) \geq \mathbb{E}_Q I_y(\mu_j(y)) - \frac{1}{4} \delta_0 a^{-4} - o(a^{-4}),$$

a contradiction. The low failure event is contradicted in the same way by (8.15), the deterministic upper bound on h_X , and (8.17). All remainders are uniform: $a \asymp_{L_1} r$, $\tau \geq c_{L_1} a^w$, and the smallest power in (8.16) is $.004 - 2.8w > 0$. The constants were fixed before letting $r_* \rightarrow 0$. $\square$

8.4 Moments for the integrated exceptional cost

The inverse estimates concern each deterministic spatial point. Their use in the next section requires integrable costs for failed comparisons, not a single event on which all points succeed.

Lemma 8.4 (Conditional-density and reaction moments). *Fix $j < n$, put $r = r_j$ and $R_+ = 2r$, and let $r \leq a_y \leq 2L_1 R_+$. Under the original joint law $P = \mathbb{P}$,*

$$\|\mu_j(y)\|_2 \leq C t_y^{-3} a_y^{-3}, \quad \|f_y(H_j(y))\|_2 \leq C a_y^{-6} (1 + \tau_y^{-3/2}). \quad (8.18)$$

Consequently, if F_y is either failure event from Lemma 8.3, then

$$\int_{\{r \leq a_y \leq 2L_1 R_+\}} \mathbb{E}[\mathbf{1}_{F_y}(\mu_j(y) + f_y(H_j(y)))] dy \leq CMr^{20-3-3w}. \quad (8.19)$$

The same bound holds with F_y replaced by $\{y \in \mathcal{E}\}$, and the expected integral over all of $\mathbb{R}^3$ of $r^{-2} \sum_Q B_Q \mathbf{1}_{Q'}$ is bounded by the right side as well.

Proof. Conditional Jensen, the master-kernel support, and Lemma 4.4 give

$$\|\mu_j(y)\|_2 \leq Ct_y^{-3} \|\#\{i : x_i \in B(y, a_y)\}\|_2 \leq Ct_y^{-3} a_y^{-3}.$$

The tail bound implies $\|(H_j(y)_+)^{3/2}\|_2 \leq Ca_y^{-6}$. For the nuclear part of the reaction, weighted convexity and the uniform integral of $|z - R_m|^{-3/2}$ on a radius- $2t_y$ ball give

$$\int \mathcal{K}_z(y) V_c(z)^{3/2} dz \leq Ct_y^{-3} a_y^{-9/2} t_y^{3/2} = Ca_y^{-6} \tau_y^{-3/2}.$$

This proves (8.18). Cauchy-Schwarz against $\mathbb{P}(F_y) < r^{40}$ contributes r^{20} . Use $a_y \asymp_{L_1} r$, $\tau_y \geq c_{L_1} r^w$, and (8.4) to obtain (8.19). The arbitrarily high polynomial bounds in Lemma 8.2 give the other assertions, choosing their exponent larger if necessary. All integrations are licensed by the joint measurable versions already constructed. $\square$

9 Outward propagation and the excess-charge bound

The inverse comparisons now allow us to replace a reaction term by the conditional electron density on one more layer of local scales. We perform this replacement before forgetting the observations at the current scale. The failed comparisons contribute a nonnegative error density. Its expected integral, rather than the probability of a simultaneous comparison throughout space, is the quantity that we shall sum.

We use the scales $r_j = 2^j r_0$, $r_n = r_*$ and the retained data $\mathcal{F}_j$ of Section 8. The data $\mathcal{F}_n$ are trivial. Write μ_j for their conditional smoothed density. At each step we take a local maximum of the previous comparison field and the observed field H_j , after lowering them by different constants. The previous field supplies the nonlinear reaction; H_j supplies the actual density through its Poisson equation. The unequal shifts let the inverse comparisons prove the missing source inequality for each branch that can attain the maximum.

After the last observations are forgotten, the resulting positive exterior field will bound the smoothed electron mass in $\{a < r_*\}$ by Z plus the accumulated error. The exterior-count estimate of Lemma 6.2 controls the remaining smoothed mass, while the error has integral $o(M)$. These are the two contributions to the final excess-charge estimate.

The invariant at scale $r = r_j$ is a continuous locally H^1 offset u and a nonnegative bounded compactly supported error density d_j such that

$$\Delta u \geq -4\pi\nu_s + 4\pi\mathbf{1}_{\{a < r\}}\mu_j - 4\pi d_j + \mathbf{1}_{\{a \geq r\}}f_y(u) \quad \text{in distributions,} \quad (9.1)$$

and

$$b_r \leq u \leq C_{\text{inv}} a^{-4} \quad \text{on } \{a \geq r\}, \quad u = b_r \quad \text{on } \{a > L_1 r\}. \quad (9.2)$$

Here and below an inequality between spatial functions is asserted for every positive-density retained path, using the versions of Lemma 5.1. The functions u, d_j are jointly measurable in the

retained data and space. We also maintain deterministic bounds, for the fixed physical system and step, on the modulus of continuity and the H^1 norm of u on every compact set, and on the size and support of d_j . These additional bounds justify conditional integration. The integral estimate for d_j will instead have universal constants.

9.1 Replacing one layer of reaction by density

For this subsection put $r = r_j$, $R_+ = 2r$, and $H = H_j$. All parameters are fixed in the order specified in Section 8. We record the inequalities that enter the argument:

$$\lambda_1 < \frac{\gamma}{4}, \quad 16\delta_0 + 2e_0 < \lambda_1 - \lambda_2, \quad \delta_0 < \lambda_2, \quad 2e_0 < c'(2L_1)^{-4}. \quad (9.3)$$

The constant $c' > 0$ is the margin in $b_r \geq h_b + c'a^{-4}$. Set $\eta = e_0 R_+^{-4}$ and $T_y = f_y/(4\pi)$.

For the current data, denote the high and low failure sets of Lemma 8.3 by

$$\begin{aligned} \mathcal{B}^h &= \{y : \mu_j(y) \geq T_y(h_b(y)), \quad H(y) < I_y(\mu_j(y)) - \delta_0 a_y^{-4}\}, \\ \mathcal{B}^l &= \{y : \mu_j(y) \leq T_y(h_{\text{top}}(y)), \quad H(y) > I_y(\mu_j(y)) + \delta_0 a_y^{-4}\}. \end{aligned}$$

These sets are used only within the participating band $r \leq a \leq 2L_1 R_+$. Let Π , B_Q , Q' and $\mathcal{E}$ be the penalty, its cube coefficients, the enlarged cubes and their bad union from Lemma 8.2. In particular $\Pi = 0$ off $\mathcal{E}$ and $H - \Pi \leq C_{\text{cap}} a^{-4}$ throughout this band.

Proposition 9.1 (One outward step). *Suppose that u, d_j satisfy the inductive conditions above, including (9.1)–(9.2), at scale $r = r_j$. There are an offset u' and a nonnegative error density $\tilde{d}$, measurable with respect to the same data $\mathcal{F}_j$, that satisfy these inequalities at scale R_+ with μ_j still in the core term. One may take*

$$\begin{aligned} \tilde{d} &= d_j + Cr^{-2} \sum_Q B_Q \mathbf{1}_{Q'} \\ &\quad + \mathbf{1}_{\{r \leq a < R_+\}} \mu_j \mathbf{1}_{\mathcal{E} \cup \mathcal{B}^h} \\ &\quad + \mathbf{1}_{\{R_+ \leq a \leq 2L_1 R_+\}} T_y(H) \mathbf{1}_{\mathcal{E} \cup \mathcal{B}^l}, \end{aligned} \quad (9.4)$$

where C is a universal constant large enough to dominate the penalty Laplacian.

Proof. Define two shifted offsets

$$u_1 = u - \lambda_1 R_+^{-4}, \quad u_2 = H - \lambda_2 R_+^{-4} - \Pi.$$

Use the following open cover to define u' :

open region	u'
$a < (1 + \kappa)R_+$	$\max(u_1, u_2)$
$(1 + \kappa/2)R_+ < a < 2L_1 R_+$	$\max(u_1, u_2, b_{R_+})$
$a > L_1 R_+$	b_{R_+}

(9.5)

The definitions agree on overlaps. On the lower overlap, Lemma 7.7 and (9.3) give

$$u_1 \geq b_r - \lambda_1 R_+^{-4} > b_{R_+}.$$

On the upper overlap the shifted cap inequalities, ensured by the choice of L_1 , put both u_1 and u_2 below b_{R_+} . Thus the local maxima glue to a continuous locally H^1 function. No assertion about

the measure or regularity of a level set of a is required. The new exterior bounds follow from the same comparisons: for $R_+ \leq a < (1 + \kappa)R_+$ the lower bound is supplied by u_1 , and elsewhere the barrier is an included branch.

We verify the differential inequality locally. At a fixed point in one of the open regions, discard branches more than η below the maximum. There is a neighborhood on which these discarded branches cannot attain the maximum, while every retained branch is within 2η of it. We show that each retained branch satisfies the common inequality

$$\Delta v \geq -4\pi\nu_s + 4\pi\mathbf{1}_{\{a < R_+\}}\mu_j - 4\pi\tilde{d} + \mathbf{1}_{\{a \geq R_+\}}f_y(v) \quad (9.6)$$

on that neighborhood.

For u_1 , the old core inequality suffices on $a < r$, and monotonicity of f_y gives the required reaction on $a \geq R_+$. On the gain region $r \leq a < R_+$, the error in (9.4) pays μ_j at points of $\mathcal{E} \cup \mathcal{B}^h$. At every other gain point, $\Pi = 0$ and near activity yields

$$H \leq u - (\lambda_1 - \lambda_2)R_+^{-4} + 2\eta.$$

Here $h_b \leq u \leq h_{\text{top}}$. If $\mu_j > T_y(u)$, then the high comparison applies and $I_y(\mu_j) \geq u$, including the case where the inverse is clipped at its upper endpoint. It would follow that

$$H \geq u - \delta_0 a^{-4} \geq u - 16\delta_0 R_+^{-4},$$

contrary to (9.3). Thus $4\pi\mu_j \leq f_y(u)$. The old reaction supplies precisely the new core term on the gain region.

For u_2 , the identity $\Delta H = -4\pi\nu_s + 4\pi\mu_j$ gives the required core term on all of $a < R_+$, while the first increment in (9.4) pays the Laplacian of Π . Consider the remaining points where u_2 is retained. The lower splice or the included barrier gives

$$u_2 \geq b_{R_+} - 2\eta \geq h_b + c'a^{-4} - 2e_0 R_+^{-4} > h_b.$$

The last inequality uses $a \leq 2L_1 R_+$ and (9.3). On $\mathcal{E} \cup \mathcal{B}^l$ the last error term pays the reaction because $u_2 \leq H$ and f_y is nondecreasing. Otherwise $\Pi = 0$ and the cap gives $H < h_{\text{top}}$. If $\mu_j < T_y(u_2)$, the low comparison applies and $I_y(\mu_j) \leq u_2$, also when the inverse is clipped at its lower endpoint. We would obtain

$$H \leq u_2 + \delta_0 a^{-4} \leq u_2 + \delta_0 R_+^{-4},$$

contradicting $H = u_2 + \lambda_2 R_+^{-4}$. Therefore $4\pi\mu_j \geq f_y(u_2)$, as required.

The barrier branch appears only where $a > R_+$, and Lemma 7.6 gives (9.6) there. The old error density and every additional error are nonnegative, so they can only weaken the required lower bound.

These comparisons cover the exact level sets as well: $a = r$ belongs to the gain region and uses the old reaction, whereas $a = R_+$ belongs to the new exterior and uses its reaction. All right-side comparisons hold almost everywhere on the chosen neighborhood and therefore upgrade each retained branch's distributional inequality to (9.6). The source ν_s is a smooth bounded density for the fixed system, and all branches here are continuous and locally H^1 . The reaction is also bounded on bounded ranges of its argument: for a fixed system, $t_{\min} := \inf_z t_z > 0$ and $V_c \in L_c^{5/2}$ give, for every $A > 0$,

$$\sup_{y, |h| \leq A} f_y(h) \leq C t_{\min}^{-3} \int V_c^{3/2} + C A^{3/2} \sup_y R1(y) < \infty.$$

Thus the first, nonsingular part of Lemma 5.4, followed by a partition of unity, proves the inequality for u' on all of $\mathbb{R}^3$. No vanishing of the reaction near the original nuclei is needed in this step. $\square$

9.2 Averaging the data and summing the error

The next step explains why the nonlinear reaction survives the removal of observations. It also records the integrability needed in that operation.

Lemma 9.2 (Forgetting the current observations). *Set*

$$u_{j+1} = \mathbb{E}[u' \mid \mathcal{F}_{j+1}], \quad d_{j+1} = \mathbb{E}[\tilde{d} \mid \mathcal{F}_{j+1}].$$

Then u_{j+1}, d_{j+1} satisfy (9.1)–(9.2) at scale r_{j+1} , with conditional density μ_{j+1} . They retain continuity and local H^1 regularity for the offset and bounded compact support for the nonnegative error density.

Proof. For the fixed system there are finitely many scales and cubes, and all master smoothing widths have a positive minimum. Lemma 5.1 therefore gives deterministic bounds for the conditional densities, their potentials and their gradients. The offsets and error densities in the assertion are jointly measurable: maxima, clipped inverses and failure flags preserve measurability, and each cube supremum may be taken over a fixed countable dense subset.

Inductively the branches have deterministic continuity moduli and local H^1 bounds on each compact set. Their open definitions agree on overlaps; a finite subcover of a compact set supplies a Lebesgue number and hence a modulus for the glued function. The gradient of a finite maximum is almost everywhere one of the branch gradients. Conditional averaging consequently preserves continuity. It preserves the local H^1 bound by testing weak derivatives against compactly supported vector fields, using Fubini and Cauchy–Schwarz. The error terms have deterministic bounded size and support for the fixed system and step, so the same is true after averaging.

Integrate the distributional inequality of Proposition 9.1 against a fixed nonnegative compact test and then against the explicit positive-parent conditional law of Lemma 5.1. The preceding bounds justify Fubini. The tower property gives $\mathbb{E}[\mu_j \mid \mathcal{F}_{j+1}] = \mu_{j+1}$, while convexity gives

$$\mathbb{E}[f_y(u') \mid \mathcal{F}_{j+1}] \geq f_y(\mathbb{E}[u' \mid \mathcal{F}_{j+1}]).$$

This is the required lower bound for the reaction. The exterior bounds and the exterior equality are deterministic and pass to the average. $\square$

Lemma 9.3 (Integrated error). *Starting with $u = b_{r_0}$ and $d_0 = 0$, the preceding construction reaches scale $r_n = r_*$ and produces a deterministic error density with*

$$\int d_n \leq CMr_*^{17-3w} = o(M) \quad \text{along the contradiction sequence.} \quad (9.7)$$

The constant is independent of the number of scales and of the nuclear configuration.

Proof. Since $r_0 < Z^{-1/3} \leq a$, Lemma 7.6 gives the initial invariant everywhere; there is no initial core region.

At scale $r = r_j$, apply Lemma 8.4 to the three increments in (9.4). The high-failure estimate pays μ_j on the gain region; the low-failure estimate pays $T_y(H) = f_y(H)/(4\pi)$ on the remaining participating region. The same Lemma pays membership in $\mathcal{E}$ and the penalty Laplacian. For $\mathcal{B} \in \{\mathcal{B}^h, \mathcal{B}^l\}$, use $\mathbf{1}_{\mathcal{E} \cup \mathcal{B}} \leq \mathbf{1}_{\mathcal{E}} + \mathbf{1}_{\mathcal{B}}$. All these integrands are nonnegative, so restriction to the indicated subregions only improves their bounds. We obtain

$$\mathbb{E} \int (\tilde{d} - d_j) \leq CMr^{17-3w}.$$

Conditional averaging preserves the unconditional expected integral of a nonnegative density. Because $\beta = 17 - 3w > 0$,

$$\sum_{j=0}^{n-1} r_j^\beta = r_*^\beta \sum_{h=1}^n 2^{-h\beta} \leq \frac{r_*^\beta}{2^\beta - 1}.$$

At the last step the data are trivial, so d_n is deterministic. This proves (9.7), uniformly in n . The argument has used the failure probability separately at each spatial point and then integrated it; it has imposed no simultaneous good-data event in space. $\square$

9.3 The total-charge comparison

We have reached the unconditional density μ_n and replaced the reaction by this density throughout $\{a < r_*\}$, at total error $o(M)$. A final potential comparison converts this statement to a bound on the number of electrons in that region.

Proof of Theorem 1.1. Use the contradiction sequence and sector eigenstates of Section 8. Its terminal scale is $r_* = (B_0 M / K_{\text{exc}})^{1/3}$. The universal B_0 will exceed the fixed count constant below. All deterministic parameters and tolerances are chosen before taking the sequence limit. The smallness requirements of the preceding lemmas then hold for all its sufficiently late members, uniformly over their scales.

At the final step put

$$U = V_c + u_n, \quad S = \nu - \mathbf{1}_{\{a < r_*\}} \mu_n + d_n.$$

The signed measure S has compact support. In fact $\{a < r_*\}$ lies in a finite union of balls around the nuclei, the error density has compact support, and the nuclei are finite. Since $\Delta V_c = -4\pi\nu + 4\pi\nu_s$, the final invariant implies

$$\Delta(U - K * S) \geq \mathbf{1}_{\{a \geq r_*\}} f_y(u_n) \geq 0.$$

The nuclear singularities cancel in the explicit identity

$$U - K * S = u_n - V_s + K * (\mathbf{1}_{\{a < r_*\}} \mu_n) - K * d_n. \quad (9.8)$$

The right-hand side is continuous and locally H^1 on all of $\mathbb{R}^3$: the last two potentials have bounded compactly supported densities, and the other terms have the stated regularity from the construction.

Far from the nuclei, (9.2) gives $U = B_{r_*} = W_0 + q_0 - \Gamma_{r_*}$, which tends to zero by the deterministic field estimates. The compact source potential $K * S$ also tends to zero. For every $\epsilon > 0$, the function $(U - K * S - \epsilon)_+$ consequently has compact support and is an admissible H^1 test in the subharmonic inequality. Testing gives a nonpositive squared gradient norm, so this positive part vanishes. Letting $\epsilon \downarrow 0$ proves

$$U \leq K * S \quad \text{away from the nuclear points.}$$

On every sufficiently large sphere $U = B_{r_*} \geq .9W_0 > 0$. The spherical mean of the potential of a compactly supported signed measure, on a sphere centered at the origin and enclosing its support, equals its total mass divided by the sphere radius. Hence

$$0 < \int dS = Z - N + \int_{\{a \geq r_*\}} \mu_n + \int d_n. \quad (9.9)$$

Only positivity on a far sphere is used here; no asymptotic charge formula for W_0 is needed.

At the last scale the data have been completely averaged away, so

$$\mu_n(y) = \mathbb{E}_\Psi \sum_i \mathcal{K}_{x_i}(y).$$

If a kernel centered at x contributes to a point with $a_y \geq r_*$, then its support and the Lipschitz scale estimate imply

$$a_y \leq a_x + \frac{|x - y|}{L} \leq a_x \left(1 + \frac{c_1 r_*^w}{L}\right).$$

For small r_* this forces $a_x \geq r_*/2$. Each kernel has integral one, and Lemma 6.2 therefore gives

$$\int_{\{a \geq r_*\}} \mu_n \leq \mathbb{E}_\Psi \#\{i : a(x_i) \geq r_*/2\} \leq CM r_*^{-3}.$$

Combining this with (9.7) and (9.9) yields

$$K_{\text{exc}} < CM r_*^{-3} + o(M) = \frac{C}{B_0} K_{\text{exc}} + o(M).$$

Choose $B_0 > 2C$, increasing it also to ensure the initial scale choice in Section 8. Since $K_{\text{exc}}/M \rightarrow \infty$, the last inequality is impossible. This proves the existence of one universal finite constant bounding the excess per nucleus in every strictly bound sector.

For an arbitrary strictly bound sector, the largest strictly bound sector of the same system has at least as many electrons, so the same bound applies. For every integer $N > Z + CM$, strict adjacent binding is therefore impossible. Sector monotonicity gives $E_N \leq E_{N-1}$, and exclusion of the strict inequality gives $E_N = E_{N-1}$. Taking $M = 1$ proves both atomic assertions for every real nuclear charge $Z \geq 1$. $\square$

10 Conditional density and field comparisons

We now specialize to one nucleus at the origin and use $V(y) = Z/|y|$, $d_y = |y|$, and the atomic distance scale $a_y = d_y/L$ with $L = 10^5$. The comparisons in this section allow every real $Z \geq 1$; the neutral-sector conclusions below use integer charges.

Fix $0 \leq D_0 < \infty$ and a normalized ground state Ψ in a sector with $n \leq 3Z$ and $E_n \leq E + D_0$. The ground-state assumption in this section is used in the multiplier identity (2.4); the patch comparisons themselves apply to the resulting states with an energy offset. All probabilities below initially refer to the raw position-and-spin law of Ψ , enlarged by the explicitly specified independent random variables. The common likelihood and event-cost estimates of Section 5 now give a simultaneous comparison throughout an atomic annulus. This strengthens the pointwise molecular comparison by using the absence of nuclear singularities from these annuli.

10.1 Nested observations and master smoothing

Choose $0 < \varepsilon < 1$ and $0 < s < 1$, and suppose Z is large enough that $2r_0 \leq s$. Set

$$r_0 = \varepsilon Z^{-1/3}, \quad r_j = 2^j r_0, \quad m = \max\{j : r_j \leq s\}, \quad \ell_j = r_j^{1.01} \quad (j < m). \quad (10.1)$$

Thus $s/2 < r_m \leq s$. At scale $j < m$ observe the positions $x_i + \ell_j U_{ji}$ and then forget the particle labels separately in each array. All Cartesian coordinates of all U_{ji} , at all scales, are independent with density

$$\zeta(u) = c_\zeta \exp\left(-\frac{1}{1-u^2}\right) \mathbf{1}_{\{|u|<1\}}.$$

Represent the unordered array by its lexicographically sorted tuple Y_j . Ties have probability zero. Write $\mathcal{F}_j = \sigma(Y_j, \dots, Y_{m-1})$ and let $\mathcal{F}_m$ be trivial. These sigma-fields decrease with j .

Fix $w = 10^{-5}$ and $0 < c_1 < (10L)^{-1}$. With the same smooth radial packet g as in Section 3, define

$$\begin{aligned} D_x^0 &= \max(d_x, r_0), & t_x &= c_1 D_x^0 \min(D_x^0, s)^w, & \mathcal{K}_x(y) &= g_{t_x}(y - x)^2, \\ \mu_j(y) &= \mathbb{E} \left[\sum_{i=1}^n \mathcal{K}_{x_i}(y) \middle| \mathcal{F}_j \right], & H_j &= V - K * \mu_j, & \mathcal{R}f(y) &= \int \mathcal{K}_z(y) f(z) \, dz. \end{aligned} \quad (10.2)$$

The lower clamp in D_x^0 is retained also for particles close to the nucleus; the upper clamp is only on the factor raised to w . In particular, $t_x \geq t_{\min} := c_1 r_0^{1+w} > 0$ and t is Lipschitz with constant at most $c_1(1+w)s^w$.

Use the likelihood-ratio versions of Lemma 5.1 for these arrays and kernels. They give joint measurability, fixed-system derivative bounds, $\mu_j \geq 0$, $\int \mu_j = n$, and

$$\mathbb{E}[\mu_j \mid \mathcal{F}_{j+1}] = \mu_{j+1}, \quad \mathbb{E}[H_j \mid \mathcal{F}_{j+1}] = H_{j+1}. \quad (10.3)$$

The offset $H_j - V$ is locally Lipschitz and

$$\Delta H_j = -4\pi\nu + 4\pi\mu_j. \quad (10.4)$$

Lemma 5.2 applies with baseline E_n and offset D_0 . In particular, an event of probability $p > 0$ at index j has tilted energy at most $E + D_0 + C\ell_j^{-2} \log^5(e/p)$. We retain this full baseline dependence throughout the argument.

10.2 The simultaneous inverse comparison

Proposition 10.1 (Conditional density determines the field). *There is a universal constant C_{cap} with the following property. Fix $L_1 \geq 2$, $0 < h_l < h_h < \infty$, and a sufficiently small $\xi > 0$; it suffices to require $\xi < h_l/16$. There exists $s_* = s_*(D_0, L_1, h_l, h_h, \xi) > 0$ such that, whenever $0 < s \leq s_*$, at each $j < m$, with $r = r_j$, there is an event $\mathcal{G}_j \in \mathcal{F}_j$ satisfying $\mathbb{P}(\mathcal{G}_j^c) \leq Cr^{25}$ on which, simultaneously for $r \leq d_y \leq 4L_1r$ and $h_l \leq h \leq h_h$,*

$$\begin{aligned} d_y^6 \mu_j(y) > kh^{3/2} &\implies d_y^4 H_j(y) \geq h - \xi, \\ d_y^6 \mu_j(y) < kh^{3/2} &\implies d_y^4 H_j(y) \leq h + \xi. \end{aligned} \quad (10.5)$$

On the same event, $H_j(y) \leq C_{\text{cap}} d_y^{-4}$ throughout that band. The constant C may depend on the fixed parameters, whereas C_{cap} does not depend on D_0, L_1, h_l, h_h, ξ . All estimates are uniform in Z , the ground state, j , and m .

Proof. We first control the observed count and the positive field throughout the band. At a fixed point and height, a hypothetical failure then defines a tilted state. Its fresh patch compares the observed density with the patch field, including the contribution of rare negative fields; the two expectation towers turn this into a contradiction. Finally, spatial and height nets give the simultaneous statement. Every occurrence of $o(1)$ below tends to zero as $s \downarrow 0$, uniformly for $r_0 \leq r = r_j \leq s$ and the stated bands, after the fixed parameters in the proposition have been chosen.

Kernel geometry. Put $a = a_y = d_y/L$, $t = t_y$ and $\tau = t/a$. If $\mathcal{K}_z(y) \neq 0$, the Lipschitz bound on t gives $|z - y| \leq t_z \leq t_y + Cs^w|z - y|$, hence $|z - y| \leq 2t_y$. On this support, $t_z/t_y = 1 + O(s^w)$. Comparison with the fixed-width kernel $g_t(y - z)^2$ therefore gives

$$\int \mathcal{K}_z(y) \, dz = 1 + O(s^w), \quad |\mathcal{K}_z(y)| \leq Ct^{-3}, \quad |\nabla_y \mathcal{K}_z(y)| + |\nabla_z \mathcal{K}_z(y)| \leq Ct^{-4}. \quad (10.6)$$

The center derivative is a weak derivative where t_z has a corner; its stated bound also gives the needed Lipschitz estimate. For $r/2 \leq d_y \leq 8L_1r$ one has $a \asymp_{L_1} r$ and

$$c_{L_1}a^w \leq \tau \leq Cs^w. \quad (10.7)$$

For example, when $d_y \geq r_0$, the formula is $\tau = c_1L \min(d_y, s)^w$; on the indicated band $\min(d_y, s) \geq c_{L_1}d_y$. The case $d_y < r_0$ is comparable, since $d_y \geq r_0/2$.

An observed-count event. Let M_j be the number of entries of Y_j with radii in $[r/4, 12L_1r]$. For a sufficiently large fixed C_{L_1} ,

$$\mathbb{P}\{M_j > C_{L_1}r^{-3}\} \leq r^{40}. \quad (10.8)$$

Indeed, if the event had probability $p > r^{40}$, its tilted raw state from Lemma 5.2 would have offset

$$\delta \leq D_0 + Cr^{-2.02} \log^5(e/r^{40}).$$

Each configuration compatible with the event has more than $C_{L_1}r^{-3}$ raw points in $[r/8, 13L_1r]$, because observation displacements are at most $\sqrt{3}\ell_j = o(r)$. But (4.45) bounds its expected count by $C'_{L_1}(r^{-3} + \sqrt{\delta}r) \leq 2C'_{L_1}r^{-3}$ for small s , a contradiction if $C_{L_1} > 2C'_{L_1}$. Call the complementary event in (10.8) $\mathcal{C}_j$. For data in $\mathcal{C}_j$, every compatible configuration has at most $C_{L_1}r^{-3}$ centers relevant to the master kernels on $r/2 \leq d_y \leq 8L_1r$. Conditional averaging and (10.6) imply there

$$\mu_j(y) \leq Cr^{-6-3w}, \quad |\nabla \mu_j(y)| \leq Cr^{-7-4w}. \quad (10.9)$$

A universal spatial field cap. Fix y in the same enlarged band and tilt an event $A' = \{H_j(y) > b d_y^{-4}\}$ of positive probability p . The original-law tower gives

$$\mathbb{E}[H_j(y) \mid A'] = V(y) - \mathbb{E}_{\Psi_{A'}} \sum_i (K * \mathcal{K}_{x_i})(y) \leq \mathbb{E}_{\Psi_{A'}} J_y.$$

To justify the last inequality, a master smear whose potential at y differs from the point potential has $|x_i - y| < t_{x_i}$ and hence $|x_i - y| \leq 2t_y < Aa_y$ for small s . All terms outside that exclusion therefore have their exact point potential, and dropping the remaining electron terms gives J_y . By (4.39) without an initial cut and Lemma 5.2,

$$b \leq C_2 + C_{L_1}r^{2.49} \log^{5/2}(e/p). \quad (10.10)$$

Here C_2 is universal: the local field term in these units is $Cd_y^4a_y^{-4} = CL^4$, and the contribution $Cd_y^{7/2}\sqrt{D_0}$ is at most one after decreasing s . Solving (10.10) for p shows that for $b \geq 2C_2$,

$$\mathbb{P}\{d_y^4 H_j(y) > b\} \leq e \exp\{-c_{L_1}(b - C_2)^{2/5} r^{-0.996}\}.$$

Tail integration consequently gives, for each fixed $M > 0$,

$$\mathbb{E}(d_y^4 H_j(y) - 2C_2)_+ \leq C_{M,L_1} r^M.$$

Subharmonicity permits a simultaneous conclusion without a Z -dependent derivative estimate. Apply the ball submean inequality with radius $d_y/10$ for $3r/4 \leq d_y \leq 6L_1r$. The averaging ball is contained in $\{r/2 \leq d_z \leq 8L_1r\}$ and $d_z \asymp d_y$ there, so

$$d_y^4 H_j(y) \leq C_3 + Cr^{-3} \int_{\{r/2 \leq d_z \leq 8L_1r\}} (d_z^4 H_j(z) - 2C_2)_+ dz.$$

The constant C_3 is universal. The integral term has expectation $O_{M,L_1}(r^M)$, since its integration volume is $O_{L_1}(r^3)$. Markov's inequality proves that, outside an event of probability $C_{M,L_1}r^M$,

$$H_j(y) \leq C_{\text{cap}} d_y^{-4} \quad (3r/4 \leq d_y \leq 6L_1r), \quad C_{\text{cap}} = C_3 + 1. \quad (10.11)$$

This establishes the claimed order of dependence of the cap.

A fixed spatial point and height. Fix now $r \leq d_y \leq 4L_1r$ and $h \in [h_l/2, 2h_h]$. Consider either failure of (10.5) with tolerance $\xi/4$, intersected with $\mathcal{C}_j$. If its probability were at least r^{42} , call that event A' and write $Q = \mathbb{P}(\cdot | A')$. The raw marginal is the state $\Psi_{A'}$ of Lemma 5.2, with

$$\delta \leq D_0 + Cr^{-2.02} \log^5(e/r^{42}) \leq a^{-7+.02} \quad (10.12)$$

for small s . Apply the refined patch (4.42) to this raw state at y , with $\beta = a^{-2}$ and $b = \beta a$. Extend Q by fresh cut labels sampled conditional on the raw positions, exactly as in that patch. Let X be these patch data, let ρ_X^c be its conditional core density, and let ρ, W, σ be its minimizing density, screened field, and fine-smear density. Thus

$$W = V - K * \rho_X^c - K * \rho, \quad \rho = kW_+^{3/2} \text{ on the patch}, \quad \mathbb{E}_Q \left[D(\sigma - \rho) + \int W_- \sigma \right] \leq Ca^{-7+.02}.$$

The roles of these densities and their laws are distinct. The pair μ_j, H_j keeps its original-observation definition in (10.2). Under the event law Q , the fresh patch gives the conditional core density ρ_X^c and its field $\Phi_X = V - K * \rho_X^c$; σ is the fine smear of retained out particles, whereas ρ is the auxiliary minimizing density and $W = \Phi_X - K * \rho$. The field comparison is made after taking Q -expectations in (10.19) below.

Each compatible raw configuration matches the observed array Y_j after a permutation, with displacement at most $\sqrt{3}\ell_j$ per position. Both configurations and their conditional averages at those same data therefore differ, when tested against $x \mapsto \mathcal{K}_x(y)$, by at most $Cr^{-3}\ell_j t^{-4}$. This follows from the count bound and the center Lipschitz estimate in (10.6); only entries within $2t + O(\ell_j)$ of y can contribute. All such raw particles are retained by the fresh patch. Its radial fine smear moves a retained point by at most $\sqrt{3}b$, adding at most $Cr^{-3}bt^{-4}$ to the error. Consequently, on every compatible path, including paths where the Coulomb error is large,

$$|\mathcal{R}\sigma(y) - \mu_j(y)| \leq Cr^{-3}(\ell_j + b)t^{-4} \leq Ca^{-6}(a^{.01}\tau^{-4} + a^{.2}\tau^{-4}) = o(a^{-6}). \quad (10.13)$$

In the comparison with the conditional average, one may first compare each raw sum with $\sum_i \mathcal{K}_{Y_{j,i}}(y)$, then average this same bound using the original posterior law. The array is fixed in both comparisons, so no equality of posterior laws is being used.

Let $G = \{D(\sigma - \rho) \leq a^{-7+.01}\}$. Then $Q(G^c) \leq Ca^{.01}$. The test function $z \mapsto \mathcal{K}_z(y)$ has support in $B(y, 2t)$ and gradient norm at most $Ct^{-5/2}$. Coulomb duality gives on G

$$|\mathcal{R}(\sigma - \rho)(y)| \leq Ca^{-7/2+.005}t^{-5/2} = Ca^{-6}a^{.005}\tau^{-5/2} = o(a^{-6}). \quad (10.14)$$

The positivity of the two powers used so far is explicit: $.01 - 4w = .00996$ and $.005 - (5/2)w = .004975$.

On G , a high-density antecedent forces

$$W(y) \geq (h - \xi/16)d_y^{-4}. \quad (10.15)$$

For if the reverse strict inequality held, the oscillation bound (4.44) on $B(y, 2t)$ would give $W(z) \leq (h - \xi/32)d_y^{-4}$ there for sufficiently small s . This remains uniform when $W(y)$ is arbitrarily negative: the upper estimate is $W(y) + C\tau(a^{-4} + W(y)_-)$, and $v \mapsto v + C\tau v_-$ is increasing for $C\tau < 1$. Using the patch equation and $\int \mathcal{K}_z(y) dz = 1 + O(s^w)$ would then give

$$\mathcal{R}\rho(y) \leq k(h - \xi/32)^{3/2}d_y^{-6}(1 + O(s^w)) < kh^{3/2}d_y^{-6} - c_{h_l, h_h, \xi}a^{-6},$$

contradicting (10.13)–(10.14). Similarly, on G a low-density antecedent forces

$$W(y) \leq (h + \xi/16)d_y^{-4}. \quad (10.16)$$

Indeed the contrary is positive, so the lower oscillation estimate gives $W(z) \geq (h + \xi/32)d_y^{-4}$ on the contributing ball; its reaction density contradicts the low antecedent with the same strict margin. Constants from $d_y = La$ are fixed before s is decreased.

The negative field on the exceptional patch event. The upper bound (10.16) passes to expectation with an $o(a^{-4})$ error, by the patch bound $W(y) \leq Ca^{-4}$ and $Q(G^c) \leq Ca^{.01}$. To pass the lower bound to expectation we need more than this small probability. Under the high-density antecedent, (10.13) holds on all paths and gives $\mathcal{R}\sigma(y) \geq ca^{-6}$, since $h \geq h_l/2 > 0$. The support and upper bound of the master kernel imply the deterministic mass bound

$$M_\sigma := \int_{B(y, 2t)} \sigma \geq ca^{-6}t^3.$$

For $Y = W(y)_-$, (4.44) gives throughout this ball $W(z)_- \geq (1 - C\tau)Y - C\tau a^{-4}$. Taking $C\tau \leq 1/2$, integrating against σ , and using the mass bound yields

$$Y \leq C(a^{-6}t^3)^{-1} \int W_- \sigma + C\tau a^{-4}.$$

Multiply by $\mathbf{1}_{G^c}$ and use the refined patch budget under the same law Q . We obtain

$$\mathbb{E}_Q[\mathbf{1}_{G^c}W(y)_-] \leq Ca^{-4}(a^{.02}\tau^{-3} + \tau) = o(a^{-4}), \quad (10.17)$$

because $.02 - 3w = .01997 > 0$. Thus the high and low antecedents give, respectively,

$$\mathbb{E}_Q W(y) \geq (h - \xi/16)d_y^{-4} - o(a^{-4}), \quad \mathbb{E}_Q W(y) \leq (h + \xi/16)d_y^{-4} + o(a^{-4}). \quad (10.18)$$

Transfer between the two conditioning procedures. Apply Lemma 5.3 with $V_s = V$, $V_c = 0$ and $h_X = W$. Its two towers are taken under the original observation law and the fresh tilted patch law, respectively. The refined patch provides all its inputs: the raw density bound and Coulomb budget are (4.42), the deleted count bound is (4.26), and $\rho \leq Ca^{-6}$ on the contributing ball by (4.25). Therefore

$$|\mathbb{E}_Q(H_j - W)(y)| \leq Ca^{-4}\{\tau^{1/5} + \beta^{1/5} + \sqrt{\beta} + a^{.01}\tau^{-1/2} + \tau^{1/2}\} = o(a^{-4}). \quad (10.19)$$

Here $\beta = a^2$ and (10.7) makes every positive-power error vanish. The source-free potential remainder actually has the sharper power τ^2 , so the common upper bound is sufficient. No equality of posterior laws is used in this transfer.

On a high-failure event, its defining inequality is $d_y^4 H_j(y) < h - \xi/4$, whereas (10.18) and (10.19) give $d_y^4 \mathbb{E}_Q H_j(y) \geq h - \xi/16 - o(1)$, a contradiction for small s . On a low-failure event the inequalities are reversed and give the same contradiction. Thus each such failure intersected with $\mathcal{C}_j$ has probability less than r^{42} . Including (10.8), the two implications at a fixed point and height, with tolerance $\xi/4$, fail with probability at most Cr^{40} .

Spatial and height nets. Choose a net of the closed band $\{r \leq d_y \leq 4L_1 r\}$ with spacing at most r^2 and at most $C_{L_1} r^{-3}$ points. For example, select one band point in each intersecting cube of side $r^2/2$; the cube count follows from the volume of its $O(r^2)$ enlargement. Choose a finite height net in $[h_l/2, 2h_h]$ with mesh at most $\xi/64$, including its endpoints. Exclude every point-and-height failure, the count failure, and the cap failure with $M = 40$ in (10.11). A union bound gives exceptional probability Cr^{37} , and hence at most Cr^{25} for $r < 1$.

On the retained event, (10.9) shows that $d_y^6 \mu_j(y)$ varies by at most $Cr^{1-4w} = o(1)$ between a band point and a net point within Cr^2 . To control the field, work in a ball of radius cr around a band point, contained in the enlarged cap band. On this ball B , let $Q = K * (\mathbf{1}_B \mu_j)$. Then $H_j + Q$ is harmonic on B . The value and gradient of Q are bounded by Cr^{-4-3w} and Cr^{-5-3w} , respectively, so $H_j + Q$ is bounded above by Cr^{-4-3w} . Its nonnegative distance from that upper bound has, by the Poisson formula on a smaller concentric ball, oscillation bounded by its central value times the relative distance. It follows that

$$|H_j(z) - H_j(y)| \leq Cr(r^{-4-3w} + H_j(y)_-) \quad (|z - y| \leq Cr^2). \quad (10.20)$$

This estimate also controls the change of the normalized field $d_y^4 H_j(y)$, since $d_z/d_y = 1 + O(r)$. In particular, if its value at y is below $h - \xi$, its value at z is below $h - 3\xi/4$ for small s . Arbitrarily negative values cause no problem: the relevant upper estimate has the form $v + Crv_- + O(r^{1-3w})$, increasing in v when $Cr < 1$. If the value at y exceeds $h + \xi$, it is positive and the cap and (10.20) make its value at z greater than $h + 3\xi/4$.

Suppose now a high-density implication failed at some (y, h) . At a nearby spatial net point choose a height-net value $h' \in [h - \xi/4, h - \xi/8]$. The positive lower height bound and the $o(1)$ density variation preserve the high antecedent at h' , while the field is below $h - 3\xi/4 < h' - \xi/4$. This is an excluded net failure. For a low-density failure choose $h' \in [h + \xi/8, h + \xi/4]$; its antecedent is preserved and the field exceeds $h + 3\xi/4 > h' + \xi/4$. The height intervals lie inside $[h_l/2, 2h_h]$ by the stipulated smallness of ξ . This proves simultaneity and finishes the proposition. $\square$

We record also an unconditional estimate used to pay for a failed band. For fixed L_1 and $r \leq d_y \leq 4L_1 r$, the support and size of the master kernel, conditional Jensen, and (4.45) in the original state give

$$\|\mu_j(y)\|_2 \leq Ct_y^{-3} \|\#\{i : r/2 < d_{x_i} < 8L_1 r\}\|_2 \leq Ct_y^{-3} r^{-3} \leq Cr^{-6-3w}. \quad (10.21)$$

The middle inequality uses the offset D_0 and sufficiently small s ; its constant is uniform in the state and scale.

The order of choices is relevant. The fixed screening constants, g, c_1, w , and the universal cap C_{cap} come first. One may then choose L_1, h_l, h_h, ξ and a bounded offset D_0 , and finally make s small enough for every estimate above. Thereafter any sufficiently large Z with $2r_0 \leq s$ is permitted. In particular no step requires $D_0 = 0$, and the number of dyadic scales introduces no loss in the constants.

11 Outward propagation and an electron deficit

The conditional comparisons let us extend a prescribed electron source from the scale $Z^{-1/3}$ to a small radius independent of Z . The result applies to ground states whose energy has a bounded offset from the minimum over all particle numbers.

Proposition 11.1 (Deficit inside a fixed small ball). *There is a universal constant $c_{\text{def}} > 0$ with the following property. For every $D_0 \in [0, \infty)$ there is $s_*(D_0) > 0$ such that, for each $0 < s \leq s_*(D_0)$, there is an integer $Z_*(D_0, s)$ for which every integer $Z \geq Z_*(D_0, s)$ and every normalized sector ground state Ψ satisfying*

$$n \leq 3Z, \quad q_n[\Psi] = E_n \leq E + D_0$$

obey

$$Z - \mathbb{E}_\Psi \#\{i : |x_i| < s/4\} \geq c_{\text{def}} s^{-3} > 1. \quad (11.1)$$

All thresholds are independent of the sector and of the choice of its ground state. In particular, admissible fixed radii s can be taken arbitrarily small.

To prove the deficit, we construct fields u_j satisfying a lower bound for their Laplacian. Alongside the nuclear term $-4\pi\nu$, this bound uses the conditional density term $4\pi\mu_j$ inside radius r_j and the model reaction $4\pi k(u_j)_+^{3/2}$ outside. Each outward step replaces that reaction by the actual density on one more annulus, then forgets the current observation array. An error density records the source omitted on exceptional data. A positive lower barrier and a small expected error will force a positive residual charge at the terminal scale. The construction also gives the following result at every intermediate stopping index.

Proposition 11.2 (Intermediate atomic subsolutions). *There are universal constants $\varepsilon, B, C_{\text{inv}} > 0$ and $L_1 > 2$ with the following property. Fix $D_0 < \infty$, then take $s > 0$ sufficiently small and integer Z sufficiently large as in Proposition 11.1. For every sector ground state in that proposition use the common observations and master packets (10.1)–(10.2), with $r_0 = \varepsilon Z^{-1/3}$ and $s/2 < r_m \leq s$. For each $0 \leq j \leq m$ there exist jointly measurable $\mathcal{F}_j$ -measurable fields u_j and densities p_j^{err} with the following properties. Put $d = |y|$, $r = r_j$, and $f(t) = 4\pi k(t_+)^{3/2}$, and define*

$$b_r(y) = Bd^{-4} \left(1 - \frac{r}{8d}\right) \quad (d \geq r).$$

$$\begin{aligned} \Delta u_j &\geq -4\pi\nu + 4\pi \mathbf{1}_{\{d < r\}} \mu_j - 4\pi p_j^{\text{err}} + \mathbf{1}_{\{d \geq r\}} f(u_j), \\ b_r \leq u_j &\leq C_{\text{inv}} d^{-4} \quad (d \geq r), \quad u_j = b_r \quad (d > L_1 r). \end{aligned} \quad (11.2)$$

The differential inequality is global in distributions. The offsets $u_j - V$ are continuous and locally Lipschitz on all of $\mathbb{R}^3$; $p_j^{\text{err}} \geq 0$ is bounded and supported in $|y| \leq r_j$. For each fixed system and step, the offsets have deterministic size and Lipschitz bounds on every compact set; the errors have deterministic size bounds. Universally,

$$\mathbb{E} \int p_j^{\text{err}} \leq Cr_0^{32} + C \sum_{i < j} r_i^{19/2-3w} \leq C \left(r_0^{32} + r_j^{19/2-3w} \right), \quad w = 10^{-5}. \quad (11.3)$$

In particular this is at most $C(s^{32} + s^{19/2-3w})$. The expectations cannot in general be omitted for $j < m$; the terminal fields are deterministic. The same master kernels are used at every intermediate stopping index. The constants $\varepsilon, B, C_{\text{inv}}, L_1$ are independent of any additional inverse-comparison parameters. Such a comparison uses these same observations whenever its own smallness condition on s is met.

Proof of Propositions 11.2 and 11.1. Use the observations and master packets (10.1)–(10.2), with ε fixed during initialization below and Z large enough that $m \geq 1$. The resulting densities and fields satisfy

$$\mathbb{E}[\mu_j \mid \mathcal{F}_{j+1}] = \mu_{j+1}, \quad \mathbb{E}[H_j \mid \mathcal{F}_{j+1}] = H_{j+1}. \quad (11.4)$$

Write $\kappa = 1/8$. The barrier b_r in the proposition is positive on $d \geq r$ and, for sufficiently small fixed $B > 0$, satisfies

$$\Delta b_r = Bd^{-6}(12 - 20\kappa r/d) \geq \frac{19}{2}Bd^{-6} \geq f(b_r). \quad (11.5)$$

For example, $4\pi k\sqrt{B} \leq 19/2$ suffices. If $R = 2r$, then

$$b_r - b_R = B\kappa rd^{-5} \geq \gamma R^{-4} \quad (R \leq d \leq 1.1R), \quad \gamma = \frac{B\kappa}{2(1.1)^5} > 0. \quad (11.6)$$

Initialization. With probability at least $1 - Cr_0^{35}$, the initial conditional density satisfies

$$P_0(y) := K * (\mathbf{1}_{\{d < r_0\}} \mu_0)(y) \leq 0.1 Z/r_0 \quad (r_0 \leq |y| \leq 1.1r_0). \quad (11.7)$$

To prove this, fix a point y in this annulus and let $A_y = \{P_0(y) > 0.05Z/r_0\}$, an event of $\mathcal{F}_0$. Suppose that $p = \mathbb{P}(A_y) > r_0^{40}$. By Lemma 5.2, the raw marginal conditioned on this event is the squared law of an antisymmetric state of offset at most

$$D_0 + Cr_0^{-2.02} \log^5(e/r_0^{40}).$$

For the fixed ε this is at most $Z^{7/3}$ when Z is large enough. The global estimate (3.3) therefore bounds the tilted density ρ_{A_y} by $\int \rho_{A_y}^{5/3} \leq CZ^{7/3}$, with a universal C .

Only packet centers with $|x| < 2r_0$ can contribute to the truncated density in P_0 : a center outside this ball has $t_x \leq c_1 s^w |x| < |x|/2$, so its packet misses $B(0, r_0)$. A truncated packet has potential at most that of the full packet, which by radial smearing is at most $K(y - x)$. The conditional tower identity and Hölder's inequality consequently give

$$\begin{aligned} \mathbb{E}[P_0(y) \mid A_y] &\leq \int_{|x| < 2r_0} \frac{\rho_{A_y}(x)}{|y - x|} dx \\ &\leq CZ^{7/5} r_0^{1/5} = C\varepsilon^{6/5} Z/r_0. \end{aligned}$$

Choose ε small enough that the last coefficient is less than 0.05. This contradicts the definition of A_y , so $\mathbb{P}(A_y) \leq r_0^{40}$.

The minimum packet width is $t_{\min} = c_1 r_0^{1+w}$. Integrating $|y - z|^{-2}$ against a bounded packet gives the deterministic bound

$$\|\nabla P_0\|_\infty \leq Cnt_{\min}^{-2} \leq CZr_0^{-2-2w}.$$

A net in the annulus of spacing $O(r_0^2)$ has $O(r_0^{-3})$ points. The union bound gives failure probability $O(r_0^{37})$, and the variation between a point and its net point, divided by Z/r_0 , is $O(r_0^{1-2w}) = o(1)$. For large Z this proves (11.7), with the stated weaker exponent 35.

Let G_0 denote the event in (11.7), and set

$$\mu^{\text{in}} = \mathbf{1}_{G_0} \mathbf{1}_{\{d < r_0\}} \mu_0, \quad p_0^{\text{err}} = \mathbf{1}_{G_0^c} \mathbf{1}_{\{d < r_0\}} \mu_0.$$

On $d < 2r_0$ use the branch

$$v = V - K * \mu^{\text{in}} - 0.7Z/r_0 + C_3(Z/r_0)^{3/2}d^2.$$

Choose C_3 universally so that $6C_3 \geq 4\pi k 2^{3/2}$, and then choose ε sufficiently small to meet the preceding tilt condition and $4C_3\varepsilon^{3/2} \leq 0.05$. The quadratic term on this ball is at most $0.05Z/r_0$, since $Zr_0^3 = \varepsilon^3$. For $r_0 \leq d < 2r_0$ we thus have $v \leq 2Z/r_0$ and

$$\Delta v = -4\pi\nu + 4\pi\mu^{\text{in}} + 6C_3(Z/r_0)^{3/2}, \quad 6C_3(Z/r_0)^{3/2} \geq f(v).$$

On $r_0 \leq d < 1.06r_0$, the good-event bound (and the easier bound $\mu^{\text{in}} = 0$ on its complement) give

$$v \geq (1/1.06 - 0.1 - 0.7)Z/r_0 > 0.14Z/r_0.$$

On $1.9r_0 < d < 2r_0$ we instead have $v \leq (1/1.9 - 0.7 + 0.05)Z/r_0 < 0$. Fix B satisfying the barrier condition and $0 < B \leq 0.1\varepsilon^3$. Then $v > b_{r_0}$ on the lower overlap, whereas $v < 0 < b_{r_0}$ on the upper overlap. Define

$$u_0 = \begin{cases} v, & d < 1.06r_0, \\ \max(v, b_{r_0}), & r_0 < d < 2r_0, \\ b_{r_0}, & d > 1.9r_0. \end{cases}$$

The definitions agree on their open overlaps. Lemma 5.4 proves (11.2) initially; near $d = r_0$ the branch v alone attains the maximum, and its displayed Laplacian supplies both required densities. The exterior upper bound holds, for example, when $C_{\text{inv}} \geq \max(B, 32\varepsilon^3)$. Also, using $n \leq 3Z = 3\varepsilon^3 r_0^{-3}$,

$$\mathbb{E} \int p_0^{\text{err}} \leq n\mathbb{P}(G_0^c) \leq Cr_0^{32}. \quad (11.8)$$

Order of the remaining constants. The constants C_3, ε, B have now been fixed universally. Increase C_{inv} to exceed also the universal cap C_{cap} in Proposition 10.1. After fixing γ in (11.6), choose $0 < \lambda_2 < \lambda_1 < \gamma/2$. Choose $L_1 > 2$ so large that $C_{\text{inv}}L_1^{-4} < \lambda_2$. This ensures

$$C_{\text{inv}}d^{-4} - \lambda_2 R^{-4} < b_R \quad (d > L_1 R).$$

Next fix $0 < h_l < B/4$ and $h_h > C_{\text{inv}}$, and then choose $e_0, \xi > 0$ sufficiently small that

$$2e_0(2L_1)^4 < B/4, \quad 16\xi + 2e_0 < \lambda_1 - \lambda_2, \quad \xi < \lambda_2, \quad (11.9)$$

with ξ also in the range of Proposition 10.1. Only now restrict s according to that proposition, D_0 , and the other smallness conditions in this proof. Finally take Z large according to D_0, s and the initialization requirements. Thus no constant is chosen in terms of a later scale or a particular state.

One outward step. Suppose that (11.2) holds at $r = r_j$, write $R = 2r$, and abbreviate $u = u_j$. On the good band event of Proposition 10.1, for the band $r \leq d \leq 4L_1 r = 2L_1 R$, introduce

$$v_1 = u - \lambda_1 R^{-4}, \quad v_2 = H_j - \lambda_2 R^{-4}.$$

Define a new function on a fixed open cover by

region	u'
$d < 1.08R$	$\max(v_1, v_2)$
$1.04R < d < 2L_1 R$	$\max(v_1, v_2, b_R)$
$d > L_1 R$	b_R .

(11.10)

On bad band data omit v_2 in every row. In both cases set

$$\tilde{p} = p_j^{\text{err}} + \mathbf{1}_{\{\text{bad}\}} \mathbf{1}_{\{r \leq d < R\}} \mu_j. \quad (11.11)$$

On the lower overlap, (11.6) gives $v_1 \geq b_r - \lambda_1 R^{-4} > b_R$. On the upper overlap, the caps on u and, on good data, H_j put both shifted branches below b_R . The definitions therefore agree. For $R \leq d < 1.08R$ the lower bound $u' \geq b_R$ follows already from v_1 ; farther out the barrier is included explicitly. The upper bound $u' \leq C_{\text{inv}} d^{-4}$ and the equality $u' = b_R$ for $d > L_1 R$ follow from the same caps.

We prove the distributional inequality at radius R before averaging. The open-neighborhood formulation is as follows. At a point y_0 in one of the open sets of (11.10), let M be the maximum of its candidate family and put $\eta = e_0 R^{-4}$. Keep precisely those candidates v with $M(y_0) - v(y_0) \leq \eta$, and choose a maximizer v_* at y_0 . Continuity and finiteness give an open neighborhood U of y_0 , inside the chosen open set, on which every discarded candidate is strictly below v_* and every retained candidate satisfies

$$M - v < 2e_0 R^{-4}. \quad (11.12)$$

The retained family has maximum M throughout U . At the origin make this construction with the continuous offsets by V ; the family there contains only branches having that common singularity.

Each retained branch satisfies on the whole of this neighborhood the common inequality

$$\Delta v \geq -4\pi\nu + 4\pi \mathbf{1}_{\{d < R\}} \mu_j - 4\pi \tilde{p} + \mathbf{1}_{\{d \geq R\}} f(v). \quad (11.13)$$

For v_1 , retain its old distributional inequality and compare its right side with the new one almost everywhere. On $d < r$ the old source suffices. On $d \geq R$, use $f(u) \geq f(v_1)$. On the gain region $r \leq d < R$, bad data are paid by the added error in (11.11). On good data, v_2 is a candidate there, so (11.12) gives

$$H_j \leq u - (\lambda_1 - \lambda_2 - 2e_0) R^{-4}.$$

Moreover $h_l < 7B/8 \leq d^4 u \leq C_{\text{inv}} < h_h$. If $4\pi\mu_j > f(u)$, the high implication in (10.5), at height $h = d^4 u$, would give

$$H_j \geq u - \xi d^{-4} \geq u - 16\xi R^{-4},$$

contrary to (11.9). Hence $4\pi\mu_j \leq f(u)$, and the old reaction supplies the added source.

For v_2 , which occurs only on good data, start from its identity on all of U ,

$$\Delta v_2 = -4\pi\nu + 4\pi\mu_j.$$

This supplies (11.13) on $d < R$, since $\tilde{p} \geq 0$. At any remaining point where v_2 is retained, the lower splice or the included barrier gives $M \geq b_R$, and $d \leq 2L_1 R$. Thus

$$d^4 v_2 \geq B(1 - \kappa R/d) - 2e_0 (d/R)^4 \geq 7B/8 - 2e_0 (2L_1)^4 > 5B/8 > h_l.$$

The cap also gives $d^4 v_2 \leq C_{\text{cap}} < h_h$. If $4\pi\mu_j < f(v_2)$, the low implication in (10.5), with $h = d^4 v_2$, would imply

$$H_j \leq v_2 + \xi d^{-4} \leq v_2 + \xi R^{-4} < v_2 + \lambda_2 R^{-4} = H_j,$$

which is impossible. Consequently $4\pi\mu_j \geq f(v_2)$ there. The barrier branch occurs only where $d > R$ and satisfies the common inequality by (11.5).

These verifications compare zero-order densities almost everywhere on the same open set U ; they do not restrict or differentiate a weak inequality across an activity set or a sphere. In particular

they apply on a neighborhood crossing $d = R$. The common reaction is $F(y, t) = \mathbf{1}_{\{d \geq R\}} f(t)$, which meets Lemma 5.4's hypotheses after canceling V near the origin. That lemma proves (11.13) for the local maximum. These local conclusions give the inequality globally by a partition of unity. We have proved the desired invariant at radius R with density μ_j , function u' , and error $\tilde{p}$.

Forgetting the current observations. Define the next offset and error by conditional integration,

$$u_{j+1} = V + \mathbb{E}[u' - V \mid \mathcal{F}_{j+1}], \quad p_{j+1}^{\text{err}} = \mathbb{E}[\tilde{p} \mid \mathcal{F}_{j+1}].$$

We give the regularity justification as well as the formal calculation. For the fixed system $t_{\min} > 0$, and integration of each packet gives deterministic bounds

$$\|\mu_j\|_\infty \leq Cnt_{\min}^{-3}, \quad \|K * \mu_j\|_\infty \leq Cnt_{\min}^{-1}, \quad \|\nabla K * \mu_j\|_\infty \leq Cnt_{\min}^{-2}.$$

The same potential bounds hold for the initially truncated density. The explicit initial branches therefore have deterministic local Lipschitz bounds for their offsets. Subtracting a constant and taking a finite maximum preserve such bounds. Where b_R occurs, its offset $b_R - V$ is smooth and separated from the origin. The fixed open cover and agreement on overlaps preserve local bounds after gluing. The errors are bounded, nonnegative, and supported in a deterministic ball. Induction supplies all the claimed bounds at every step, without requiring them to be uniform in Z .

All functions are jointly measurable in data and position. Indeed the conditional densities have the likelihood versions described in Section 10; the initialization flag is a supremum of a continuous function on a fixed compact annulus, testable on a countable dense set; and the band flags and finite maxima are measurable. Conditional integration of functions with deterministic local Lipschitz bounds preserves those bounds. It also commutes with weak derivatives by Fubini. Nonnegative smooth tests in a countable dense family first give simultaneous almost-sure weak inequalities; local bounds extend them to every test. Thus the following conditional integration is legitimate as an inequality of distributions.

Integrate (11.13) conditionally on $\mathcal{F}_{j+1}$. The tower property supplies μ_{j+1} , while convexity gives

$$\mathbb{E}[f(u') \mid \mathcal{F}_{j+1}] \geq f(V + \mathbb{E}[u' - V \mid \mathcal{F}_{j+1}]).$$

The deterministic bounds $b_R \leq u' \leq C_{\text{inv}} d^{-4}$ and the equality outside $L_1 R$ pass to this average. Positivity, boundedness, and support of the error do as well. This proves (11.2) at step $j + 1$.

Summing the lost source. For $r \leq d_y < 2r$, every master packet contributing to $\mu_j(y)$ has its center in a fixed annulus comparable to r , and its width is comparable to t_y . Conditional Jensen and the annular count estimate (4.45), applied with offset at most D_0 , yield

$$\|\mu_j(y)\|_2 \leq Ct_y^{-3} r^{-3} \leq Cr^{-6-3w}.$$

Here $r \leq s \leq 1$ and s is small according to D_0 , so $\max(r^{-3}, 1) + \sqrt{D_0}r \leq Cr^{-3}$. The bad band probability is at most Cr^{25} by Proposition 10.1. Cauchy-Schwarz in data and integration over the gain annulus give

$$\mathbb{E} \int (\tilde{p} - p_j^{\text{err}}) \leq Cr^{25/2} r^3 r^{-6-3w} = Cr^\alpha, \quad \alpha = \frac{19}{2} - 3w > 0.$$

Conditional averaging preserves the expected integral. Stopping the sum before any index j and using (11.8) proves (11.3), with the same master kernels and without replacing an expectation by a pathwise bound. At the terminal index the geometric progression of radii gives

$$\int p_m^{\text{err}} = \mathbb{E} \int p_m^{\text{err}} \leq Cr_0^{32} + C \sum_{j=0}^{m-1} r_j^\alpha \leq C(s^{32} + s^\alpha). \quad (11.14)$$

The last error is deterministic because $\mathcal{F}_m$ is trivial. The constants are independent of the number of scales and of Z .

From the potential to the raw electron count. Put $r = r_m$, $u = u_m$, and introduce the compactly supported signed measure

$$S = \nu - \mathbf{1}_{\{d < r\}} \mu_m \, dy + p_m^{\text{err}} \, dy.$$

The invariant implies $\Delta(u - K * S) \geq 0$. Its singularities cancel in the identity

$$u - K * S = (u - V) + K * (\mathbf{1}_{\{d < r\}} \mu_m) - K * p_m^{\text{err}},$$

whose right side is continuous and locally H^1 on all of $\mathbb{R}^3$. The difference tends to zero at infinity: $u = b_r$ outside $L_1 r$, and the potential of the finite compact source S tends to zero. For each $a > 0$, the positive part $(u - K * S - a)_+$ has compact support and is an admissible H^1 test in its subharmonic inequality. Testing gives a nonpositive squared gradient integral, hence this positive part vanishes. Letting $a \downarrow 0$ proves $u \leq K * S$.

On the sphere $d = 2r$, the invariant gives $u \geq b_r(2r) = 15Br^{-4}/256$. The spherical mean of $K * S$ there equals the total mass of S divided by $2r$, since S is supported in $\{d \leq r\}$. Consequently

$$Z - \int_{d < r_m} \mu_m + \int p_m^{\text{err}} \geq \frac{15B}{128} r_m^{-3}. \quad (11.15)$$

No sign condition on the measure S is needed for this mean identity.

Finally μ_m is the unconditional master-smoothed density. If $|x| < s/4$, then $r_0 \leq s/2$ and $D_x^0 \leq s/2$, so $t_x \leq c_1(s/2)s^w < s/4$. The packet at x is therefore wholly contained in $B(0, s/2) \subset B(0, r_m)$. Since every packet has mass one,

$$\mathbb{E}_\Psi \# \{i : |x_i| < s/4\} \leq \int_{d < r_m} \mu_m.$$

Combining this with (11.15), (11.14), and $r_m \leq s$ gives

$$Z - \mathbb{E}_\Psi \# \{i : |x_i| < s/4\} \geq \frac{15B}{128} s^{-3} - C(s^{32} + s^\alpha).$$

Take $c_{\text{def}} = 15B/256$ and decrease $s_*(D_0)$ so that the error is at most $c_{\text{def}} s^{-3}$ and $c_{\text{def}} s^{-3} > 1$ for every $0 < s \leq s_*(D_0)$. All remaining requirements are met by increasing $Z_*(D_0, s)$. This proves the proposition. $\square$

12 Returning to the neutral sector

The deficit estimate was proved for ground states whose energy is within a fixed distance of the minimum over all particle numbers. We first apply it at that minimum. A screened deletion comparison will then put the neutral and singly ionized ground states in the same class.

We record explicitly the harmonic-field estimate used in the deletion and again in the final tail argument. For a configuration $\omega = (x_i, \sigma_i)_{i=1}^n$ and $h > 0$, write

$$B_h(x; \omega) = V(x) - \sum_{|x_i| < h} K(x - x_i). \quad (12.1)$$

This field is harmonic on $\{|x| > h\}$, configuration by configuration.

Lemma 12.1 (Conditional fields on annuli). *Fix positive constants $\alpha < \alpha' < \beta' < \beta$ and $\alpha_0 < \beta_0$. Let $u > 0$, $0 < h \leq \alpha u/2$, and let ψ be a normalized δ -state with $n \leq 3Z$, where $\delta < \infty$ is arbitrary. Let Y be the data from a cut whose out support lies in $\{\alpha_0 u < |x| < \beta_0 u\}$ and whose squared gradient cost is at most Cu^{-2} . The cut may also be absent, in which case Y is trivial. There is a jointly measurable version*

$$H_Y(x) = \mathbb{E}_\psi[B_h(x) \mid Y]$$

which is harmonic on $\{\alpha u < |x| < \beta u\}$ for almost every datum. Put

$$M(Y) = \sup_{\alpha' u \leq |x| \leq \beta' u} (H_Y(x))_+.$$

Then

$$\|M\|_2 \leq C \left\{ \frac{\max(u^{-3}, 1) + \sqrt{\delta}u}{u} + \sum_{j=0}^{\infty} \frac{\max((2^j h)^{-3}, 1) + \sqrt{\delta} 2^j h}{\max(u, 2^j h)} \right\}. \quad (12.2)$$

The constant depends only on the fixed annular ratios and cut-gradient constant, in addition to the fixed screening constants L, A . In particular it does not depend on n, Z, δ, h, u .

Proof. For x in the broad annulus, every source with $|x_i| < h$ lies outside $B(x, Aa_x)$: indeed $h \leq |x|/2$ and $A/L < 1/2$. Consequently

$$B_h(x) = J_x + \sum_{\substack{|x_i| \geq h \\ |x_i - x| \geq Aa_x}} K(x - x_i). \quad (12.3)$$

If $2^j h \leq |x_i| < 2^{j+1} h$ and the summand occurs, then

$$K(x - x_i) \leq \frac{C}{\max(u, 2^j h)}.$$

For radii comparable to or below u this follows from the exclusion $|x_i - x| \geq Aa_x \geq A\alpha u/L$; for radii greater than $2\beta u$ it follows from $|x_i - x| \geq |x_i|/2$.

The initial-cut hypotheses of Proposition 4.5 hold uniformly here. Cover its support by a fixed finite number of cells B_z with $a_z \asymp u$, the constants being allowed to depend on L and the annular ratios. For each target x in the broad annulus these cell scales are comparable to a_x . The gradient bound can be written as $\sum_z D_z^2 a_z^{-2} \mathbf{1}_{B_z}$, with $D_z^2 \leq C(a_z/u)^2$. Since $a_z m_z \geq 1$ and the screening parameter $P \geq 1$,

$$\frac{D_z^2}{a_z m_z} \leq C \leq C_0 \sqrt{P}.$$

Enlarging the fixed cover constant C_0 suffices for every finite δ . Thus (4.39), conditional Jensen, the triangle inequality in L^2 , and (4.45) applied to (12.3) give, pointwise in the spatial variable,

$$\|(H_Y(x))_+\|_2 \leq C \left\{ \frac{\max(u^{-3}, 1) + \sqrt{\delta}u}{u} + \sum_{j \geq 0} \frac{\max((2^j h)^{-3}, 1) + \sqrt{\delta} 2^j h}{\max(u, 2^j h)} \right\}.$$

The series converges: its large- j terms are bounded by a constant times $2^{-j} h^{-1} + \sqrt{\delta} 2^{-j/2} h^{-1/2}$.

For completeness, conditional fields here have simultaneous spatial versions. On every compact subset of $\{|x| > h\}$, the kernels in (12.1) and each fixed number of their derivatives have deterministic bounds for the fixed finite system. Conditional integration therefore gives continuous

harmonic functions, with measurable dependence on the data. One may first use the conditional law of configurations supplied by the cut and then differentiate under that integral. For a harmonic function H , its positive part is subharmonic. Balls of radius cu about points of the smaller annulus remain in the broad annulus, so their mean-value inequalities imply

$$\sup_{\alpha'u \leq |x| \leq \beta'u} H(x)_+ \leq Cu^{-3} \int_{\alpha u < |z| < \beta u} H(z)_+ \, dz.$$

Minkowski's integral inequality and the pointwise bound prove (12.2). Suprema can be taken on countable dense sets, which also proves their measurability. $\square$

Two forms of this estimate will be used. With fixed $t > 0$, $h = t/4$, $u = 2^\ell t$, and $\delta = 0$, it gives for the positive conditional supremum on $u \leq |x| \leq 2u$ the bound

$$\|M\|_2 \leq C_t \frac{1 + \ell}{u}. \quad (12.4)$$

Indeed the terms with $2^j h \leq u$ have total numerator at most $C_t(1 + \ell)$, and the remaining series is $O_t(u^{-1})$. With $h = u/8$, $u \geq 1$, and the smaller annulus $u/2 \leq |x| \leq 4u$, the broad annulus $u/4 < |x| < 6u$ is admissible and gives

$$\|M\|_2 \leq C(u^{-1} + \sqrt{\delta} u^{-1/2}). \quad (12.5)$$

In the first application the broad annulus can be $u/2 < |x| < 3u$. The non-strict equality $h = \alpha u/2$ causes no boundary issue, since the broad annulus is open and still separated from the sources.

Lemma 12.2 (An upper bound on an inner deficit). *For each fixed $t > 0$ and every normalized δ -state with $n \leq 3Z$,*

$$Z - \mathbb{E} \# \{i : |x_i| < t/4\} \leq C_t(1 + \sqrt{\delta}). \quad (12.6)$$

Proof. Use Lemma 12.1 with no cut, $h = t/4$, and $u = t$. Its convergent series is bounded by $C_t(1 + \sqrt{\delta})$. The spherical average of the expected field $B_{t/4}$ on $|x| = t$ equals

$$\frac{Z - \mathbb{E} \# \{i : |x_i| < t/4\}}{t},$$

by the spherical mean of the Coulomb kernel. Bounding this average from above by the positive supremum proves the assertion, with the fixed factor t absorbed into C_t . $\square$

Proposition 12.3 (A bounded offset for neutral sectors). *There are universal constants $C, Z_0 < \infty$ such that, for $Z \geq Z_0$,*

$$E_{Z-1} \leq E + C. \quad (12.7)$$

Proof. Take a normalized ground state Ψ_* in sector N_* , where $Z \leq N_* \leq 3Z$ and $E_{N_*} = E$, as provided by Lemma 2.6. Proposition 11.1, with $D_0 = 0$, supplies a fixed sufficiently small $t > 0$ such that

$$\mathbb{E}_{\Psi_*} [Z - \# \{i : |x_i| < 2t\}] > 1$$

for all sufficiently large Z . Choose a smooth square partition $c^2 + o^2 = 1$, with $c = 1$ on $|x| \leq t$, $c = 0$ on $|x| \geq 2t$, and $|\nabla c|^2 + |\nabla o|^2 \leq Ct^{-2}$ supported in the intervening annulus. Let X be its full out data and set $q^c = Z - n_c$, where n_c is the number of core particles. Since $n_c \leq \# \{i : |x_i| < 2t\}$, we have $\mathbb{E} q^c > 1$.

We first show that the favorable event $G = \{n_c \leq Z - 1\}$ has probability bounded below independently of Z . Independently of the cut labels, observe one unordered array with the smooth compact noise law of Lemma 5.2, choosing its fixed width so that each spatial displacement is less than $t/8$. Put

$$q^{\text{obs}} = Z - \#\{\text{array entries in } B(0, t/2)\}.$$

Every entry counted here comes from a particle of radius less than $5t/8$, which must be core. Thus $q^c \leq q^{\text{obs}}$ on the enlarged probability space. Conversely every particle of radius less than $t/4$ produces an entry in $B(0, t/2)$.

For an event $A_\lambda = \{q^{\text{obs}} > \lambda\}$ of probability $p > 0$, the single-array version of (5.4) produces a normalized state Ψ_{A_λ} whose raw configuration law is the law conditioned on that event and whose offset satisfies

$$\delta \leq C_t \log^5(e/p).$$

Every compatible configuration obeys $Z - \#\{|x_i| < t/4\} > \lambda$. Hence Lemma 12.2 gives

$$\lambda \leq C_t(1 + \log^{5/2}(e/p)).$$

For sufficiently large λ this implies $p \leq e \exp(-c_t \lambda^{2/5})$; the inequality is also harmless when $p = 0$. Integrating the tail yields

$$\mathbb{E}(q^c)_+^2 \leq \mathbb{E}(q^{\text{obs}})_+^2 \leq C_t, \quad \mathbb{P}(G) = \mathbb{P}(q^c > 0) \geq \frac{(\mathbb{E}(q^c)_+)^2}{\mathbb{E}(q^c)_+^2} \geq c_t > 0. \quad (12.8)$$

Here the equality of events uses the integer-valued core count, and $\mathbb{E}(q^c)_+ \geq \mathbb{E}q^c > 1$.

We next estimate the average energy of the core slices. The localization error is $O_t(1)$ by (4.45) at offset zero. In the attractive term of (3.6), retain only the core repulsion from particles with radius less than $t/4$, which are core with probability one. Conditional integration followed by averaging the original cut labels gives

$$\mathbb{E} \sum_{i \text{ out}} \Phi_X(x_i) \leq \mathbb{E}_{\Psi_*} \sum_i o(x_i)^2 B_{t/4}(x_i). \quad (12.9)$$

This comparison of expectations uses the raw inner particles; it does not identify fields from different core slices.

Divide the nonzero terms on the right into shells $u \leq |x_i| < 2u$, $u = 2^\ell t$, $\ell \geq 0$. For each shell use an auxiliary cut of the original state, full out on that shell, supported in $u/2 < |x| < 4u$, with gradient cost Cu^{-2} , and call its data Y . All weighted shell positions and their weights $o(x_i)^2$ are then recorded. Conditional integration of the raw field, with the harmonic version of Lemma 12.1, bounds the shell expectation by $\mathbb{E}[n_u M(Y)]$, where $n_u = \#\{u \leq |x_i| < 2u\}$. Equations (4.45) and (12.4) give

$$\mathbb{E}[n_u M] \leq \|n_u\|_2 \|M\|_2 \leq C_t \frac{1 + \ell}{2^\ell t}.$$

The constants are independent of the shell index and this series is summable. For each fixed system the original attraction and repulsion expectations are finite by the form bounds, so the shell decomposition is legitimate; alternatively one can first use finitely many shells and pass to the limit. Dropping the nonnegative out kinetic energy and out-out repulsion from (3.6), and using (12.9), now proves

$$0 \leq \mathbb{E}(e_X - E) \leq C_t. \quad (12.10)$$

Every slice satisfies $e_X \geq E$, and on G sector monotonicity gives $e_X \geq E_{n_c} \geq E_{Z-1}$. Therefore

$$c_t(E_{Z-1} - E) \leq \mathbb{P}(G)(E_{Z-1} - E) \leq \mathbb{E}(e_X - E) \leq C_t,$$

which proves (12.7). This argument has used only the zero-offset ground state and the general-offset screening estimates; no neutral offset or ionization gap was assumed. $\square$

By monotonicity, $E_Z \leq E_{Z-1} \leq E + C$. We may consequently apply Proposition 11.1 again, now with this fixed value of D_0 , to every normalized neutral ground state. There are universal $s_* > 0$ and Z_1 such that

$$\int_{|x| \geq s_*} \rho_{\Psi_Z}(x) \, dx > \frac{1}{2} \quad (Z \geq Z_1). \quad (12.11)$$

In fact the deficit proposition gives a tail greater than one at an appropriate such radius. Absolute continuity of the density makes the choice between strict and non-strict radial endpoints immaterial.

13 A uniform positive ionization gap

The energy bound of Section 12 makes a distant insertion trial effective in every singly ionized atom. Averaging the insertion center recovers its unit unscreened charge; a complementary localized state controls the attraction lost when a hole is made for the orbital.

Proposition 13.1 (Positive ionization energy). *There are universal constants $c > 0$ and $Z_2 < \infty$ such that*

$$E_{Z-1} - E_Z \geq c \quad (Z \geq Z_2). \quad (13.1)$$

Proof. For $Z \geq \max(Z_0, 2)$, take a normalized ground state Φ of $H_{Z-1,Z}$. It exists by Lemma 2.5, and its offset from E is bounded by Proposition 12.3. In this proof $S \geq 1$ will be a large universal constant, fixed at the end, and $b = S^{3/5}$. Angle brackets denote normalized Lebesgue averaging over the centers $\mathcal{A}_S = \{y : S \leq |y| \leq 2S\}$.

Use the real normalized radial packet g_b from Section 3. Choose smooth real functions c_y, o_y with $c_y^2 + o_y^2 = 1$, $c_y = 0$ on $B(y, 2b)$, $c_y = 1$ outside $B(y, 3b)$, and

$$|\nabla c_y|^2 + |\nabla o_y|^2 \leq Cb^{-2} \mathbf{1}_{B(y, 3b)}.$$

Set

$$F_y = \prod_{i=1}^{Z-1} c_y(x_i), \quad n_y^{\text{near}} = \#\{i : |x_i - y| < 3b\}, \quad N_y = \mathbb{E}_\Phi n_y^{\text{near}}, \quad A_y = \mathbb{E}_\Phi F_y^2.$$

The unnormalized core $F_y \Phi$ has no particles in $B(y, 2b)$. Wedge this core with the normalized one-spin orbital $g_b(\cdot - y)$; the resulting unnormalized Z -electron vector has squared norm A_y . Spatial separation makes the terms assigning different particle labels to the two groups orthogonal. It also makes its energy equal to the core energy, plus the orbital energy times A_y , plus the direct cross repulsion. In particular, with the raw-configuration field

$$H_y^{\text{add}} = \int g_b(x - y)^2 \left(V(x) - \sum_{i=1}^{Z-1} K(x - x_i) \right) dx,$$

the multiplier identity (2.4) and the variational principle give

$$\begin{aligned} (E_Z - E_{Z-1})A_y &\leq \frac{1}{2}\mathbb{E}_\Phi|\nabla F_y|^2 + \frac{\|\nabla g\|_2^2}{2b^2}A_y - \mathbb{E}_\Phi[F_y^2 H_y^{\text{add}}] \\ &\leq Cb^{-2}(1 + N_y) - \mathbb{E}_\Phi[F_y^2 H_y^{\text{add}}]. \end{aligned} \quad (13.2)$$

This inequality also holds if $A_y = 0$, since its derivation uses the unnormalized trial and its zero norm.

For every raw configuration, averaging the full insertion field gives

$$\langle H_y^{\text{add}} \rangle \geq \frac{1}{2S}. \quad (13.3)$$

To verify this without a symmetry assumption on Φ , fix the center radius $r \in [S, 2S]$. The nuclear packet average is Z/r since $b < r$. For any fixed electron position z , the spherical average in y of its packet potential is

$$\int g_b(v)^2 \frac{1}{\max(r, |z - v|)} dv \leq \frac{1}{r}.$$

This follows by first averaging $K(y + v - z)$ over $|y| = r$ and then integrating v . There are $Z - 1$ electrons, so the averaged field is at least $1/r \geq 1/(2S)$. Radial averaging proves (13.3).

The mean number of particles near a proposed center is small. For large S , $3b < S/2$, and a particle can lie in $B(y, 3b)$ for some $y \in \mathcal{A}_S$ only if $S/2 \leq |x_i| \leq 3S$. The fraction of such centers for any particle is at most Cb^3/S^3 . Fubini and (4.45), with the bounded offset of Φ , yield

$$\langle N_y \rangle \leq C(b/S)^3 \mathbb{E}_\Phi \#\{i : S/2 \leq |x_i| \leq 3S\} \leq C(b/S)^3(1 + \sqrt{S}) \leq CS^{-7/10}. \quad (13.4)$$

Writing $p_y = 1 - A_y = \mathbb{E}_\Phi(1 - F_y^2)$, the product inequality $1 - \prod_i c_y(x_i)^2 \leq \sum_i (1 - c_y(x_i)^2)$ gives $p_y \leq N_y$.

It remains to bound the potentially positive field lost on the discarded configurations. The symmetric multiplier $G_y = \sqrt{1 - F_y^2}$ is Lipschitz, even where it vanishes, and

$$|\nabla G_y|^2 \leq Cb^{-2}n_y^{\text{near}}. \quad (13.5)$$

For a direct justification, take all products of c_y and o_y in the particle variables except the all-core product. Their Euclidean norm is G_y . The squared gradient of a norm is bounded by the sum of the squared gradients of its components; the full product partition has squared gradient sum $\sum_i (|\nabla c_y|^2 + |\nabla o_y|^2)(x_i)$. This proves (13.5) and the stated Lipschitz property.

If $p_y > 0$, put $\Phi_y^{\text{loss}} = G_y \Phi / \sqrt{p_y}$. This is a normalized antisymmetric form-domain vector. By (2.4), (12.7), and (13.5), its offset satisfies

$$\delta_y \leq C + Cb^{-2}N_y/p_y. \quad (13.6)$$

Choose S large enough also that $b < AS/L$. At each center y in the annulus, the nucleus and every electron included in J_y lie outside the radial packet. Their packet averages equal their point potentials at y . All omitted electron terms enter with a negative sign, whence $H_y^{\text{add}} \leq J_y$ pointwise in the configuration. Using (4.39) with no initial cut, now in the state Φ_y^{loss} , gives

$$\begin{aligned} \mathbb{E}_\Phi[(1 - F_y^2)H_y^{\text{add}}] &= p_y \mathbb{E}_{\Phi_y^{\text{loss}}} H_y^{\text{add}} \\ &\leq Cp_y(S^{-1} + \sqrt{\delta_y} S^{-1/2}) \\ &\leq C \left[p_y(S^{-1} + S^{-1/2}) + b^{-1}S^{-1/2}\sqrt{p_y N_y} \right]. \end{aligned} \quad (13.7)$$

Here we may assume $S/L \geq 1$; L is fixed. If $p_y = 0$, the left side is zero and the same bound holds. All expectations in this calculation are finite for a fixed system by the form bounds (the packet potential is bounded as well).

Since $p_y \leq N_y$, (13.4) implies

$$\langle \mathbb{E}_\Phi[(1 - F_y^2)H_y^{\text{add}}] \rangle \leq C \left(S^{-17/10} + S^{-6/5} + S^{-9/5} \right) = o(S^{-1}).$$

Also

$$Cb^{-2}(1 + \langle N_y \rangle) \leq C(S^{-6/5} + S^{-19/10}) = o(S^{-1}).$$

These estimates explain the width choice: any $b = S^a$ with $1/2 < a < 2/3$ would make the kinetic term and the largest averaged field loss smaller than S^{-1} . All these limits are uniform in Z and in the ground state Φ . Average (13.2) and write $F_y^2 H_y^{\text{add}} = H_y^{\text{add}} - (1 - F_y^2)H_y^{\text{add}}$. Equations (13.3) and (13.7) then show, for one sufficiently large universal choice of S ,

$$(E_Z - E_{Z-1})\langle A_y \rangle \leq -\frac{1}{4S}.$$

In particular $\langle A_y \rangle > 0$. As it is at most one, dividing gives $E_{Z-1} - E_Z \geq 1/(4S)$. This proves (13.1). $\square$

Proof of Corollary 1.3. In the notation $E_Z(n) = E(n, Z)$ of the corollary, Proposition 12.3 gives universal $C_0, Z_0 < \infty$ such that, for $Z \geq Z_0$,

$$I_1(Z) = E_Z(Z-1) - E_Z(Z) \leq E_Z(Z-1) - \inf_{n \geq 0} E_Z(n) \leq C_0.$$

The first inequality uses $\inf_{n \geq 0} E_Z(n) \leq E_Z(Z)$. Proposition 13.1 gives universal $c_0 > 0$ and $Z_2 < \infty$ such that $I_1(Z) \geq c_0$ for $Z \geq Z_2$.

Choose an integer $Z_* \geq \max\{2, Z_0, Z_2\}$. For each integer $1 \leq Z < Z_*$, the finite-sector Coulomb forms are bounded below and admit finite-energy trial states, so $E_Z(Z-1)$ and $E_Z(Z)$ are finite; the vacuum energy is $E_Z(0) = 0$. Lemma 2.5, applied with $n = Z$, gives $I_1(Z) > 0$. Thus all the finitely many remaining gaps lie in $(0, \infty)$. Set

$$c = \min\{c_0, I_1(1), \dots, I_1(Z_* - 1)\}, \quad C = \max\{C_0, I_1(1), \dots, I_1(Z_* - 1)\}.$$

These constants satisfy $0 < c \leq C < \infty$. They are universal because Z_* is universal and the exceptional gaps belong to finitely many atoms in the fixed normalization. The large- Z bounds and these choices prove the assertion for every integer $Z \geq 1$. $\square$

14 The outer tail and the half-electron radius

We now combine the positive ionization gap with a weighted shell estimate. The conditional field is estimated after changing the law by the shell mass itself; the measurability of this mass in the cut data is what avoids an additional particle-count factor.

Proposition 14.1 (Uniform exterior mass). *There exist universal $C, R_0, Z_3 < \infty$ such that every normalized neutral ground state with $Z \geq Z_3$ satisfies*

$$\int_{|x| > R} \rho_{\Psi_Z}(x) \, dx \leq \frac{C}{R} \quad (R \geq R_0). \quad (14.1)$$

Proof. Fix any such ground state $\Psi = \Psi_Z$ and write $\mathbb{E}$ for its raw configuration expectation. By Proposition 12.3 its offset is bounded by a universal constant, and by Proposition 13.1 its ionization gap is at least $c > 0$. For $u \geq 1$, choose a smooth radial cutoff $\chi_u(r) = \chi(r/u)$, where $0 \leq \chi \leq 1$, $\chi = 1$ on $[1, 2]$, and χ is supported in $[1/2, 4]$. Put

$$w_i = \chi_u(|x_i|)^2, \quad S_u = \sum_{i=1}^Z w_i, \quad m_u = \mathbb{E} S_u, \\ n_{\text{adj}} = \#\{i : u/2 \leq |x_i| \leq 4u\}, \quad N_u = \mathbb{E} n_{\text{adj}}.$$

In particular $m_u \leq N_u$. Define

$$\mathcal{B}_u(x) = V(x) - \sum_{|x_j| < u/8} K(x - x_j) = B_{u/8}(x).$$

Only its values on the support of χ_u will be used, so the selected electron never occurs among these inner sources.

Testing the weak eigen-equation by $S_u \Psi$, and decomposing the result into its i -th weighted terms, gives a useful lower bound. For fixed x_i and spin, the other $Z - 1$ variables are antisymmetric; their energy is at least E_{Z-1} times their squared norm. The selected electron's kinetic pairing is

$$\operatorname{Re} \frac{1}{2} \int \nabla_i \bar{\Psi} \cdot \nabla_i (w_i \Psi) = \frac{1}{2} \int w_i |\nabla_i \Psi|^2 - \frac{1}{4} \int (\Delta_i w_i) |\Psi|^2 \geq -Cu^{-2} \mathbb{E} \mathbf{1}_{\{u/2 \leq |x_i| \leq 4u\}}.$$

The integration by parts follows first for smooth form approximants; w_i and its first two derivatives are bounded, so it passes to the form domain. Keeping only the selected electron's repulsion from inner sources, and discarding its other nonnegative pair terms, therefore gives

$$cm_u \leq Cu^{-2} N_u + \mathbb{E} \sum_i w_i \mathcal{B}_u(x_i). \quad (14.2)$$

Suppose first that $m_u > 0$. The normalized antisymmetric state

$$\Psi_u^* = \sqrt{S_u/m_u} \Psi$$

is form admissible. Indeed $\sqrt{S_u}$ is the Euclidean norm of the vector $(\chi_u(|x_i|))_i$, hence is Lipschitz. Where $S_u > 0$,

$$|\nabla \sqrt{S_u}|^2 = \frac{\sum_i \chi_u(|x_i|)^2 |\nabla_i \chi_u(|x_i|)|^2}{S_u} \leq Cu^{-2},$$

its gradient vanishes almost everywhere on its zero set. In particular this is bounded by $Cu^{-2} n_{\text{adj}}$. Identity (2.4) shows that its offset satisfies

$$\delta_u \leq C + Cu^{-2} N_u / m_u < \infty. \quad (14.3)$$

There is no assumed positive lower bound on m_u .

We reveal every position carrying a positive shell weight so that S_u is determined by the data. Use an auxiliary cut which is full out on $u/2 \leq |x| \leq 4u$, supported in $u/4 < |x| < 8u$, and has squared gradient cost at most Cu^{-2} . Let ω denote the raw configuration, let ℓ denote these cut labels, and let $Y = Y(\omega, \ell)$ be all recorded out data. If $Q(d\ell | \omega)$ is the conditional label kernel, use exactly this kernel for both states. Their joint laws are then

$$P(d\omega, d\ell) = |\Psi(\omega)|^2 d\omega Q(d\ell | \omega), \\ P^*(d\omega, d\ell) = \frac{S_u(\omega)}{m_u} P(d\omega, d\ell). \quad (14.4)$$

Integration in ω includes the spin sum. Every particle with positive w_i is recorded by this cut, so

$$S_u(\omega) = s(Y) := \sum_{i \text{ recorded}} \chi_u(|x_i|)^2.$$

This equality includes the random labels in the transition regions: particles there have weight zero. It follows from (14.4) that

$$\frac{dP_Y^*}{dP_Y} = \frac{s(Y)}{m_u}, \quad \mathbb{E}_{P^*}[F | Y] = \mathbb{E}_P[F | Y] \quad \text{on } \{s > 0\} \quad (14.5)$$

for every bounded measurable F , and then for integrable F by truncation wherever the conditional expectations are defined. For example, multiply the asserted equality by $s(Y)$ and integrate over an arbitrary data event to check it directly. At the slice level the same fact says that $\sqrt{S_u/m_u}$ is constant in the surviving core variables and cancels when their wavefunction is normalized. The set $s = 0$ has zero P^* -mass and contributes nothing to the original weighted sum.

Let

$$h_Y(x) = \mathbb{E}_P[\mathcal{B}_u(x) | Y], \quad M(Y) = \sup_{u/2 \leq |x| \leq 4u} (h_Y(x))_+.$$

On the broad annulus $u/4 < |x| < 6u$ all the inner sources are separated from the evaluation point. Lemma 12.1 supplies continuous harmonic versions of these fields. Applying (14.5) first at a countable dense set of points and then using continuity makes the conditional fields for P and P^* equal on the entire broad annulus, outside one data-null set on $\{s > 0\}$. The two data laws are equivalent on that set. Thus the equality applies at recorded random positions as well as to the positive supremum.

Use (12.5) in the state Ψ_u^* . It gives

$$\|M\|_{L^2(P^*)} \leq C(u^{-1} + \sqrt{\delta_u} u^{-1/2}). \quad (14.6)$$

This is an application of the screening estimate for arbitrary finite offsets, not of its small-offset refinement. In particular the cut condition is uniform even if m_u is very small: as checked in Lemma 12.1, its cell coefficients obey $D_z^2/(a_z m_z) \leq C$ for every δ_u , because $a_z m_z \geq 1$.

To see explicitly how the weighted count is used, introduce the data-measurable measure

$$\eta_Y = \sum_{i \text{ recorded}} w_i \delta_{x_i}, \quad \eta_Y(\mathbb{R}^3) = s(Y).$$

Conditional integration and (14.4) now yield

$$\begin{aligned} \mathbb{E} \sum_i w_i \mathcal{B}_u(x_i) &= \mathbb{E}_P \int h_Y \, d\eta_Y \\ &= m_u \mathbb{E}_{P^*} \left[\frac{1}{s(Y)} \int h_Y \, d\eta_Y \right] \\ &\leq m_u \mathbb{E}_{P^*} M \leq C m_u (u^{-1} + \sqrt{\delta_u} u^{-1/2}). \end{aligned} \quad (14.7)$$

The quotient may be defined as zero on $s = 0$. This exact change of measure accounts for all shell weights, so there is no further particle-count factor in the last bound.

Substituting (14.3) into (14.7), and then into (14.2), gives

$$\begin{aligned} c m_u &\leq C u^{-2} N_u + C m_u (u^{-1} + u^{-1/2}) + C u^{-3/2} \sqrt{m_u N_u} \\ &\leq C (u^{-2} + u^{-3/2}) N_u + C m_u (u^{-1} + u^{-1/2}). \end{aligned}$$

The second line uses $m_u \leq N_u$. When $m_u = 0$ it holds trivially without introducing a biased law. For all sufficiently large u , absorb the last term into the left side. Applying (4.45) in the original state, whose offset is uniformly bounded, gives $N_u \leq C(1 + \sqrt{u})$ and hence

$$\mathbb{E}\#\{i : u \leq |x_i| \leq 2u\} \leq m_u \leq C/u. \quad (14.8)$$

Finally, sum over $u = 2^j R$, $j \geq 0$. The density gives zero mass to the intervening spheres, and $\sum_{j \geq 0} (2^j R)^{-1} = 2/R$. This proves (14.1). $\square$

Completion of the proof of Theorem 1.2. For all sufficiently large Z , (12.11) supplies a fixed $s_* > 0$ for which the exterior mass exceeds $1/2$. Monotonicity of that mass as a function of the radius implies $R(\Psi_Z) \geq s_*$. Proposition 14.1 supplies a fixed large R_* for which the exterior mass is at most $1/2$, and hence $R(\Psi_Z) \leq R_*$.

For clarity we give uniform arguments over the entire ground-state space for each of the finitely many remaining charges. Fix such a Z . The strict gap $E_Z < E_{Z-1}$ and the compactness conclusion in Lemma 2.5 show that every sequence of normalized neutral ground states has a strongly L^2 convergent subsequence. In particular their configuration laws are uniformly tight. To see directly why compactness gives the required uniform tail, suppose otherwise and choose radii $R_k \rightarrow \infty$ and normalized ground states Ψ_k with

$$\int_{|x| > R_k} \rho_{\Psi_k}(x) \, dx > \frac{1}{2}.$$

After a subsequence $\Psi_k \rightarrow \Psi$ strongly in L^2 . The bounded multiplication operator $\sum_i \mathbf{1}_{\{|x_i| > R_k\}}$ has norm at most Z , so its expectations in Ψ_k and Ψ differ by at most $2Z\|\Psi_k - \Psi\|_2$. Its expectation in the fixed Ψ tends to zero by dominated convergence, a contradiction. Thus a finite upper radius works uniformly for this Z .

For the lower bound, the kinetic estimate (3.3), with the fixed finite offset $E_Z - E$, gives

$$\int \rho_{\Psi_Z}^{5/3} \leq C_Z$$

uniformly over its normalized ground states. Hölder's inequality therefore yields

$$\int_{|x| < r} \rho_{\Psi_Z}(x) \, dx \leq |B(0, r)|^{2/5} \left(\int \rho_{\Psi_Z}^{5/3} \right)^{3/5} \leq C_Z r^{6/5}.$$

Choose a positive r_Z making this less than $Z - 1/2$. The exterior mass then exceeds $1/2$, uniformly over the ground states, so $R(\Psi_Z) \geq r_Z$.

Taking the minimum of s_* and these finitely many positive lower radii, and the maximum of R_* and the finitely many upper radii, gives universal $0 < c \leq C < \infty$. All density measures used above are absolutely continuous. Their exterior masses are consequently continuous in the radius, start at Z , and tend to zero. Thus the infimum defining the half-electron radius has the stated bounds for every normalized neutral ground state, as required. $\square$

15 Interfaces for atomic asymptotics

The estimates used in questions about ionization energies and outer radii have different offset requirements. We collect their precise scope here; the complete proofs are in the preceding sections.

Arbitrary energy offsets. For a finite-sector state of energy at most $E + \delta$, the exact cut identity, coherent insertion, and fine-density lower bounds are Lemmas 3.4, 3.5, and 3.6. In atomic distance geometry, Proposition 4.5 and Corollary 4.9 allow every finite $\delta \geq 0$; their count unit is $\max(a^{-3}, 1) + \sqrt{\delta a}$. For sectors $n \leq 3Z$, Lemma 12.1 keeps this dependence in its complete convergent series for conditional inner fields. The fine comparison Lemma 4.6 has the additional restriction $\delta \leq a^{-7+.02}$. These are distinct assertions.

Events of a sector ground state. For a ground state of energy E_n , Lemma 5.2 bounds the cost above E_n by $C\ell_j^{-2} \log^5(e/p)$, independent of the number of particles or arrays. The offset above E is therefore $E_n - E + C\ell_j^{-2} \log^5(e/p)$. This exact baseline dependence is available when the offset grows. The fixed-offset comparison Proposition 10.1, by contrast, first fixes a finite $D_0 \geq E_n - E$, then takes s sufficiently small according to D_0 and the stated band and accuracy parameters. It is not a uniform comparison at arbitrary growing D_0 .

Common observations and intermediate fields. The master-kernel formula and its universal constants in (10.2) do not depend on the inverse-comparison heights or accuracy. To use several comparisons with the same conditional density and field, choose s to satisfy all their corresponding smallness conditions in Proposition 10.1, then fix the resulting packet widths and observation arrays. This applies to any collection admitting such a common s ; for finitely many fixed requests, the minimum of their thresholds suffices. Proposition 11.2 uses exactly these packets at every intermediate index; its error bound (11.3) is in expectation before the last array is forgotten. Its differential inequality holds globally after cancellation of the nuclear singularity, its lower barrier persists at every index, and its exterior equality holds beyond $L_1 r_j$. The construction may be stopped at any prefix without changing the master smoothing.

Classical comparison and the neutral offset. The Thomas–Fermi minimizer and point-source comparison required for coefficient identification are Lemmas 7.1 and 7.2, valid for every positive real point charge. The quantum offset needed for neutral states is Proposition 12.3. It follows from zero-offset deletion and is uniform over all neutral ground states by sector monotonicity. These are independent inputs: the excess-charge conclusion itself does not replace either the Thomas–Fermi estimates or the neutral-sector argument.

References

- [1] R. Benguria and Elliott H. Lieb. The most negative ion in the Thomas–Fermi–von Weizsäcker theory of atoms and molecules. *Journal of Physics B: Atomic and Molecular Physics*, 18(6): 1045–1059, 1985. doi: 10.1088/0022-3700/18/6/006.
- [2] Charles L. Fefferman and Luis A. Seco. Asymptotic neutrality of large ions. *Communications in Mathematical Physics*, 128(1):109–130, 1990. doi: 10.1007/BF02097048.
- [3] Enrico Fermi. Un metodo statistico per la determinazione di alcune proprietà dell’atomo. *Rendiconti dell’Accademia Nazionale dei Lincei*, 6:602–607, 1927.
- [4] Dirk Hundertmark, Nikolaos Pattakos, and Marvin Raimund Schulz. On the excess charge problem of atoms, 2025. URL <https://arxiv.org/abs/2504.18487>. Version 1, 25 April 2025.

- [5] Victor Ivrii. Asymptotics of the ground state energy of heavy molecules and related topics, 2012. arXiv:1210.1132v2, 28 March 2017.
- [6] Mathieu Lewin. Geometric methods for nonlinear many-body quantum systems. *Journal of Functional Analysis*, 260(12):3535–3595, 2011. doi: 10.1016/j.jfa.2010.11.017.
- [7] Mathieu Lewin. Some open mathematical problems concerning charged quantum particles. *Comptes Rendus. Physique*, 26:369–380, 2025. doi: 10.5802/crphys.249.
- [8] Elliott H. Lieb. Bound on the maximum negative ionization of atoms and molecules. *Physical Review A*, 29(6):3018–3028, 1984. doi: 10.1103/PhysRevA.29.3018.
- [9] Elliott H. Lieb and Barry Simon. Thomas–Fermi theory revisited. *Physical Review Letters*, 31(11):681–683, 1973. doi: 10.1103/PhysRevLett.31.681.
- [10] Elliott H. Lieb and Barry Simon. The Thomas–Fermi theory of atoms, molecules and solids. *Advances in Mathematics*, 23(1):22–116, 1977. doi: 10.1016/0001-8708(77)90108-6.
- [11] Elliott H. Lieb and Walter E. Thirring. Bound for the kinetic energy of fermions which proves the stability of matter. *Physical Review Letters*, 35(11):687–689, 1975. doi: 10.1103/PhysRevLett.35.687. Erratum: *Physical Review Letters* 35 (1975), 1116.
- [12] Elliott H. Lieb, Israel M. Sigal, Barry Simon, and Walter Thirring. Asymptotic neutrality of large- Z ions. *Physical Review Letters*, 52(12):994–996, 1984. doi: 10.1103/PhysRevLett.52.994.
- [13] Elliott H. Lieb, Israel M. Sigal, Barry Simon, and Walter Thirring. Approximate neutrality of large- Z ions. *Communications in Mathematical Physics*, 116(4):635–644, 1988. doi: 10.1007/BF01224904.
- [14] Phan Thành Nam. New bounds on the maximum ionization of atoms. *Communications in Mathematical Physics*, 312(2):427–445, 2012. doi: 10.1007/s00220-012-1479-y.
- [15] OpenAI. Generalized ionization energies for full Coulomb atoms. OpenAI Math Release preprint [OAI:Generalized-ionization-energies-for-full-Coulomb-atoms-September-24-2026](#), 2026.
- [16] OpenAI. Generalized outer-electron radii of neutral Coulomb atoms. OpenAI Math Release preprint [OAI:Generalized-outer-electron-radii-of-neutral-Coulomb-atoms-September-24-2026](#), 2026.
- [17] Michel Rumin. Balanced distribution-energy inequalities and related entropy bounds. *Duke Mathematical Journal*, 160(3):567–597, 2011. doi: 10.1215/00127094-1444305.
- [18] Luis A. Seco, Israel M. Sigal, and Jan Philip Solovej. Bound on the ionization energy of large atoms. *Communications in Mathematical Physics*, 131(2):307–315, 1990. doi: 10.1007/BF02161416.
- [19] Barry Simon. Schrödinger operators in the twenty-first century. In A. Fokas, A. Grigoryan, T. Kibble, and B. Zegarlinski, editors, *Mathematical Physics 2000*, pages 283–288. Imperial College Press, London, 2000. doi: 10.1142/9781848160224_0014.
- [20] Jan Philip Solovej. Proof of the ionization conjecture in a reduced Hartree–Fock model. *Inventiones mathematicae*, 104:291–311, 1991. doi: 10.1007/BF01245077.

- [21] Jan Philip Solovej. The ionization conjecture in Hartree–Fock theory. *Annals of Mathematics*, 158(2):509–576, 2003. doi: 10.4007/annals.2003.158.509.
- [22] Jan Philip Solovej. A new look at Thomas–Fermi theory. *Molecular Physics*, 114(7–8): 1036–1040, 2016. doi: 10.1080/00268976.2015.1130273.
- [23] L. H. Thomas. The calculation of atomic fields. *Proceedings of the Cambridge Philosophical Society*, 23(5):542–548, 1927. doi: 10.1017/S0305004100011683.
- [24] G. M. Zhislin. A study of the spectrum of the Schrödinger operator for a system of several particles. *Trudy Moskovskogo Matematicheskogo Obshchestva*, 9:81–120, 1960. URL <https://www.mathnet.ru/eng/mmo97>. Erratum: volume 10 (1961), 437.
- [25] G. M. Zhislin and A. G. Sigalov. The spectrum of the energy operator for atoms with fixed nuclei on subspaces corresponding to irreducible representations of the group of permutations. *Izvestiya Akademii Nauk SSSR. Seriya Matematicheskaya*, 29(4):835–860, 1965. URL <https://www.mathnet.ru/eng/im3076>.